Probability & Stochastic Processes

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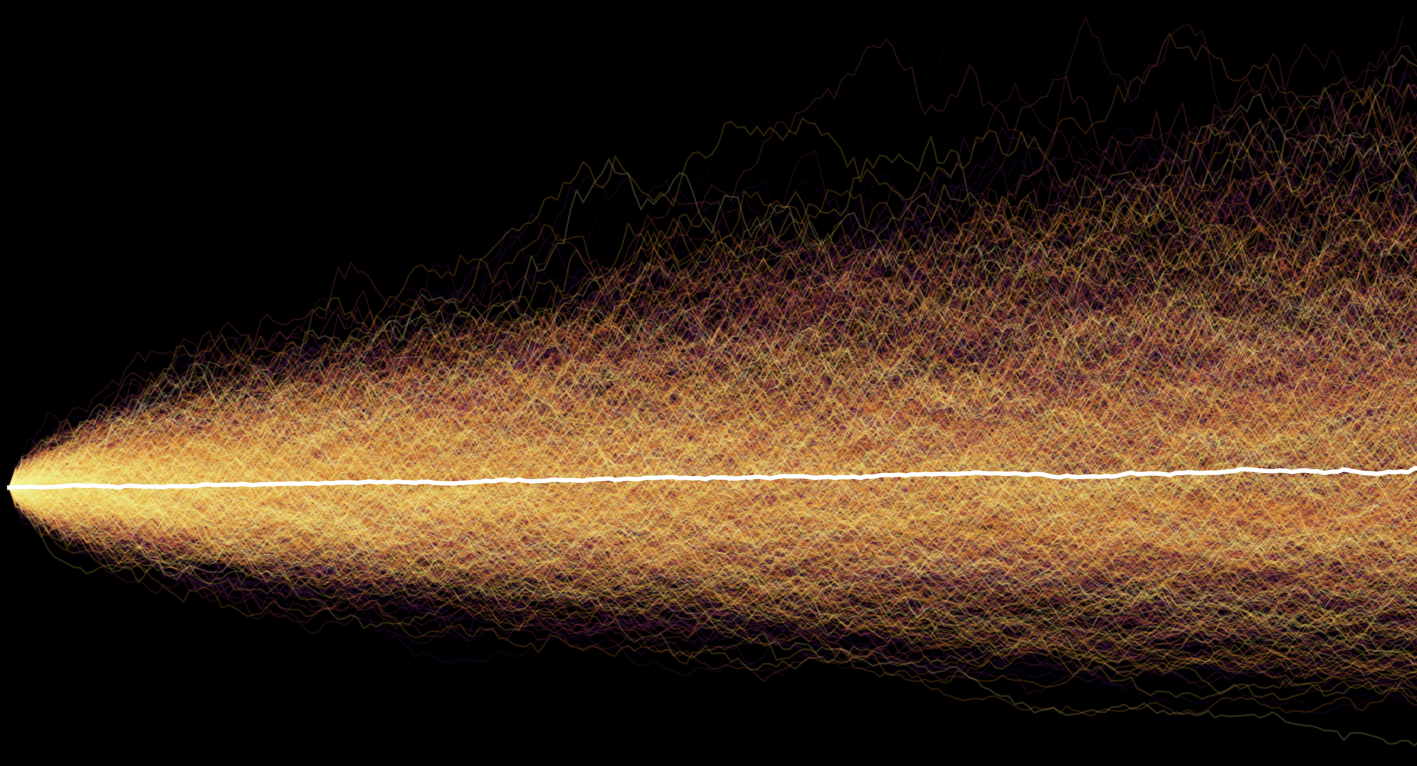


Simulated Geometric Brownian Motions

Preface

This book is intended as a concise reference for the central concepts and results of probability and stochastic processes. It is not designed for instruction or self-contained learning, but as a structured handbook for readers who already possess familiarity with the material and require a clear, reliable way to refresh definitions, recall key formulas, and connect ideas in applied settings. The emphasis throughout is on essential structure and final expressions rather than derivations or examples, with brief conceptual notes included only where they support interpretation or use.

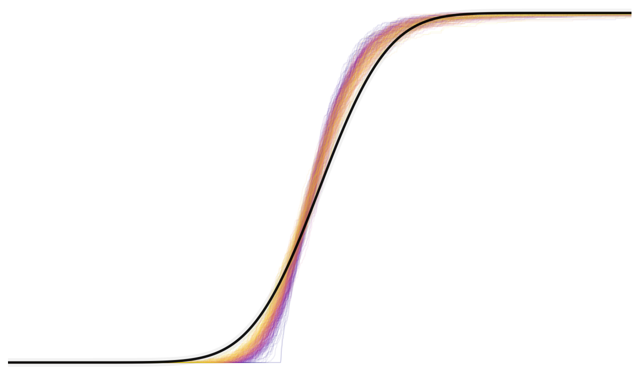
The material is organized linearly, beginning with probability theory and progressing into stochastic processes, reflecting the natural conceptual dependency between the two. Topics are presented with an emphasis on canonical formulas, limiting results, and modeling primitives that frequently arise in quantitative finance, financial engineering, machine learning, and related applied fields. Visualizations are used in place of worked examples to reinforce intuition and highlight behavior that is most relevant in practice.

The content is compiled and synthesized from coursework and publicly available sources. While the underlying theory is standard, this work represents a deliberate effort to curate, structure, and distill the material into a reference format optimized for rapid lookup and practical application. All figures and visualizations were generated programmatically. This document is an evolving work and reflects an ongoing process of refinement; as such, errors or omissions may remain.

Contents

Part I Probability

Probability is the language used whenever outcomes are uncertain but decisions still must be made. It appears wherever data are noisy, systems are complex, and perfect information is unavailable, which is to say, almost everywhere. From finance and engineering to machine learning, physics, and economics, probability provides the structure that allows uncertainty to be modeled, quantified, and acted upon.



What makes probability compelling is not that it predicts individual outcomes, but that it reveals order beneath randomness. Seemingly erratic behavior gives rise to stable patterns when viewed at the right scale. Aggregates behave differently from individuals, averages become reliable, and variability itself follows precise laws. These regularities make it possible to reason about risk, infer hidden structure from data, and design systems that perform reliably in uncertain environments.

This section develops the mathematical machinery that supports those ideas. The concepts introduced here form the foundation for inference, estimation, stochastic modeling, and learning from data. Probability does not remove uncertainty, but it makes uncertainty manageable, and, in many settings, exploitable.

Chapter 1: Basics of Probability Theory

Probability theory begins with a precise description of what outcomes are possible, what collections of outcomes can be meaningfully distinguished, and how likelihood is assigned to those collections. Random variables then arise as numerical observables defined on this structure, linking abstract events to measurable quantities. This chapter collects the basic set–theoretic and measure–theoretic foundations on which all subsequent constructions rest.

1 Sample Space and Events

Sample Space. Let X denote the underlying set of all possible outcomes of an experiment. In probability theory this set is called the *sample space* and is typically denoted by

$$\Omega \equiv X.$$

Each element $x \in \Omega$ represents one possible realization of the experiment.

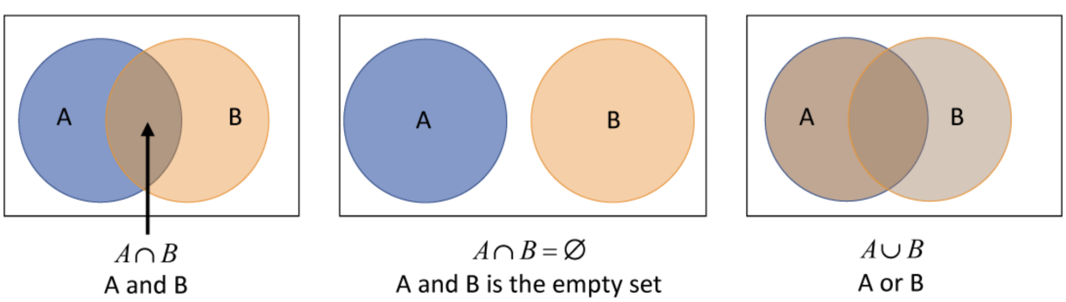
Power Set. The collection of all subsets of X is called the *power set* and is denoted by

$$2^X = \{A : A \subseteq X\}.$$

Thus every set A satisfying $A \subseteq X$ belongs to 2^X. Elements of 2^X are themselves sets of outcomes.

Set Operations. For events $A, B \subseteq X$, the basic operations are:

- Union (*or*):
  $$A \cup B = \{x \in X : x \in A \text{ or } x \in B\}.$$
  This represents the event that at least one of A or B occurs.
- Intersection (*and*):
  $$A \cap B = \{x \in X : x \in A \text{ and } x \in B\}.$$
  This represents the event that both A and B occur.
- Complement (*not*):
  $$A^c = \{x \in X : x \notin A\}.$$
  This represents the event that A does not occur.



These operations turn collections of events into an algebraic structure. A $\sigma-$algebra is precisely a collection of subsets of X that is closed under these operations (in the countable sense for unions), ensuring that logical combinations of observable events are themselves observable.

Events. An *event* is a subset of the sample space, i.e. a set $A \subseteq X$. However, in general not every subset is assigned a probability. Only those subsets that belong to a specified collection of measurable sets are considered legitimate events.

2 $\sigma-$Algebras

A probability model must specify not only the set of all possible outcomes, but also which subsets of outcomes are distinguishable given the available information. This information structure is encoded by a $\sigma-$algebra, which determines the collection of events to which probabilities can consistently be assigned and on which conditioning and limits are well-defined.

$\sigma-$Algebra (CUT). Let $\mathcal{F} \subset 2^X$ be a subcollection of the power set. $\mathcal{F}$ is called a *$\sigma-$algebra* (or $\sigma-$field) if it satisfies the three CUT properties:

1. Closed under **C**omplement: If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.
2. Closed under Countable **U**nions: If $A_1, A_2, \ldots \in \mathcal{F}$, then
$$\bigcup_{k=1}^{\infty} A_k \in \mathcal{F}.$$
3. **T**otal Set:
$$X \in \mathcal{F}.$$

Measurable Space. The pair $(X, \mathcal{F})$ is called a *measurable space* if $\mathcal{F} \subset 2^X$ is a $\sigma-$algebra.

Events as Measurable Sets. A set $A \subseteq X$ is called an *event* (or is said to be *measurable*) if

$$A \in \mathcal{F},$$

where $\mathcal{F}$ is a $\sigma-$algebra on X.
Thus, probability theory does not assign probabilities to arbitrary subsets of X, but only to those belonging to the information structure $\mathcal{F}$.
Equivalently, closure under complements and countable unions implies closure under countable intersections by De Morgan's laws.

Why $\sigma-$Algebras Are Required. Probability is defined on collections of events that are stable under limiting operations. Countable unions and intersections arise naturally from:

- Sequences of experiments and repeated trials,
- Limits of events (e.g. $\limsup$ and $\liminf$),

- Long–run behavior and almost sure convergence,
- Conditioning and Bayes' updates on increasingly refined information.

Closure under countable operations guarantees that probabilities remain well-defined when passing to limits, forming complements, or conditioning on accumulated information.

Information Interpretation. A σ−algebra represents an *information structure.* Each $A \in \mathcal{F}$ corresponds to a yes/no question about the outcome whose answer is observable. Larger σ−algebras encode finer information; smaller ones encode coarser information.

Filtration. In stochastic processes, information evolves over time. A *filtration* is an increasing family of σ−algebras

$$\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}_t \subseteq \cdots$$

where $\mathcal{F}_t$ represents all events observable up to time t.
This formalizes the notion of accumulating information and will later underlie the definitions of adapted processes, martingales, stopping times, and conditional expectation in continuous time.

3 Probability Measure and Axioms

Once an information structure is fixed, probability is introduced as a numerical assignment to events that is consistent with logical operations and limiting procedures. A probability measure formalizes this assignment by imposing axioms that ensure coherence under disjoint unions, complements, and infinite aggregation of events.

Probability Measure (Kolmogorov Axioms / CAT). Let $(X, \mathcal{F})$ be a measurable space, where $\mathcal{F} \subset 2^X$ is a σ−algebra. A function

$$P : \mathcal{F} \to [0, 1]$$

is called a *probability measure* if it satisfies the Kolmogorov axioms (CAT):

1. **C**ountable **A**dditivity: For any sequence of pairwise disjoint events $A_1, A_2, \ldots \in \mathcal{F}$,

$$P\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} P(A_k).$$

   If $A \cap B = \varnothing$, then

$$P(A \cup B) = P(A) + P(B).$$

2. **T**otal Set X gets unit mass:

$$P(X) = 1.$$

Equivalently, these conditions are the Kolmogorov axioms: nonnegativity, normalization, and countable additivity.

Probability Space. The triple

$$(\Omega, \mathcal{F}, P)$$

is called a *probability space*, where:

- Ω is the sample space,
- $\mathcal{F}$ is a σ–algebra of events (CUT),
- P is a probability measure on $\mathcal{F}$ (CAT).

Continuity Properties of Probability. These properties describe how probabilities behave under set inclusion and under limits of increasing or decreasing sequences of events. They formalize the compatibility between probability and infinite set operations, which underlies almost sure convergence, long–run events, and conditioning on accumulating information.
Monotonicity. If $A \subseteq B$, then

$$P(A) \le P(B).$$

Continuity from Below. For an increasing sequence $A_1 \subseteq A_2 \subseteq \cdots$,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} P(A_n).$$

Continuity from Above. For a decreasing sequence $A_1 \supseteq A_2 \supseteq \cdots$ with $P(A_1) < \infty$,

$$P\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} P(A_n).$$

These properties justify taking limits of probabilities of sequences of events and ensure that probability is compatible with infinite operations, long–run behavior, and conditioning.

4 Random Variables as Measurable Maps

While events describe yes/no outcomes, most models require numerical quantities such as counts, times, or prices. Random variables formalize this by mapping each outcome in the sample space to a real number in a way that is consistent with the underlying information structure, allowing probabilities to be assigned to numerical statements and thresholds.

Random Variable. Let $(\Omega, \mathcal{F})$ be a measurable space. A *random variable* is a measurable function

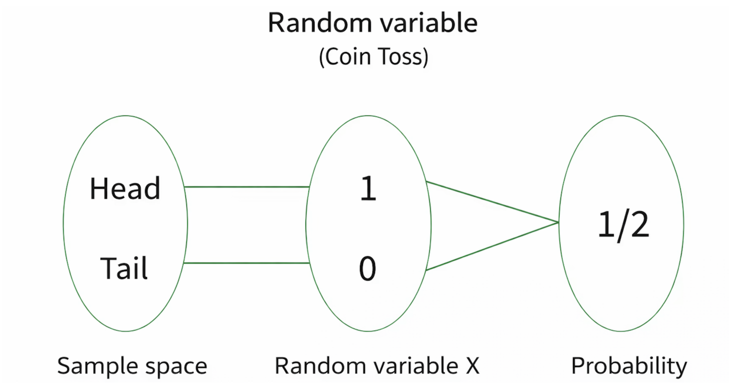
$$X : \Omega \to \mathbb{R},$$

meaning that for every Borel set $B \subset \mathbb{R}$,

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}.$$

Equivalently, all events of the form $\{X \le x\}$ must belong to $\mathcal{F}$ for every $x \in \mathbb{R}$.

A random variable assigns a numerical value to each outcome of the experiment in a way that is compatible with the information structure. Measurability ensures that numerical statements about the outcome correspond to observable events and can therefore be assigned probabilities.



Events Generated by a Random Variable. For any real number x, the set

$$\{X \le x\} = \{\omega \in \Omega : X(\omega) \le x\}$$

is an event in $\mathcal{F}$. Such threshold events generate the cumulative distribution function and encode all probabilistic information about X.

Types of Random Variables. The nature of X is determined by the structure of its induced measure P_X:

- *Discrete:* P_X is supported on a countable set and is described by a PMF.
- *Continuous:* P_X is absolutely continuous with respect to Lebesgue measure and is described by a PDF.
- *Mixed:* P_X has both discrete and continuous components.

Thus, discreteness or continuity is not a property of the mapping X itself, but of the probability measure it induces on $\mathbb{R}$.

Chapter 2: Events, Information, & Conditioning

When we talk about probability, we are really talking about events, how they relate to one another, and how information alters their likelihood. At its core, this subject concerns the structure of uncertainty: what it means for occurrences to be compatible, independent, or conditioned on further knowledge. The formulas collected here express these relationships in their most compact form. They serve as the basic mechanisms by which probability is assigned, combined, and updated, independent of any specific application.

1 Probability Rules and Independence

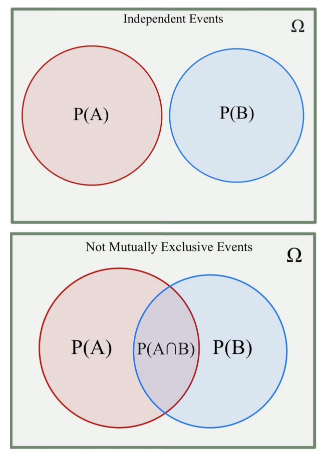
Independence. Two events A and B are *independent* if

$$P(A \cap B) = P(A)\,P(B).$$

This equality expresses that the occurrence of one event carries no information about the other: the probability of the intersection factors into the product of the individual probabilities. Equivalently,

$$P(B \mid A) = P(B) \quad \text{and} \quad P(A \mid B) = P(A),$$

whenever the conditional probabilities are defined.



Conditional Probability. For events A and B with $P(A) > 0$, the conditional probability of B given A is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$

Conditioning restricts attention to the portion of the sample space where A occurs, and renormalizes probabilities accordingly.

Law of Total Probability. If $\{H_k\}$ is a partition of the sample space Ω, then

$$P(E) = \sum_k P(H_k)\,P(E \mid H_k).$$

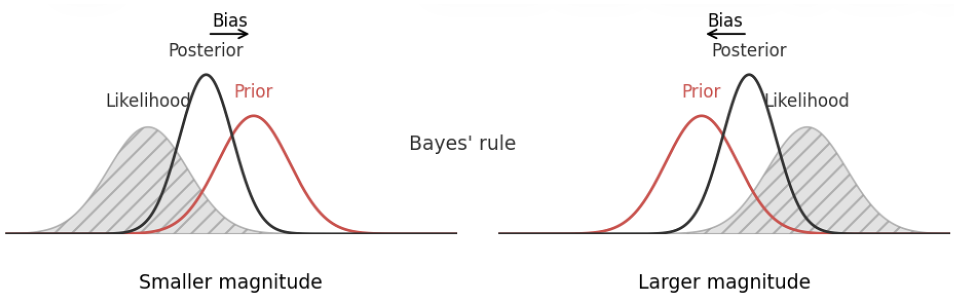
This expresses the probability of E as a weighted combination over disjoint scenarios. Each term corresponds to the contribution of E within H_k, scaled by $P(H_k)$.
Special case: when $\Omega = A \cup A^c$,

$$P(B) = P(A)\,P(B \mid A) + P(A^c)\,P(B \mid A^c).$$

Bayes' Theorem. For a partition $\{H_k\}$ of Ω and an event E with $P(E) > 0$,

$$P(H_j \mid E) = \frac{P(E \mid H_j)\, P(H_j)}{\sum\limits_k P(E \mid H_k)\, P(H_k)}.$$

Conceptually, Bayes' theorem re-weights the prior probabilities $\{P(H_j)\}$ by how compatible each hypothesis is with the observed evidence E, via the likelihoods $\{P(E \mid H_j)\}$, and then re-normalizes so that the updated probabilities $\{P(H_j \mid E)\}$ again sum to one.



Bayes shifts beliefs toward data & disagreement between prior and likelihood shows up as bias

Example. If $\{H_1, H_2\}$ partitions Ω and

$$P(H_1 \mid E) = \frac{a}{a+b}, \qquad P(H_2 \mid E) = \frac{b}{a+b},$$

then a and b represent the relative contributions of H_1 and H_2 to the likelihood of E. The posterior probabilities are obtained by normalizing these contributions.

Issue Spotting Sequence (PUTCB). This sequence provides a structured checklist for identifying which probability tools are needed in a given problem. The mnemonic *PUTCB* stands for *"Party Unconditionally To Conquer Bayes"*.

P:	Partition?	$\{H_k\}$ or $A \cup A^c$
U:	Unconditional probability?	$P(B)$
T:	Total probability:	$P(E) = \sum_k P(H_k)\, P(E \mid H_k)$
C:	Conditional probability:	$P(B \mid A) = \dfrac{P(A \cap B)}{P(A)}$
B:	Bayes' theorem:	$P(H_j \mid E) = \dfrac{P(E \mid H_j) P(H_j)}{\sum_k P(E \mid H_k) P(H_k)}$ with $P(H_j)$ (prior), $P(E \mid H_j)$ (likelihood), $P(H_j \mid E)$ (posterior).

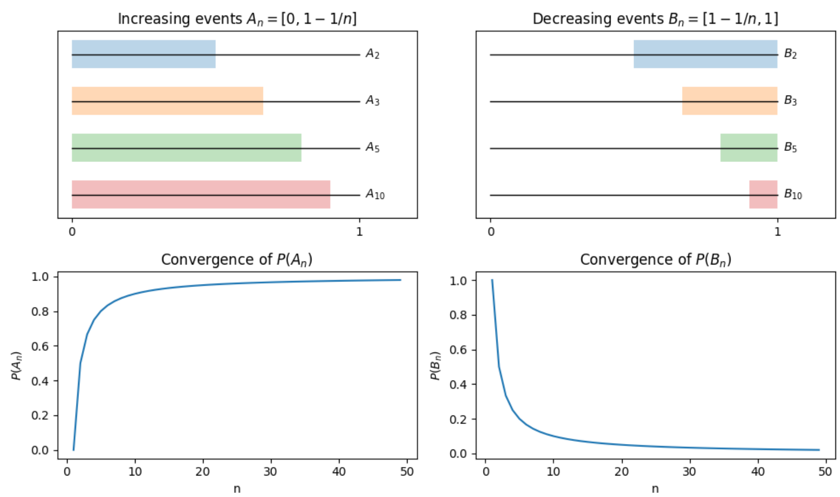
2 Long-Run Events & Borel–Cantelli

Probability Limits. Sequences of events play the role of limits for sets, describing what eventually happens as n increases. For a sequence $\{A_n\}$,

$$\lim_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} A_n \quad \text{if } A_k \subseteq A_{k+1} \quad \text{(increasing sequence)},$$

$$\lim_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} A_n \quad \text{if } A_k \supseteq A_{k+1} \quad \text{(decreasing sequence)}.$$

In the increasing case, the limit collects everything that eventually occurs; in the decreasing case, it retains only what persists forever.



Increasing and decreasing events converge in both sets and probabilities.

Continuity of Probability. Continuity of probability states that limits of events and limits of their probabilities are compatible for monotone sequences. If $\{A_n\}$ is increasing, then

$$P\left(\lim_{n\to\infty} A_n\right) = \lim_{n\to\infty} P(A_n) = P\left(\bigcup_{n=1}^{\infty} A_n\right).$$

If $\{A_n\}$ is decreasing, then

$$P\left(\lim_{n\to\infty} A_n\right) = \lim_{n\to\infty} P(A_n) = P\left(\bigcap_{n=1}^{\infty} A_n\right).$$

Thus, for monotone sequences, taking limits and taking probabilities commute.

Boole's Inequality. Boole's inequality provides a general upper bound on the probability of a union of events:

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} P(A_n).$$

The probability of at least one A_n occurring cannot exceed the sum of their individual probabilities; overlapping events only make the union smaller, never larger.

Infinitely Often Events. A sequence of events $\{A_k\}$ is said to occur *infinitely often (i.o.)* if

$$\forall n \geq 1,\ \exists k \geq n \text{ such that } A_k \text{ occurs,}$$

which is represented by the event

$$\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k.$$

This event captures the long-run behavior that A_k keeps happening for arbitrarily large indices k, rather than eventually stopping.

Borel–Cantelli Lemma. Let $\{A_n\}$ be a sequence of events.

1. If $\sum_{n=1}^{\infty} P(A_n) < \infty$, then

$$P\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right) = 0.$$

   In this case, the total probability mass assigned to the A_n is finite, and the events occur only finitely many times almost surely.

2. If $\sum_{n=1}^{\infty} P(A_n) = \infty$ and the events $\{A_n\}$ are independent, then

$$P\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right) = 1.$$

   Here, the accumulated probability is infinite and, under independence, the events occur infinitely often almost surely.

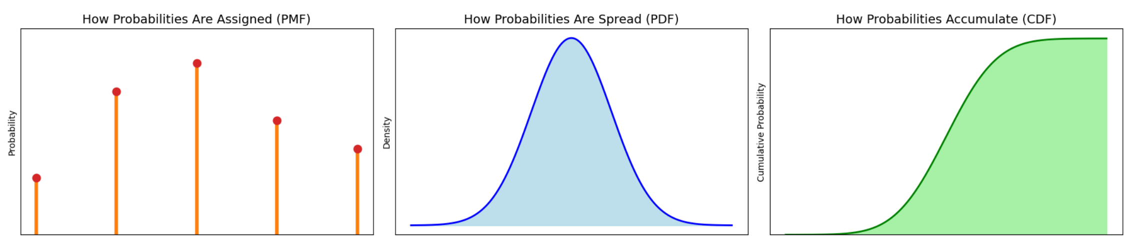
The lemma thus distinguishes sequences that die out from those that persist indefinitely.

Chapter 3: Distributions

Randomness is not only about whether events occur, but also about how frequently they occur and in what patterns. A probability distribution assigns weight to possible outcomes and thereby determines the characteristic *shape* of uncertainty. Some distributions cluster tightly around a typical value, while others possess heavy tails in which extreme events, though rare, dominate long–run risk and variability. For example, in heavy–tailed models a single observation can outweigh thousands of typical ones, and in memoryless waiting–time models the expected remaining time is unchanged no matter how long one has already waited. Distributions therefore encode far more than averages: they describe the geometry of chance itself, governing concentration, asymmetry, and the frequency of rare but consequential events.

1 Distribution Functions

A random variable can be described in several equivalent but conceptually distinct ways. The appropriate representation depends on whether the support is discrete, continuous, or mixed. The three fundamental objects are the *probability mass function*, the *probability density function*, and the *cumulative distribution function*.



The PMF assigns probability to isolated points, the PDF distributes probability continuously so that areas under the curve represent interval probabilities, and the CDF records the accumulated probability up to a given threshold, increasing from 0 to 1 as x grows.

Probability Mass Function (PMF). For a *discrete* random variable X with countable support $\mathcal{X}$, the probability mass function assigns probability to each attainable value,

$$f_X(x) = P(X = x), \qquad x \in \mathcal{X}.$$

It satisfies the normalization and nonnegativity conditions

$$f_X(x) \geq 0, \qquad \sum_{x \in \mathcal{X}} f_X(x) = 1.$$

Probabilities of events are obtained by summing the PMF over the corresponding subsets of the support. In this way, the PMF describes how the total probability mass is allocated among the individual outcomes of a discrete experiment.

Cumulative Distribution Function (CDF). The cumulative distribution function of a random variable X is defined by

$$F_X(x) = P(X \le x), \qquad x \in \mathbb{R}.$$

It is a nondecreasing, right–continuous function with boundary limits

$$\lim_{x \to -\infty} F_X(x) = 0, \qquad \lim_{x \to +\infty} F_X(x) = 1.$$

For discrete distributions, F_X is a step function with jumps at the support points, while for continuous distributions it increases smoothly. In all cases, the CDF represents the accumulated probability up to the level x, and therefore quantifies the probability that the random variable has not exceeded a given threshold.

Probability Density Function (PDF). A distribution is *absolutely continuous* if there exists a function f such that its CDF can be written as

$$F_X(x) = \int_{-\infty}^{x} f(t)\, dt.$$

In this case the density satisfies

$$f(x) = F_X'(x), \qquad f(x) \ge 0, \qquad \int_{-\infty}^{\infty} f(x)\, dx = 1.$$

The PDF characterizes how probability is distributed along the real line, and probabilities of intervals are obtained by integration:

$$P(a \le X \le b) = \int_{a}^{b} f(x)\, dx.$$

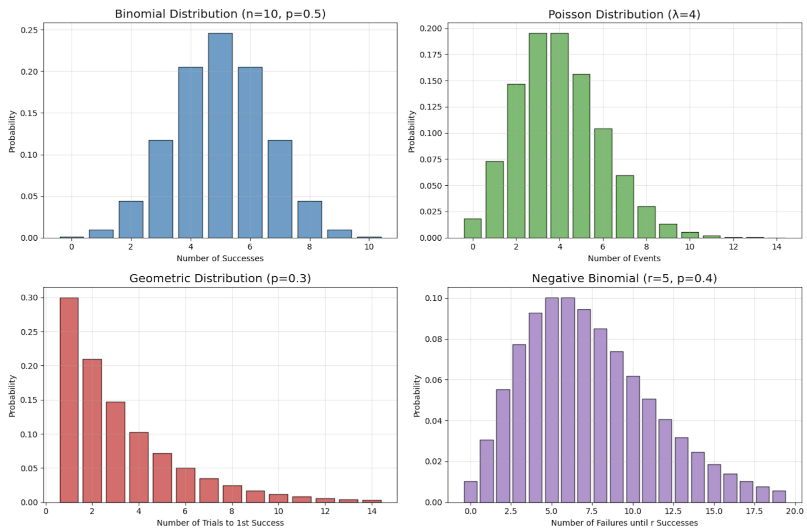
Thus, while individual points have zero probability, the density indicates how tightly probability mass is concentrated locally, with areas under the curve corresponding to probabilities of ranges. A global density exists only for absolutely continuous distributions; discrete or mixed distributions may not admit a PDF, but every random variable nevertheless possesses a well-defined CDF.

Table 1: Probability Distribution Functions

Object	**Support Type**	**Formula**	**Use**
PMF	Discrete	$f_X(x) = P(X = x)$	Probabilities by summation
PDF	Continuous	$f_X(x) = F_X'(x)$	Probabilities by integration
CDF	All	$F_X(x) = P(X \le x)$	Universal description of distribution

2 Core Discrete Distributions

Discrete distributions assign probability to individual points rather than intervals, so uncertainty is organized over a countable support in which each outcome can, in principle, be enumerated. Such models arise naturally in experiments based on repeated trials, counting events, or sampling from finite populations, and many of the standard distributions are linked through limiting arguments or structural relationships.



Bernoulli Distribution (p)**.** The Bernoulli distribution models a single binary experiment with two possible outcomes, commonly labeled success and failure, occurring with probabilities p and $1-p$, respectively. Its support is $\{0, 1\}$, where the value 1 typically represents success. This distribution serves as the fundamental building block for most discrete count models based on repeated trials.

$$P(X = x) = p^x(1-p)^{1-x}, \qquad x \in \{0, 1\}.$$

Binomial Distribution (n, p)**.** The Binomial distribution describes the total number of successes in n independent Bernoulli(p) trials. Its support is $\{0, 1, \dots, n\}$, and its shape depends strongly on p: it is symmetric when $p = 1/2$, right–skewed for $p < 1/2$, and left–skewed for $p > 1/2$. For large n with small p it is well approximated by a Poisson distribution with parameter $\lambda = np$, and for large np and $n(1-p)$ it is well approximated by a Normal distribution with mean np and variance $np(1-p)$.

$$P(X = x) = \binom{n}{x} p^x(1-p)^{n-x}, \qquad x = 0, 1, \dots, n.$$

Geometric Distribution (p)**.** The Geometric distribution models the number of trials required until the first success occurs in a sequence of independent Bernoulli(p) experiments. It is the only discrete distribution possessing the memoryless property, $P(X > m + n \mid X > m) = P(X > n)$, and is therefore characterized by a highly right–skewed shape with a long tail.

$$P(X = x) = (1 - p)^{x-1} p, \qquad x = 1, 2, \ldots$$

Negative Binomial Distribution (r, p)**.** The Negative Binomial distribution describes the number of trials required to observe r successes in independent Bernoulli(p) experiments. It generalizes the Geometric distribution, which corresponds to the special case $r = 1$, and is frequently used to model overdispersed count data whose variance exceeds the mean.

$$P(X = x) = \binom{x-1}{r-1} p^r (1 - p)^{x-r}, \qquad x = r, r + 1, \ldots$$

Poisson Distribution (λ)**.** The Poisson distribution models the number of events occurring in a fixed interval of time or space when events arrive independently at a constant rate λ. It is additive under independent superposition and serves as a limiting form of the Binomial distribution for rare events. For large λ, its shape is well approximated by a Normal distribution with mean and variance both equal to λ.

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \qquad x = 0, 1, 2, \ldots$$

Hypergeometric Distribution (N_1, N_2, n)**.** The Hypergeometric distribution counts the number of successes in a sample of size n drawn *without replacement* from a finite population containing N_1 successes and N_2 failures, with $N = N_1 + N_2$. Unlike the Binomial model, the trials are dependent, and the variance reflects a finite–population correction. When the population is large relative to the sample size, the distribution is well approximated by a Binomial(n, p) with $p = N_1/N$.

$$P(X = x) = \frac{\binom{N_1}{x}\binom{N_2}{n-x}}{\binom{N}{n}}, \qquad \max(0, n - N_2) \leq x \leq \min(n, N_1).$$

Discrete Uniform Distribution on $\{1, \ldots, m\}$**.** The discrete uniform distribution assigns equal probability to each element of a finite set, modeling complete symmetry or absence of preference among the outcomes.

$$P(X = x) = \frac{1}{m}, \qquad x = 1, 2, \ldots, m.$$

Multinomial Distribution $(n, \mathbf{p})$**.** The Multinomial distribution generalizes the Binomial model to k outcome categories. It describes the joint distribution of counts $(X_1, \ldots, X_k)$ obtained from n independent trials, where each trial results in category i with probability p_i and $\sum_{i=1}^{k} p_i = 1$. The component counts are negatively correlated, reflecting competition among categories.

$$P(X_1 = x_1, \ldots, X_k = x_k) = \frac{n!}{x_1! \cdots x_k!}\, p_1^{x_1} \cdots p_k^{x_k}, \qquad \sum_{i=1}^{k} x_i = n.$$

Table 2: Discrete Distributions and Their Moments

Distribution	**PMF** $f(x)$	**Mean** μ	**Variance** σ^2
Bernoulli $x \in \{0, 1\}$	$p^x(1-p)^{1-x}$	p	$p(1-p)$
Binomial $x = 0, 1, \dots, n$	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Geometric $x = 1, 2, \dots$	$(1-p)^{x-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Binomial $x = r, r+1, \dots$	$\binom{x-1}{r-1} p^r (1-p)^{x-r}$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$
Poisson $x = 0, 1, 2, \dots$	$\frac{\lambda^x e^{-\lambda}}{x!}$	λ	λ
Hypergeometric $x \in \mathcal{S}_{HG}$	$\frac{\binom{N_1}{x}\binom{N_2}{n-x}}{\binom{N}{n}}$	$n\frac{N_1}{N}$	$n\frac{N_1}{N}\frac{N_2}{N}\frac{N-N_1}{N-1}$
Discrete Uniform $x = 1, 2, \dots, m$	$\frac{1}{m}$	$\frac{m+1}{2}$	$\frac{m^2-1}{12}$
Multinomial $\sum_{i=1}^{k} x_i = n$	$\frac{n!}{x_1! \cdots x_k!} p_1^{x_1} \cdots p_k^{x_k}$	np_i	$np_i(1-p_i)$

Where $\mathcal{S}_{HG} = \{x \in \mathbb{Z} : \max(0, n - N_2) \le x \le \min(n, N_1)\}$.

3 Choosing a Discrete Model

The appropriate discrete distribution is determined by three structural features of the experiment: (i) whether sampling is performed with or without replacement, (ii) the number of outcome categories, and (iii) whether the data arise from repeated trials or from counting events in time or space. These distinctions induce the natural classification summarized in Table 3.

Table 3: Sampling vs Outcomes

	With Replacement	**Without Replacement**
2 Outcomes	Bernoulli Binomial Geometric Negative Binomial	Hypergeometric
>2 **Outcomes**	Multinomial	Multivariate– Hypergeometric
Event Counts	Poisson	

From a modeling perspective, the selection procedure may be summarized as follows.
Sampling mechanism:

- With replacement $\Rightarrow$ independent trials.
- Without replacement $\Rightarrow$ dependent trials.

Number of outcome categories:

- Two outcomes: Bernoulli, Binomial, Geometric, Negative Binomial (independent); Hypergeometric (dependent).
- More than two outcomes: Multinomial (independent); Multivariate Hypergeometric (dependent).

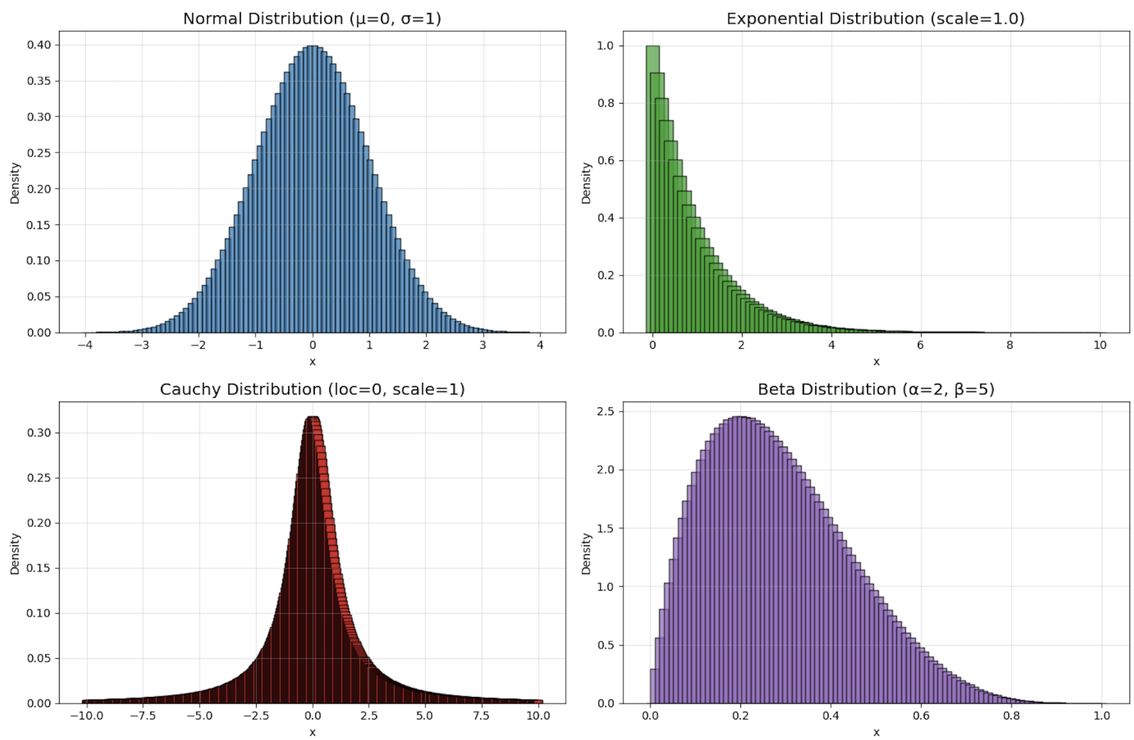
Event counts in time or space:

- Poisson distribution.

Modeling Implications. Choosing the wrong structural class can lead to systematically incorrect uncertainty quantification. For example, replacing a Hypergeometric model by a Binomial one ignores dependence induced by finite populations and underestimates variance, while modeling overdispersed counts with a Poisson law enforces an equality of mean and variance that may be violated in practice. Thus, before selecting a specific distribution, the sampling mechanism and the generative process should be identified at the structural level, not merely by matching empirical frequencies.

4 Core Continuous Distributions

Continuous distributions assign probability through a density over an interval or over the real line. Individual points have zero probability; only intervals carry mass. The defining features of a continuous model are its support, the shape of its density, and how its parameters control concentration, skewness, and tail behavior.



Uniform Distribution (a, b)**.** The Uniform distribution represents complete indifference over a finite interval, assigning equal density to all points in $[a, b]$. It is the simplest continuous model and serves as a baseline for comparison when no location within the interval is preferred. The density is constant and vanishes outside the interval.

$$f(x) = \frac{1}{b-a}, \qquad a \le x \le b.$$

Exponential Distribution (λ)**.** The Exponential distribution models the waiting time until the first event in a Poisson process with constant arrival rate λ. It is supported on $[0, \infty)$ and is strongly right–skewed, reflecting the possibility of long waiting times. Its density is

$$f(x) = \lambda e^{-\lambda x}, \qquad x \ge 0.$$

It is the only continuous distribution with the *memoryless* property, meaning that the remaining waiting time is independent of how long one has already waited:

$$P(X > s + t \mid X > s) = P(X > t).$$

Gamma Distribution (α, β). The Gamma distribution generalizes the Exponential distribution and models the waiting time until the α-th event in a Poisson process when α is an integer. More generally, it provides a flexible family on $[0, \infty)$ whose shape is controlled by α and whose scale is set by β. Many important distributions, including the Exponential and Chi–square, arise as special cases of the Gamma family. Its density is

$$f(x) = \frac{1}{\Gamma(\alpha)\beta^{\alpha}} x^{\alpha-1} e^{-x/\beta}, \qquad x \geq 0,$$

where the Gamma function is defined by

$$\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1} e^{-t} \, dt,$$

which satisfies $\Gamma(n) = (n-1)!$ for positive integers n and $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

Chi–Square Distribution (k). The Chi–square distribution is the distribution of the sum of squares of k independent standard Normal variables and plays a central role in inference for variances and quadratic forms. It is supported on $[0, \infty)$, is right–skewed for small k, and approaches a Normal shape as k increases. It is a special case of the Gamma distribution,

$$\chi_k^2 \sim \Gamma\left(\frac{k}{2}, \frac{1}{2}\right),$$

with density

$$f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}, \qquad x \geq 0.$$

Normal Distribution (μ, σ^2). The Normal distribution is the fundamental continuous law arising as the limit of sums and averages of many small independent effects, as formalized by the Central Limit Theorem. It is symmetric about its mean μ, completely determined by its mean and variance, and invariant under affine transformations. The standardization $Z = (X - \mu)/\sigma$ reduces any Normal variable to the standard form. Its density is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}.$$

The Normal Distribution is revisited in greater detail in the next section.

Beta Distribution (α, β). The Beta distribution is supported on the unit interval and models random proportions and probabilities. By varying α and β, it can represent uniform, U–shaped, symmetric, or highly skewed behavior. It is conjugate to the Bernoulli and Binomial likelihoods and is therefore fundamental in Bayesian inference for probability parameters. Its density is

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1}, \qquad 0 \leq x \leq 1,$$

where the normalizing constant $B(\alpha, \beta)$ is the Beta function,

$$B(\alpha, \beta) = \int_0^1 u^{\alpha-1}(1-u)^{\beta-1}\, du = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

Cauchy Distribution (x_0, γ)**.** The Cauchy distribution arises as the ratio of two independent standard Normal variables and exhibits extremely heavy tails. Unlike the Normal distribution, it has no finite mean or variance, and sample averages do not converge, illustrating a fundamental breakdown of the law of large numbers. Its density is

$$f(x) = \frac{1}{\pi d\left[1 + \left(\frac{x-m}{d}\right)^2\right]}, \qquad x \in \mathbb{R}.$$

Table 4: Continuous Distributions & Their Moments

Distribution	**PDF** $f(x)$	**CDF** $F(x)$	**Mean** μ	**Variance** σ^2
Beta (α,β) $0 < x < 1$	$\dfrac{\Gamma(\alpha+\beta)x^{\alpha-1}(1-x)^{\beta-1}}{\Gamma(\alpha)\Gamma(\beta)}$	$I_x(\alpha,\beta)$	$\dfrac{\alpha}{\alpha+\beta}$	$\dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Chi-square (r) $0 \le x < \infty$	$\dfrac{x^{r/2-1}e^{-x/2}}{2^{r/2}\Gamma(r/2)}$	$P\left(\dfrac{r}{2},\dfrac{x}{2}\right)$	r	$2r$
Exponential (θ) $0 \le x < \infty$	$\dfrac{e^{-x/\theta}}{\theta}$	$1 - e^{-x/\theta}$	θ	θ^2
Gamma (α,θ) $0 \le x < \infty$	$\dfrac{x^{\alpha-1}e^{-x/\theta}}{\Gamma(\alpha)\theta^\alpha}$	$P\left(\alpha,\dfrac{x}{\theta}\right)$	$\alpha\theta$	$\alpha\theta^2$
Normal (μ,σ^2) $-\infty < x < \infty$	$\dfrac{e^{-(x-\mu)^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}}$	$\Phi\left(\dfrac{x-\mu}{\sigma}\right)$	μ	σ^2
Uniform (a,b) $a \le x \le b$	$\dfrac{1}{b-a}$	$\dfrac{x-a}{b-a}$	$\dfrac{a+b}{2}$	$\dfrac{(b-a)^2}{12}$
Cauchy (m,d) $-\infty < x < \infty$	$\dfrac{1}{\pi d\left[1+\left(\dfrac{x-m}{d}\right)^2\right]}$	$\dfrac{1}{\pi}\tan^{-1}\left(\dfrac{x-m}{d}\right)+\dfrac{1}{2}$	undefined	undefined

Special functions.

Beta Function: $B(\alpha,\beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}\,dt,\quad \Gamma(s) = \int_0^\infty t^{s-1}e^{-t}\,dt$

Gamma Function: $P(s,x) = \dfrac{1}{\Gamma(s)}\int_0^x t^{s-1}e^{-t}\,dt,$

Normal CDF: $\Phi(z) = \dfrac{1}{\sqrt{2\pi}}\int_{-\infty}^z e^{-t^2/2}\,dt.$

5 Normal and Standard Normal Distributions

Normal Distribution. A random variable X is said to be Normally distributed with mean μ and variance σ^2 if

$$X \sim \mathcal{N}(\mu, \sigma^2), \qquad f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}.$$

The Normal distribution is symmetric about μ, completely determined by its first two moments, and closed under affine transformations.

Cumulative Distribution Function. The CDF of a Normal random variable has no closed form in elementary functions and is defined by

$$F_X(x) = P(X \leq x) = \frac{1}{\sqrt{2\pi}\,\sigma} \int_{-\infty}^{x} \exp\left(-\frac{1}{2}\left(\frac{w-\mu}{\sigma}\right)^2\right) dw.$$

Standardization. Any Normal random variable can be reduced to the standard form by centering and scaling:

$$Z = \frac{X-\mu}{\sigma}, \qquad Z \sim \mathcal{N}(0,1).$$

Standard Normal Distribution. The standard Normal CDF and PDF are denoted by

$$\Phi(z) = P(Z \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-w^2/2}\, dw, \qquad \phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

By symmetry,

$$\Phi(-z) = 1 - \Phi(z), \qquad P(Z > z) = 1 - \Phi(z).$$

Tables or numerical routines for Φ allow probability calculations for any Normal variable via standardization.

Jointly Gaussian Random Variables. If (X, Y) are jointly Normal with correlation coefficient ρ, then independence is equivalent to zero correlation:

$$(X, Y) \text{ jointly Gaussian} \quad \Longrightarrow \quad \rho = 0 \iff X \text{ and } Y \text{ are independent.}$$

The joint density is

$$f_{X,Y}(x,y) =$$
$$\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right\}.$$

Normal Approximations (CLT Regime). Many discrete and skewed distributions are well approximated by a Normal distribution when their parameters are large:

$$\begin{aligned}
\text{Binomial:} \quad & B(n,p) \approx \mathcal{N}(np,\ npq), \\
\text{Negative Binomial:} \quad & NB(n,p) \approx \mathcal{N}\left(\frac{n}{p},\ \frac{nq}{p^2}\right), \\
\text{Poisson:} \quad & P(\lambda) \approx \mathcal{N}(\lambda,\ \lambda), \\
\text{Gamma:} \quad & \Gamma(n,\theta) \approx \mathcal{N}(n\theta,\ n\theta^2), \\
\text{Chi–square:} \quad & \chi^2(n) \approx \mathcal{N}(n,\ 2n).
\end{aligned}$$

These approximations are consequences of the Central Limit Theorem.

6 Signal Detection and Hypothesis Testing

Signal detection theory models observation as a noisy measurement that may or may not contain a structured effect, and formulates inference as a decision between competing hypotheses based on that observation. The central objective is to design a decision rule that balances false alarms against missed detections and to quantify this tradeoff through error probabilities and the power of the test.

Hypothesis Formulation. We consider a binary testing problem in which

$$H_0 : \text{ No signal (noise only)}, \qquad X \sim \mathcal{N}(0, \sigma^2),$$

$$H_1 : \text{ Signal present}, \qquad X \sim \mathcal{N}(1, \sigma^2).$$

Thus, the two hypotheses differ by a shift in mean while sharing a common noise variance.

Decision Rule. A threshold test is used to decide between the hypotheses. Given a decision threshold T, the rule is

$$\begin{cases} X \le T \ \Rightarrow \ \text{accept } H_0 & \text{(no detection)}, \\ X > T \ \Rightarrow \ \text{accept } H_1 & \text{(signal detected)}. \end{cases}$$

Error Probabilities. Two types of errors arise naturally:

$$\alpha \triangleq P[\text{Type I Error}] = P[\text{reject } H_0 | H_0 \text{ true}] = P[X > T | H_0]$$

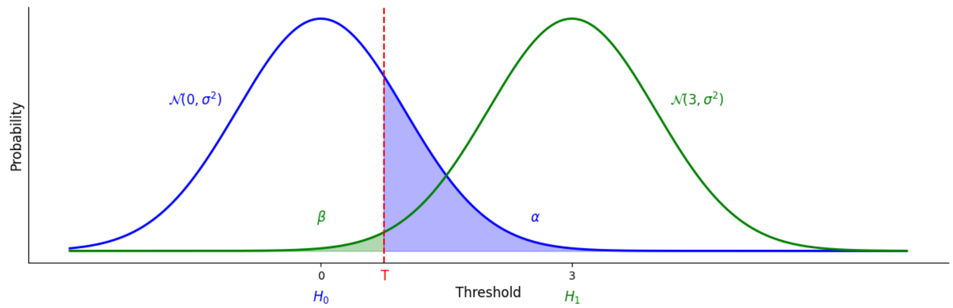
$$\beta \triangleq P[\text{Type II Error}] = P[\text{accept } H_0 | H_1 \text{ true}] = P[X \le T | H_1]$$

The quantity α is the *false alarm probability*, while β is the *miss probability*.

Power of the Test. The probability of correctly detecting the signal is

$$\text{Power} = 1 - \beta = P(\text{correct detection} \mid H_1).$$

Power increases as the two distributions separate or as the threshold is lowered, reflecting a higher likelihood of correctly declaring the presence of a signal at the expense of increased false alarms.



Normal Formulas via Standardization. Under the Gaussian assumptions, standardization yields closed-form expressions for the error probabilities:

$$\alpha = P(X > T \mid H_0) = 1 - \Phi\left(\frac{T}{\sigma}\right), \qquad \beta = P(X \le T \mid H_1) = \Phi\left(\frac{T-1}{\sigma}\right),$$

where Φ denotes the standard Normal CDF. As T increases, α decreases and β increases, making explicit the tradeoff between false alarms and missed detections.

Detectability Index. A convenient measure of hypothesis separability is

$$d' \triangleq \frac{\mu_1 - \mu_0}{\sigma} = \frac{1}{\sigma},$$

which quantifies the distance between the two Normal means in units of noise standard deviation. Larger values of d' correspond to better distinguishability and permit lower error probabilities for an appropriately chosen threshold.

Chapter 4: Moments & Dependence

Moments are the language of structure in probability. They describe not only where a distribution is centered, but how tightly it spreads, how asymmetrically it leans, and how heavy its tails may be. Dependence extends this language from single variables to pairs, capturing whether two quantities tend to rise and fall together, or whether they vary independently despite sharing the same environment. Together, moments and dependence provide a concise summary of the behavior of random variables, isolating the features that matter across models, data, and applications.

1 Expectation

Expectation captures the central tendency of a random variable, giving the average or typical magnitude around which its realizations are centered. It serves as the basic measure of location of a distribution and underlies how we summarize random outcomes and form predictions, both unconditionally and when conditioning on available information.

Expectation. The expectation $E[X]$ of a random variable X is its average value under the distribution of X:

$$\begin{aligned} E[X] &= \mu \\ &= \begin{cases} \displaystyle\sum_x x\, f_X(x), & \text{discrete,} \\ \displaystyle\int_{-\infty}^{\infty} x\, f_X(x)\, dx, & \text{continuous.} \end{cases} \end{aligned}$$

Conceptually, $E[X]$ is the "center of mass" of the distribution: more probable outcomes pull the average more strongly.

Linearity and sums. Expectation is linear, so constants and sums pass through:

$$E[aX + b] = a\,E[X] + b, \qquad E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i\, E[X_i].$$

No independence is required; this is purely an algebraic property of the integral/sum.

Products under Independence. If X and Y are independent,

$$E[XY] = E[X]\,E[Y].$$

Independence removes any interaction term: on average, the product behaves like the product of averages.

Conditional Expectation. Conditional expectation refines the average once additional information $Y = y$ is known:

$$E[X \mid Y = y] = \begin{cases} \displaystyle\sum_x x\, p_{X|Y}(x \mid y), & \text{discrete,} \\ \displaystyle\int_{-\infty}^{\infty} x\, f_{X|Y}(x \mid y)\, dx, & \text{continuous.} \end{cases}$$

For jointly distributed (X, Y),

$$E[X] = \iint x\, f_{X,Y}(x, y)\, dx\, dy = \int E[X \mid Y = y]\, f_Y(y)\, dy,$$

which links the joint density, the conditional expectation, and the marginal density of Y.

Law of Total Expectation. Averaging conditional expectations over Y recovers the unconditional mean:

$$E[X] = E_Y\big[E[X \mid Y]\big].$$

Operationally: Compute the mean of X in each scenario $Y = y$, then average these scenario-wise means according to how likely each y is.

Table 5: Expectation Toolbox

Topic	**Formula**	**Interpretation**
Definition	$E[X] = \sum_x x f_X(x)$ or $E[X] = \int x f_X(x)\, dx$	Center / average value
Linearity	$E[aX + b] = aE[X] + b$	Shift and scale pass through
Sums	$E\left[\sum_i a_i X_i\right] = \sum_i a_i E[X_i]$	Expectation distributes over sums
Independence	$X \perp Y \Rightarrow E[XY] = E[X]E[Y]$	Product of means when independent
Conditional mean	$E[X \mid Y = y]$	Mean with $Y = y$ fixed
Total expectation	$E[X] = E_Y[E[X \mid Y]]$	Average of scenario-wise means

2 Variance

Variance measures the dispersion of a random variable by quantifying how widely its possible values are spread around the mean. It reflects the intrinsic uncertainty or risk in the distribution and provides a precise way to compare the stability of different random quantities and of estimators derived from data.

Variance. The variance of X measures the typical squared deviation from its mean:

$$V[X] = \sigma_X^2 = E\big[(X - E[X])^2\big] = E[X^2] - (E[X])^2.$$

A large variance means realizations of X are widely scattered around $E[X]$; a small variance means they are tightly clustered.

Scaling and Shifts. Adding a constant does not change variability, while scaling stretches or contracts it:

$$V[aX + b] = a^2\, V[X].$$

Sums and Linear Combinations. For any collection $\{X_i\}_{i=1}^n$,

$$V\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n V[X_i] \;+\; 2 \sum_{1 \le i < j \le n} \operatorname{cov}(X_i, X_j).$$

Variance of a sum consists of the individual variances plus all pairwise covariance terms. If the X_i are independent, all covariances vanish and

$$V\Big[\sum_{i=1}^n X_i\Big] = \sum_{i=1}^n V[X_i].$$

For two variables and constants a, b,

$$V[aX + bY] = a^2 V[X] + b^2 V[Y] + 2ab\ \operatorname{cov}(X, Y),$$

which is the basic variance formula for linear combinations.

Product of Independent Variables. When X and Y are independent,

$$V[XY] = E[X^2]E[Y^2] - \big(E[X]E[Y]\big)^2,$$

expressing the variability of a product in terms of second moments and means of the factors.

Conditional Variance and Total Variance. Conditional variance quantifies spread once Y is fixed:

$$V[X \mid Y] = E\big[(X - E[X \mid Y])^2 \mid Y\big] = E[X^2 \mid Y] - \big(E[X \mid Y]\big)^2.$$

The law of total variance decomposes overall variability into within-scenario and between-scenario parts:

$$V[X] = E_Y\big[V[X \mid Y]\big] + V_Y\big(E[X \mid Y]\big).$$

Bias–Variance Decomposition for Estimators. For an estimator $\hat{\theta}$ of a parameter θ,

$$\mathrm{MSE}(\hat{\theta}) = E\big[(\hat{\theta} - \theta)^2\big] = V[\hat{\theta}] + \big(E[\hat{\theta}] - \theta\big)^2,$$

with

$$\text{Squared Bias} = \big|E[\hat{\theta}] - \theta\big|^2.$$

Here $V[\hat{\theta}]$ captures random fluctuation of the estimator, while the squared bias measures systematic offset from the true parameter.

Table 6: Variance Toolbox

Topic	**Formula**	**Interpretation**
Definition	$V[X] = E[(X - E[X])^2]$	Spread around the mean
Moment form	$V[X] = E[X^2] - (E[X])^2$	Uses first and second moments
Scaling	$V[aX + b] = a^2 V[X]$	Shifts irrelevant, scaling matters
Sums	$V\left[\sum_i X_i\right] = \sum_i V[X_i] + 2\sum_{i<j} \mathrm{cov}(X_i, X_j)$	Variances plus covariances
Independence	X_i indep $\Rightarrow V\left[\sum_i X_i\right] = \sum_i V[X_i]$	Covariances vanish
Total variance	$V[X] = E[V[X \mid Y]] + V(E[X \mid Y])$	Within + between variability
Bias–variance	$\mathrm{MSE}(\hat{\theta}) = V[\hat{\theta}] + (E[\hat{\theta}] - \theta)^2$	Random + systematic error

3 Covariance

Covariance describes how two random variables move together, indicating whether large values of one tend to be associated with large or small values of the other. It captures the direction of their linear co-movement and provides a basic measure of joint variability around their respective means.

Covariance. The covariance of (X, Y) is

$$\begin{aligned}\mathrm{cov}(X, Y) &= E\big[(X - E[X])(Y - E[Y])\big]\\ &= E[XY] - E[X]E[Y].\end{aligned}$$

It is positive when X and Y move together, negative when they move oppositely, and zero when they show no linear co-variation.

Basic Properties. Covariance behaves linearly in both arguments:

$$\mathrm{cov}(aX + b,\ cY + d) = ac\ \mathrm{cov}(X, Y).$$

It is symmetric and identifies variance as a special case:

$$\mathrm{cov}(X, Y) = \mathrm{cov}(Y, X), \qquad \mathrm{cov}(X, X) = V[X].$$

Linearity Over Sums. For linear combinations,

$$\mathrm{cov}\left(\sum_i a_i X_i,\ \sum_j b_j Y_j \right) = \sum_i \sum_j a_i b_j\ \mathrm{cov}(X_i, Y_j).$$

Variance of a sum is a sum of variances and covariances:

$$V\Big[\sum_{i=1}^n X_i \Big] = \sum_{i=1}^n V[X_i]\ +\ 2 \sum_{1 \le i < j \le n} \mathrm{cov}(X_i, X_j).$$

Relation to Product Expectation.

$$E[XY] = \mathrm{cov}(X, Y) + E[X]\, E[Y].$$

Covariance captures the deviation from $E[X]E[Y]$.

Linear Transformations. If $Y = aX + b$ then

$$\mathrm{cov}(X, Y) = a\, V[X].$$

More generally,

$$V[aX + bY] = a^2 V[X] + b^2 V[Y] + 2ab\ \mathrm{cov}(X, Y).$$

Conditional Covariance and Total Covariance. Conditioning on Z,

$$\mathrm{cov}(X, Y \mid Z) = E[XY \mid Z] - E[X \mid Z]E[Y \mid Z],$$

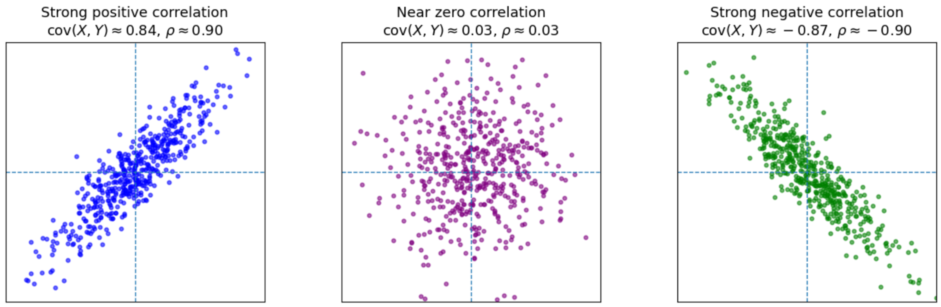
and averaging yields the total covariance decomposition:

$$\mathrm{cov}(X, Y) = E_Z\big[\mathrm{cov}(X, Y \mid Z) \big] + \mathrm{cov}_Z\big(E[X \mid Z],\ E[Y \mid Z] \big).$$

The first term is "within-Z" co-variation; the second is co-variation of conditional means.

4 Correlation

Correlation is the normalized version of covariance that measures the strength of linear dependence between two random variables on a scale from 1 to 1. By removing the effect of units and scale, it allows direct comparison of how strongly different pairs of variables are related.



The shape of each scatter shows the sign and strength of linear dependence.

Correlation. Correlation rescales covariance by the marginal standard deviations:

$$\rho_{XY} = \frac{\operatorname{cov}(X, Y)}{\sigma_X \, \sigma_Y}, \qquad -1 \le \rho_{XY} \le 1.$$

It measures direction and strength of linear dependence on a unitless scale.

$\rho_{XY} = 1$ (perfect positive), $\rho_{XY} = -1$ (perfect negative), $\rho_{XY} = 0$ (no linear association).

Relation Back to Covariance.

$$\operatorname{cov}(X, Y) = \rho_{XY} \, \sigma_X \sigma_Y.$$

Covariance captures raw co-variation; correlation captures standardized strength.

Table 7: Key Differences Between Covariance and Correlation

	Covariance	**Correlation**
What it measures	Direction of the relationship (positive, negative, or none)	Direction and strength of the relationship (scaled between -1 and 1)
Scale	Depends on the units and magnitude of X and Y	Standardized, does not depend on units
Range	Unbounded, can take any real value	Always between -1 and 1
Interpretation	Shows direction and rough magnitude of co-movement	Shows direction and strength of linear relationship
Use case	Understanding the direction of a relationship within one dataset	Comparing relationships across different datasets

5 Higher Moments & Central Moments

Moments extend the notions of mean and variance by characterizing the global shape of a distribution. Raw moments describe the average magnitude of powers of X relative to the origin, while central moments describe powers of deviations from the mean. As the order k increases, extreme realizations are increasingly amplified, so higher moments encode information about tail behavior, asymmetry, and concentration.

Raw Moments. The kth (raw) moment of a random variable X is

$$\mathbb{E}[X^k] = \begin{cases} \sum_x x^k \, p(x), & \text{discrete}, \\ \int_{-\infty}^{\infty} x^k \, f(x)\, dx, & \text{continuous}. \end{cases}$$

Raw moments locate the distribution relative to the origin and capture the scale of typical and extreme values of X.

Central Moments. The kth central moment is defined by

$$\mathbb{E}[(X-\mu)^k] = \begin{cases} \sum_x (x-\mu)^k \, p(x), & \text{discrete}, \\ \int_{-\infty}^{\infty} (x-\mu)^k \, f(x)\, dx, & \text{continuous}, \end{cases}$$

where $\mu = \mathbb{E}[X]$. Central moments measure the geometry of the distribution around its mean and therefore describe spread, asymmetry, and tail thickness rather than absolute location.
A moment of order k exists only if the corresponding expectation is finite. For heavy–tailed distributions, low–order moments may exist while higher–order ones diverge, a feature that itself signals extreme–risk dominance.

Low–Order Central Moments and Shape. The first few central moments have standard geometric interpretations:

$$\begin{aligned} &\text{1st moment:} && \mathbb{E}[X] = \mu && \textit{location}, \\ &\text{2nd central moment:} && \mathbb{E}[(X-\mu)^2] = \sigma^2 && \textit{dispersion}, \\ &\text{3rd central moment:} && \mathbb{E}[(X-\mu)^3] = \gamma_1 && \textit{skewness}, \\ &\text{4th central moment:} && \mathbb{E}[(X-\mu)^4] = \gamma_2 && \textit{kurtosis}. \end{aligned}$$

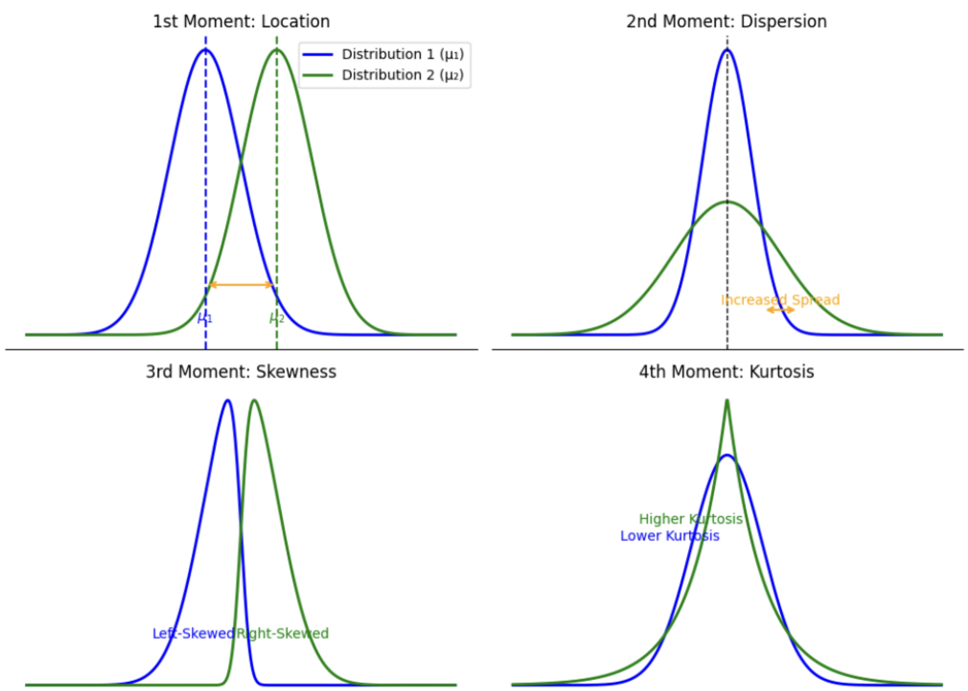
Skewness quantifies asymmetry: positive skew indicates a longer or heavier right tail, while negative skew indicates a longer or heavier left tail. Kurtosis quantifies tail heaviness and peak concentration relative to the center: larger kurtosis places more probability in the tails (and typically a sharper peak). Higher even moments increasingly weight extreme deviations and therefore emphasize tail behavior.

Standardized Moments. To remove scale, the kth central moment is normalized by σ^k,

$$\gamma_k = \frac{\mathbb{E}[(X-\mu)^k]}{\sigma^k},$$

yielding dimensionless shape measures. In particular, γ_3 is the standardized skewness and γ_4 the standardized kurtosis, allowing meaningful comparison of asymmetry and tail thickness across distributions.

Raw and central moments are thus complementary: raw moments describe how probability mass is positioned relative to zero, while central and standardized moments describe how the distribution spreads, tilts, and concentrates around its mean. As the order increases, moments become progressively more sensitive to rare events; their existence or divergence provides direct information about tail risk.



The first four moments successively characterize location, dispersion, asymmetry, and tail behavior. Higher moments emphasize extremes and may fail to exist for heavy–tailed distributions, a fact that is itself diagnostically informative.

On the next page, a table lists explicit formulas for $\mathbb{E}[X^k]$ and $\mathbb{E}[(X-\mu)^k]$ for the discrete and continuous families considered.

Table 8: kth Moments and Central Moments

Distribution	**kth moment $\mathbb{E}[X^k]$**	**kth central moment $\mathbb{E}[(X-\mu)^k]$**
Bernoulli (p)	p	$(1-p)(-p)^k + p(1-p)^k$
Binomial (n, p)	$\sum_{j=0}^{n} j^k \binom{n}{j} p^j (1-p)^{n-j}$	$\sum_{j=0}^{n} (j-np)^k \binom{n}{j} p^j (1-p)^{n-j}$
Geometric (p)	$p \sum_{x=1}^{\infty} x^k (1-p)^{x-1}$	$p \sum_{x=1}^{\infty} (x - \frac{1}{p})^k (1-p)^{x-1}$
Poisson (λ)	$e^{-\lambda} \sum_{x=0}^{\infty} \frac{x^k \lambda^x}{x!}$	$e^{-\lambda} \sum_{x=0}^{\infty} \frac{(x-\lambda)^k \lambda^x}{x!}$
Negative Binomial (r, p)	$\sum_{x=0}^{\infty} x^k \binom{r+x-1}{x} (1-p)^x p^r$	$\sum_{x=0}^{\infty} (x - \frac{r(1-p)}{p})^k \binom{r+x-1}{x} (1-p)^x p^r$
Discrete Uniform $\{1, \dots, n\}$	$\frac{1}{n} \sum_{x=1}^{n} x^k$	$\frac{1}{n} \sum_{x=1}^{n} (x - \frac{n+1}{2})^k$
Beta (α, β)	$\frac{\Gamma(\alpha+k)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+k)}$	$\sum_{j=0}^{k} \binom{k}{j} (-\frac{\alpha}{\alpha+\beta})^{k-j} \frac{\Gamma(\alpha+j)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+j)}$
Chi-square (r)	$2^k \frac{\Gamma(\frac{r}{2}+k)}{\Gamma(\frac{r}{2})}$	$\sum_{j=0}^{k} \binom{k}{j} (-r)^{k-j} 2^j \frac{\Gamma(\frac{r}{2}+j)}{\Gamma(\frac{r}{2})}$
Exponential (θ)	$k!\, \theta^k$	$\sum_{j=0}^{k} \binom{k}{j} (-\theta)^{k-j} j!\, \theta^j$
Gamma (α, θ)	$\theta^k \frac{\Gamma(\alpha+k)}{\Gamma(\alpha)}$	$\sum_{j=0}^{k} \binom{k}{j} (-\alpha\theta)^{k-j} \theta^j \frac{\Gamma(\alpha+j)}{\Gamma(\alpha)}$
Normal (μ, σ^2)	$\sum_{j=0}^{\lfloor k/2 \rfloor} \frac{k!}{(k-2j)!\, j!\, 2^j} \mu^{k-2j} \sigma^{2j}$	$\begin{cases} 0, & k \text{ odd}, \\ (k-1)!!\, \sigma^k, & k \text{ even} \end{cases}$
Uniform (a, b)	$\frac{b^{k+1}-a^{k+1}}{(k+1)(b-a)}$	$\frac{1}{b-a} \int_a^b (x - \frac{a+b}{2})^k \, dx$
Cauchy (m, d)	undefined	undefined

6 Sample Statistics & Standardization

Sample statistics are empirical analogues of population parameters. They are computed from finite data and used to estimate moments, dependence, and dimensionless shape and scale. Under suitable regularity conditions (e.g. i.i.d. sampling and finite moments), these estimators converge to their population counterparts as the sample size increases.

Throughout, we focus on *standardization* (centering and rescaling to zero mean and unit variance), which should be distinguished from *normalization*, a more general term that typically refers to rescaling data to a fixed range or unit norm without preserving moment structure.

Population and Sample Quantities. For random variables X and Y, population parameters are

$$\mu_X, \qquad \sigma_X^2, \qquad \sigma_X, \qquad \sigma_{XY},$$

while their empirical estimators based on n observations are

$$\bar{X}_n \to \mu_X, \qquad s_X^2 \to \sigma_X^2, \qquad s_X \to \sigma_X, \qquad s_{XY} \to \sigma_{XY},$$

in probability (and almost surely under stronger assumptions). The sample covariance satisfies the *Cauchy–Schwarz* bound

$$s_{XY}^2 \le s_X^2 s_Y^2.$$

Sample Moments and Dependence. Given paired samples $X_1, \dots, X_n$ and $Y_1, \dots, Y_n$, the sample mean, variance, and covariance are

$$\begin{aligned}
\bar{X}_n &= \frac{1}{n} \sum_{k=1}^{n} X_k, \\
s_X^2 &= \frac{1}{n-1} \sum_{k=1}^{n} (X_k - \bar{X}_n)^2, \\
s_{XY} &= \frac{1}{n-1} \sum_{k=1}^{n} (X_k - \bar{X}_n)(Y_k - \bar{Y}_n).
\end{aligned}$$

The normalization by $n-1$ renders s_X^2 and s_{XY} unbiased estimators of σ_X^2 and σ_{XY} in the i.i.d. setting.

Sample Mean and Sample Sum (i.i.d. Case). Let $X_1, \dots, X_n$ be i.i.d. with

$$E[X_i] = \mu_X, \qquad V[X_i] = \sigma_X^2 < \infty.$$

The sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{k=1}^{n} X_k$$

satisfies

$$E[\bar{X}_n] = \mu_X, \qquad V[\bar{X}_n] = \frac{\sigma_X^2}{n},$$

and is therefore an unbiased, consistent estimator whose variance shrinks at rate $1/n$.
The sample sum

$$S_n = \sum_{k=1}^{n} X_k$$

has

$$E[S_n] = n\mu_X, \qquad V[S_n] = n\sigma_X^2,$$

so both its mean and variance scale linearly with the sample size.

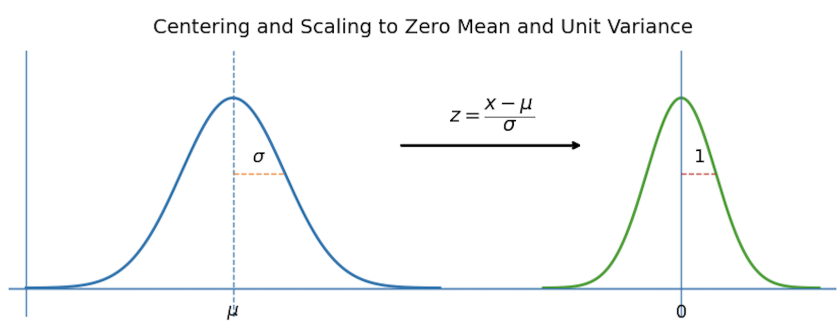
Sample Correlation. The empirical correlation coefficient is

$$r_{XY} = \frac{s_{XY}}{s_X s_Y},$$

which estimates the population correlation $\rho_{XY} = \sigma_{XY}/(\sigma_X \sigma_Y)$.

Standardization. Standardization removes location and scale by centering and rescaling a variable to have zero mean and unit variance.
For a population random variable,

$$Z = \frac{X - \mu_X}{\sigma_X}, \qquad E[Z] = 0, \qquad V[Z] = 1.$$



For observed data, the standardized sample (z–scores) is

$$z_k = \frac{X_k - \bar{X}_n}{s_X}, \qquad w_k = \frac{Y_k - \bar{Y}_n}{s_Y}.$$

These satisfy

$$\frac{1}{n}\sum_{k=1}^{n} z_k = 0, \qquad \frac{1}{n-1}\sum_{k=1}^{n}(z_k - \bar{z})^2 = 1,$$

so the empirical distribution is centered at zero and normalized to unit variance. Since $\bar{z} = 0$ by construction, the second identity may be written equivalently as

$$\frac{1}{n-1}\sum_{k=1}^{n} z_k^2 = 1.$$

Under standardization, correlation becomes the covariance of the normalized variables:

$$r_{XY} = \frac{1}{n-1}\sum_{k=1}^{n} z_k w_k.$$

Standardization therefore maps heterogeneous variables to a common, dimensionless scale, enabling direct comparison, dependence analysis, and correlation–based modeling.

Chapter 5: Limits, Averages, and Approximations

Randomness may look chaotic at small scales, but in large samples it reveals remarkably stable patterns. As data accumulate, averages settle, fluctuations follow universal laws, and uncertainty becomes mathematically predictable. This chapter develops the tools that explain this emergence of order, showing how repeated randomness gives rise to deterministic limits and why the Normal distribution appears so ubiquitously in nature, data, and models.

1 Modes of Convergence

For deterministic sequences there is a single notion of convergence, but for random variables several inequivalent notions coexist. Each mode formalizes a different sense of closeness, emphasizing large–deviation control, sample–path behavior, or moments. These distinctions matter because the law of large numbers, the central limit theorem, and related results are each stated in specific modes of convergence.

From Numerical to Stochastic Convergence. For a deterministic sequence $\{a_n\}_{n\geq 0} \subset \mathbb{R}$, convergence to $a^* \in \mathbb{R}$ means that

$$\forall\, \varepsilon > 0,\ \exists\, N \text{ such that } |a_n - a^*| < \varepsilon \quad \text{for all } n \geq N.$$

Equivalently, the sequence eventually remains arbitrarily close to its limit. When the terms a_n are replaced by random variables X_n, this pointwise notion must be reinterpreted probabilistically, leading to several distinct concepts of stochastic convergence.

Stochastic Convergence. Let $\{X_n\}_{n\geq 0}$ be a sequence of random variables defined on a common probability space, and let X be another random variable. The principal modes of convergence used in probability theory are the following.

Almost Sure Convergence. Almost sure convergence requires that, except on a set of probability zero, the entire sequence of sample paths eventually locks onto the limiting random variable and remains arbitrarily close thereafter. It is therefore a genuinely pathwise notion of convergence and represents the strongest standard mode of stochastic convergence. Formally,

$$X_n \xrightarrow{\text{a.s.}} X \quad \Longleftrightarrow \quad P\Big(\lim_{n\to\infty} X_n = X\Big) = 1.$$

Convergence in Probability. Convergence in probability demands that large deviations from the limit become increasingly unlikely, even though individual sample paths need not settle down permanently. It captures stabilization in a probabilistic, rather than pathwise, sense and is the natural mode underlying consistency of estimators. Formally,

$$X_n \xrightarrow{p} X \quad \Longleftrightarrow \quad \lim_{n\to\infty} P\big(|X_n - X| > \varepsilon\big) = 0, \qquad \forall\, \varepsilon > 0.$$

Mean–Square and L^p Convergence. These modes quantify convergence by controlling moments of the error, thereby measuring typical fluctuation size and, for larger p, increasingly strong tail suppression. They provide quantitative, metric notions of convergence in function spaces. In particular,

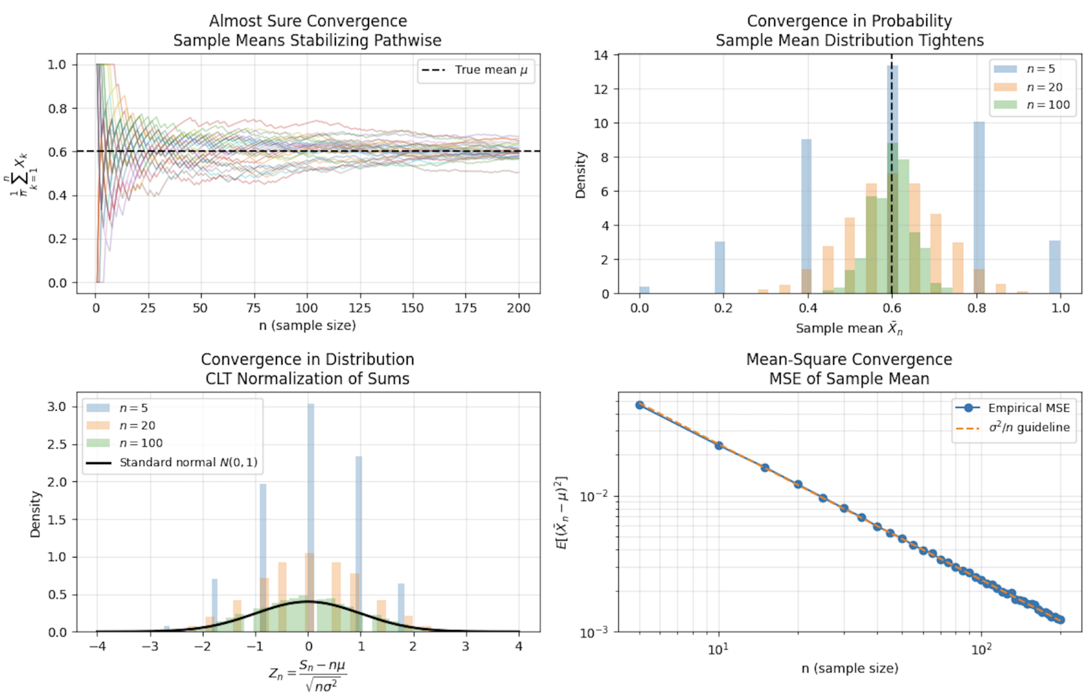
$$X_n \xrightarrow{L^2} X \iff \lim_{n\to\infty} E[(X_n - X)^2] = 0,$$

and more generally, for $p \geq 1$,

$$X_n \xrightarrow{L^p} X \iff \lim_{n\to\infty} E[|X_n - X|^p] = 0.$$

Convergence in Distribution. Convergence in distribution concerns only the limiting behavior of probability laws, not the alignment of individual realizations on a common sample space. It is therefore the weakest standard notion of convergence, yet it provides the natural framework for asymptotic theory and universal limit laws such as the central limit theorem. Formally,

$$X_n \xrightarrow{d} X \iff \lim_{n\to\infty} F_{X_n}(x) = F_X(x) \quad \text{for all continuity points of } F_X.$$



Implication Structure. The principal implication relations are

$$\text{a.s.} \Rightarrow \text{in probability} \Rightarrow \text{in distribution}, \qquad L^2 \Rightarrow \text{in probability} \Rightarrow \text{in distribution}.$$

Almost sure and mean–square convergence give the strongest control, while convergence in distribution provides the weakest notion sufficient to describe limiting laws.

2 Deviation Inequalities and Convergence

To convert qualitative convergence statements into quantitative control, one needs bounds on the probability of large deviations. Deviation inequalities relate tail probabilities to moments such as the mean and variance, and they form the basic tools behind many proofs of convergence in probability and almost sure convergence, most notably in the law of large numbers.

Markov's Inequality. Let $X \geq 0$ be a nonnegative random variable and let $a > 0$. Then

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

This bound depends only on the first moment. If the mean of X is small relative to the threshold a, then the probability that X exceeds a must be small. Markov's inequality is therefore extremely general, though often loose, and serves as a fundamental starting point for bounding tail probabilities and establishing convergence in probability.

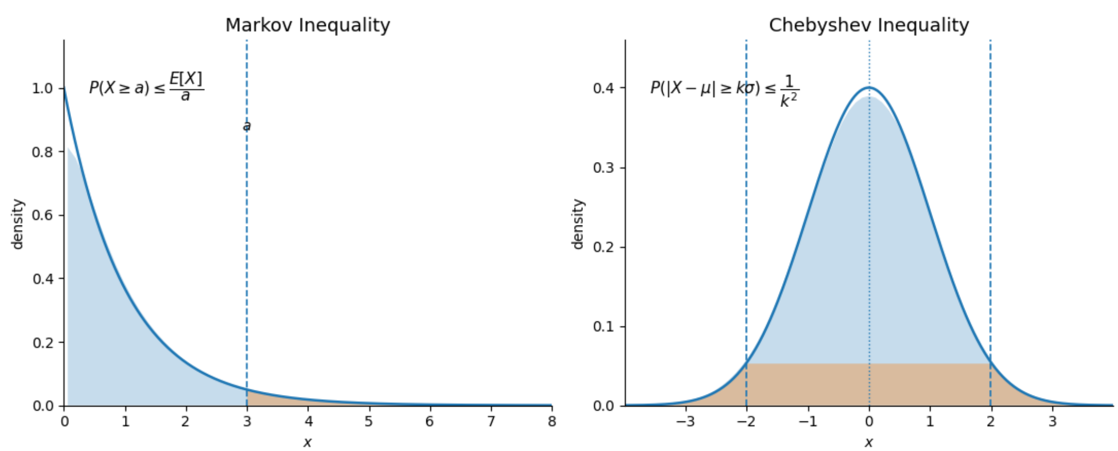
Chebyshev's Inequality. Let X be a random variable with finite mean $\mu = E[X]$ and variance $\sigma^2 = \mathrm{Var}(X)$. For any $k > 0$,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2},$$

or equivalently, for any $\varepsilon > 0$,

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Chebyshev's inequality refines Markov's bound by incorporating second–moment information and quantifying how tightly a distribution concentrates around its mean. Applied to sample averages, it yields a direct proof of the weak law of large numbers by showing that the probability of macroscopic deviations from the true mean vanishes as the sample size grows.



Left: Markov's inequality bounds this probability by $E[X]/a$, using only the mean and therefore providing a distribution–free but generally coarse estimate. Right: Chebyshev's inequality bounds this probability by $1/k^2$, ensuring that for any distribution with finite variance, most of the mass lies within a few standard deviations of the mean.

3 Law of Large Numbers

The law of large numbers formalizes the stabilization of empirical averages. Although individual observations remain random, their arithmetic mean becomes increasingly concentrated around a deterministic limit as the sample size grows, with fluctuations shrinking at the rate $n^{-1/2}$.

Setup. Let $X_1, X_2, \dots$ be i.i.d. random variables with

$$\mu = E[X_k], \qquad \sigma^2 = \operatorname{Var}(X_k) < \infty.$$

The sample mean of the first n observations is

$$\bar{X}_n = \frac{1}{n} \sum_{k=1}^{n} X_k.$$

Linearity of expectation and variance additivity give

$$E[\bar{X}_n] = \mu, \qquad \operatorname{Var}(\bar{X}_n) = \frac{\sigma^2}{n},$$

so $\bar{X}_n$ is an unbiased estimator of μ and its standard deviation decays as $\sigma/\sqrt{n}$.

Law of Large Numbers (LLN). Under suitable integrability assumptions, the sample mean converges to the population mean as the number of observations grows.

$$\bar{X}_n \longrightarrow \mu \qquad (n \to \infty).$$

The precise meaning of this convergence depends on the mode considered and leads to the following standard formulations.

Weak Law of Large Numbers (Convergence in Probability). The weak law states that large deviations of the sample mean from the true mean become increasingly unlikely as the sample size grows:

$$\bar{X}_n \xrightarrow{P} \mu \quad \iff \quad \forall\, \varepsilon > 0,\ P\big(|\bar{X}_n - \mu| > \varepsilon\big) \xrightarrow[n\to\infty]{} 0.$$

Strong Law of Large Numbers (Almost Sure Convergence). The strong law asserts that, except on a null event, the sequence of sample means converges pathwise to the true mean:

$$\bar{X}_n \xrightarrow{\text{a.s.}} \mu \quad \iff \quad P\Big(\lim_{n\to\infty} \bar{X}_n = \mu\Big) = 1.$$

This is a genuinely sample-path statement and represents the strongest classical form of the LLN.

Mean–Square Law of Large Numbers. Mean–square convergence quantifies stabilization through the decay of quadratic error:

$$\bar{X}_n \xrightarrow{L^2} \mu \quad \Longleftrightarrow \quad E\big[(\bar{X}_n - \mu)^2\big] \xrightarrow[n\to\infty]{} 0.$$

In the i.i.d. finite-variance case,

$$E[(\bar{X}_n - \mu)^2] = \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n} \longrightarrow 0,$$

so L^2 convergence holds automatically and implies convergence in probability.

Deviation Bounds and Proof of the Weak LLN. Chebyshev's inequality makes the concentration of $\bar{X}_n$ around μ explicit:

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \xrightarrow[n\to\infty]{} 0.$$

This provides a direct proof of the weak law and exhibits the rate at which macroscopic deviations of the sample mean vanish.

All versions of the law of large numbers express the same asymptotic principle: repeated averaging suppresses randomness and reveals the underlying mean. The central limit theorem sharpens this picture by describing the limiting distribution of the rescaled fluctuations $\sqrt{n}(\bar{X}_n - \mu)$.

4 Central Limit Theorem

The law of large numbers asserts that sample averages stabilize around their mean, but it does not describe the *distribution* of the remaining fluctuations. The central limit theorem (CLT) refines this result by identifying the universal limiting shape of these fluctuations after appropriate centering and scaling. This explains the pervasive role of the normal distribution, z–scores, and Gaussian approximations in probability and statistics.

Setup. Let $X_1, X_2, \ldots$ be i.i.d. random variables with

$$\mu_X = E[X_k], \qquad \sigma_X^2 = \mathrm{Var}(X_k) < \infty,$$

and define the sample mean and sum

$$\bar{X}_n = \frac{1}{n}\sum_{k=1}^{n} X_k, \qquad S_n = \sum_{k=1}^{n} X_k.$$

From the law of large numbers, $\bar{X}_n \to \mu_X$ as $n \to \infty$, and

$$\mathrm{Var}(\bar{X}_n) = \frac{\sigma_X^2}{n}.$$

The CLT characterizes the limiting distribution of the centered and rescaled quantities

$$\sqrt{n}(\bar{X}_n - \mu_X) \quad \text{and} \quad (S_n - n\mu_X)/\sqrt{n}.$$

Central Limit Theorem (CLT). Under the assumptions above,

$$Z_n = \frac{\bar{X}_n - \mu_X}{\sigma_X/\sqrt{n}} = \frac{\sqrt{n}(\bar{X}_n - \mu_X)}{\sigma_X} \xrightarrow{d} Z \sim \mathcal{N}(0,1),$$

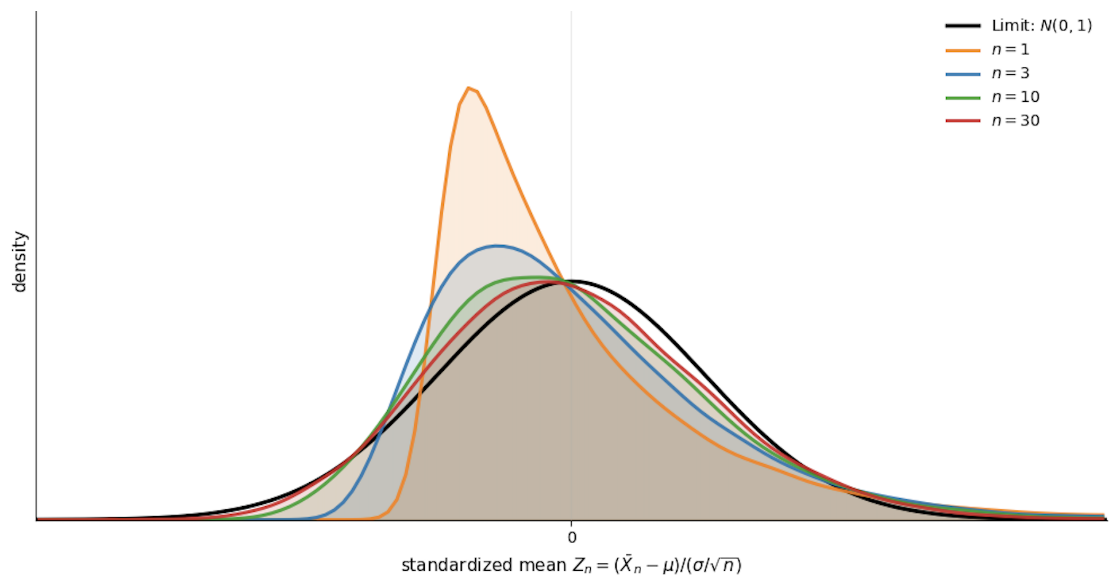
or equivalently,

$$Z_n = \frac{S_n - n\mu_X}{\sqrt{n}\,\sigma_X} \xrightarrow{d} \mathcal{N}(0,1).$$

Thus, after centering by the mean and scaling by the standard deviation, the fluctuations of sums and averages converge in distribution to a standard normal law. For moderately large n, the approximation

$$\frac{\bar{X}_n - \mu_X}{\sigma_X/\sqrt{n}} \approx Z \sim \mathcal{N}(0,1)$$

is already accurate, even when the underlying variables X_k are not themselves Gaussian.



Standardized sample means from a skewed distribution for increasing n. After centering and scaling, the empirical distributions collapse toward the standard normal law, illustrating convergence in distribution in the CLT.

Interpretation. The CLT complements the LLN in two distinct ways. The LLN identifies the deterministic limit μ_X around which $\bar{X}_n$ concentrates. The CLT describes the *shape* and *scale* of the residual randomness: fluctuations of order $\sigma_X/\sqrt{n}$ persist, and their standardized form is asymptotically Gaussian. Convergence is in distribution, not pathwise: the random variables Z_n do not converge almost surely, but their laws converge to that of a standard normal random variable.

Standardization in the CLT. The centering and scaling

$$Z_n = \frac{\bar{X}_n - \mu_X}{\sigma_X/\sqrt{n}}$$

is an instance of standardization: it removes location and scale so that the remaining fluctuations are dimensionless and of unit variance. The central limit theorem asserts that this standardized quantity converges in distribution to $\mathcal{N}(0,1)$, providing the universal normal approximation that underlies z–scores, confidence intervals, and classical hypothesis testing. Thus, standardization is the operator that maps empirical averages to their Gaussian asymptotic regime.

5 Confidence Intervals and Sample Size

The central limit theorem provides an approximate normal law for the standardized sample mean. Confidence intervals invert this approximation: rather than studying the distribution of $\bar{X}_n$ for a fixed mean μ_X, one constructs a random interval centered at $\bar{X}_n$ that contains the unknown parameter μ_X with a prescribed long–run frequency.

Setup. Assume $X_1, \ldots, X_n$ are i.i.d. with mean μ_X and known variance σ_X^2. For sufficiently large n, the CLT yields

$$Z_n = \frac{\bar{X}_n - \mu_X}{\sigma_X/\sqrt{n}} \xrightarrow{d} Z \sim \mathcal{N}(0,1),$$

so the distribution of Z_n is well approximated by the standard normal law.

Confidence Interval for the Mean (Known Variance). Let $1-\alpha$ be a desired confidence level and let $z_{\alpha/2}$ denote the $(1-\alpha/2)$–quantile of the standard normal distribution, defined by

$$P(-z_{\alpha/2} \le Z \le z_{\alpha/2}) = 1-\alpha, \qquad Z \sim \mathcal{N}(0,1).$$

Using the CLT approximation for Z_n,

$$P\left(-z_{\alpha/2} \le \frac{\bar{X}_n - \mu_X}{\sigma_X/\sqrt{n}} \le z_{\alpha/2}\right) \approx 1-\alpha.$$

Rearranging the inequalities yields

$$P\left(\bar{X}_n - z_{\alpha/2}\frac{\sigma_X}{\sqrt{n}} \le \mu_X \le \bar{X}_n + z_{\alpha/2}\frac{\sigma_X}{\sqrt{n}}\right) \approx 1-\alpha,$$

so an approximate $(1-\alpha)$ confidence interval for μ_X is

$$\left[\bar{X}_n - z_{\alpha/2}\frac{\sigma_X}{\sqrt{n}},\ \bar{X}_n + z_{\alpha/2}\frac{\sigma_X}{\sqrt{n}}\right].$$

The confidence level $1-\alpha$ is a *coverage probability*: if the sampling experiment were repeated indefinitely and a new interval constructed each time, the proportion of intervals containing the fixed parameter μ_X would approach $1 - \alpha$. The randomness lies in the interval, not in μ_X.

Sample Size Determination. Suppose a maximal half–width $\varepsilon > 0$ is prescribed, so that

$$z_{\alpha/2} \frac{\sigma_X}{\sqrt{n}} \leq \varepsilon.$$

Solving for n yields the sample size requirement

$$n \geq \left(\frac{z_{\alpha/2} \sigma_X}{\varepsilon} \right)^2.$$

Higher confidence levels (smaller α) and tighter accuracy requirements (smaller ε) both increase the required sample size, expressing the fundamental trade–off between reliability and precision.

Common Confidence Levels.

Confidence level	α	$z_{\alpha/2}$
90%	0.10	1.645
95%	0.05	1.960
99%	0.01	2.576

These quantiles translate coverage probabilities into explicit margins of error and sample size requirements under the normal approximation.

Chapter 6: Transformations & Generating Functions

Transformations and generating functions describe how probability laws respond to algebraic operations. Transformations track how distributions are reshaped under mappings of one or several random variables, preserving probability while redistributing mass and density. Generating functions encode an entire distribution into a single analytic object, turning operations on random variables into differentiation, multiplication, and limits. Together, these tools replace direct manipulation of densities with structural rules, providing a concise way to analyze sums, moments, and distributional behavior.

1 Transformations of Random Variables

Transformations describe how probability distributions change under deterministic mappings of random variables. Probability itself is conserved, but its representation, mass in the discrete case and density in the continuous case, is reshaped by the geometry of the transformation. The guiding principle is invariance of probability: events and their transformed images must carry the same probability.

Discrete Transformations. Let X be a discrete random variable with pmf $p_X(x)$, and suppose

$$Y = g(X)$$

is a one-to-one transformation. Each value y of Y corresponds uniquely to a value $x = g^{-1}(y)$ of X, so probability mass is transferred directly:

$$p_Y(y) = p_X\big(g^{-1}(y)\big).$$

When the transformation is not invertible or when the pmf of X is inconvenient to compute directly, the distribution of Y can always be obtained via the cumulative distribution function:

$$F_Y(y) = P(Y \le y) = P(g(X) \le y),$$

which remains valid regardless of the form of g.

Continuous Transformations. Let X be a continuous random variable with pdf $f_X(x)$, and let

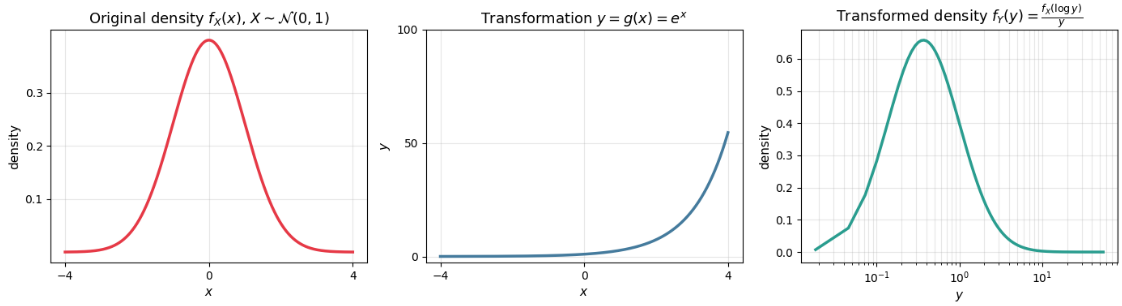
$$Y = g(X)$$

where g is monotonic and differentiable. Probability density transforms according to the change of variables:

$$f_Y(y) = f_X\big(g^{-1}(y)\big)\left|\frac{dx}{dy}\right|, \qquad x = g^{-1}(y).$$

The absolute derivative measures how intervals in x-space expand or contract under the map to y-space; stretching lowers density, compression raises it, preserving total probability.

Monotone Continuous Transformation



Monotone change of variables illustrating density reshaping under $y = g(x) = e^x$.

If g is *not* monotonic, multiple values of X may map to the same value of Y. In this case, the density of Y is obtained by summing the contributions from all preimages $\{x_k\}$ satisfying $y = g(x_k)$:

$$f_Y(y) = \sum_k f_X(x_k) \left| \frac{dx}{dy} \right|_{x=x_k}.$$

Each branch contributes additively, reflecting the fact that probability mass from different regions of the original space accumulates at the same point in the transformed space.

Joint Transformations and Change of Variables. Transformations naturally extend to multiple random variables. Let (X, Y) be a pair of continuous random variables with joint pdf $f_{X,Y}(x, y)$, and define new variables

$$U = g(X, Y), \qquad V = h(X, Y),$$

where the transformation is invertible with inverse

$$x = x(u, v), \qquad y = y(u, v).$$

The joint pdf of (U, V) is then

$$f_{U,V}(u, v) = f_{X,Y}\big(x(u, v),\ y(u, v)\big) \left| \frac{d(x, y)}{d(u, v)} \right|.$$

This formula generalizes the one-dimensional change of variables and ensures that probability assigned to regions in the (x, y)-plane matches the probability assigned to their images in the (u, v)-plane.

Jacobian Determinant. The Jacobian determinant is defined as

$$\frac{d(x, y)}{d(u, v)} = \det \begin{bmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{bmatrix}.$$

It measures the local area distortion induced by the transformation. Values greater than one indicate local expansion, while values less than one indicate compression.

Table 9: Transformation Formulas Summary

Transformation Type	**Distribution Formula**
Discrete, one-to-one	$p_Y(y) = p_X\big(g^{-1}(y)\big)$
Discrete (via CDF)	$F_Y(y) = P(Y \le y) = P\big(g(X) \le y\big)$
Continuous, monotone	$f_Y(y) = f_X\big(g^{-1}(y)\big) \left\lvert \frac{dx}{dy} \right\rvert$
Continuous, non-monotone	$f_Y(y) = \sum_k f_X(x_k) \left\lvert \frac{dx}{dy} \right\rvert_{x=x_k}, \quad y = g(x_k)$
Two-variable transformation	$f_{U,V}(u,v) = f_{X,Y}\big(x(u,v), y(u,v)\big) \left\lvert \frac{d(x,y)}{d(u,v)} \right\rvert$
Jacobian determinant	$\frac{d(x,y)}{d(u,v)} = \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}$

2 Moment Generating Functions & Characteristic Functions

Transformations describe how distributions change under mappings of random variables. Generating functions provide a complementary perspective; instead of reshaping probability densities directly, they encode an entire distribution into a single analytic object. Operations on random variables then correspond to simple algebraic operations on these functions. This shift from densities to functions is especially powerful for studying moments, sums of independent variables, and distributional limits.

Moment Generating Function (MGF). The moment generating function of a random variable X is defined as

$$M_X(s) = E\left[e^{sX}\right],$$

whenever the expectation exists in an open neighborhood of $s = 0$. If it exists, the MGF uniquely determines the distribution of X.

The defining feature of the MGF is that its derivatives at the origin recover the raw moments of the distribution:

$$E[X^k] = M_X^{(k)}(0) = \left. \frac{d^k M_X(s)}{ds^k} \right|_{s=0}.$$

Thus, the entire sequence of moments is encoded in the local behavior of $M_X(s)$ near zero.

Raw vs. Central Moments. The moments generated by $M_X(s)$ are raw moments $E[X^k]$, not central moments. Central moments measure deviations from the mean,

$$E\left[(X - E[X])^k\right],$$

and must be obtained by algebraic manipulation of raw moments. Only when $E[X] = 0$ do raw and central moments coincide.

Characteristic Function (CF). The characteristic function of X is defined as

$$\phi_X(\omega) = E\left[e^{i\omega X}\right], \qquad \omega \in \mathbb{R}.$$

Unlike the MGF, the characteristic function always exists for every random variable, regardless of tail behavior. It plays an analogous role to the MGF but operates in the complex domain and is closely related to the Fourier transform of the distribution.

Relationship Between MGFs and CFs. When both exist, the MGF and CF are analytically connected. Moments can be recovered from either representation:

$$E[X^k] = \frac{d^k}{ds^k} M_X(s)\bigg|_{s=0} = \frac{1}{i^k} \frac{d^k}{d\omega^k} \phi_X(\omega)\bigg|_{\omega=0}.$$

In practice, MGFs are convenient when they exist, while CFs provide a universally valid alternative.

Key Structural Properties. Generating functions turn operations on random variables into algebraic rules:

1. *Additivity for independent sums.* If X and Y are independent,

$$M_{X+Y}(s) = M_X(s)\, M_Y(s), \qquad \phi_{X+Y}(\omega) = \phi_X(\omega)\, \phi_Y(\omega).$$

2. *Sums of i.i.d. variables.* For $S_n = \sum_{k=1}^n X_k$ with X_k i.i.d.,

$$M_{S_n}(s) = [M_X(s)]^n.$$

3. *Sample mean.* For $\bar{X}_n = \frac{1}{n} \sum_{k=1}^n X_k$,

$$M_{\bar{X}_n}(s) = [M_X(s/n)]^n.$$

4. *Linear transformations.* If $Y = aX + b$,

$$M_Y(s) = e^{sb}\, M_X(as), \qquad \phi_Y(\omega) = e^{i\omega b}\, \phi_X(a\omega).$$

Uses of Generating Functions.

- Compact derivation of moments
- Efficient analysis of sums of independent variables
- Distributional identification via uniqueness
- Proofs of limit theorems through functional convergence

Lévy's Continuity Theorem. If the characteristic functions of a sequence $\{X_n\}$ converge pointwise to the characteristic function of X,

$$\phi_{X_n}(\omega) \to \phi_X(\omega),$$

and ϕ_X is continuous at the origin, then

$$X_n \xrightarrow{d} X.$$

An analogous statement holds for MGFs when they exist in a neighborhood of zero.

Convergence of generating functions implies convergence in distribution. This principle underlies many classical limit results, including the Central Limit Theorem, and explains why generating functions serve as a bridge between finite-sample distributions and asymptotic behavior.

Table 10: MGFs & CFs Formulas Summary

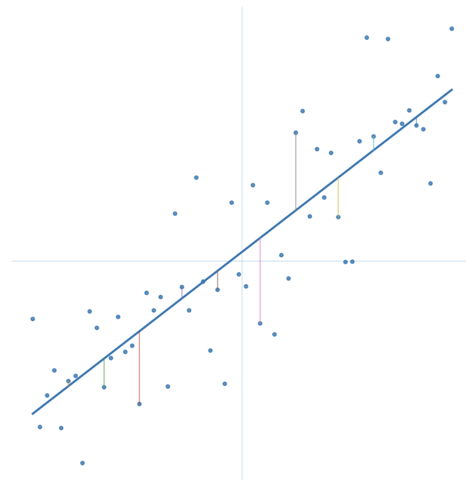
Object / Operation	**Formula**
Moment generating function	$M_X(s) = E\left[e^{sX}\right]$
Raw moments from MGF	$E[X^k] = \left.\dfrac{d^k M_X(s)}{ds^k}\right\vert_{s=0}$
Characteristic function	$\phi_X(\omega) = E\left[e^{i\omega X}\right]$
Moments from CF	$E[X^k] = \dfrac{1}{i^k} \left.\dfrac{d^k \phi_X(\omega)}{d\omega^k}\right\vert_{\omega=0}$
Lévy continuity (MGF / CF)	$M_{X_n}(s) \to M_X(s)$ or $\phi_{X_n}(\omega) \to \phi_X(\omega) \;\Rightarrow\; X_n \xrightarrow{d} X$
Independent sum	$M_{X+Y}(s) = M_X(s)\, M_Y(s)$
Sum of i.i.d. variables	$M_{S_n}(s) = [M_X(s)]^n$
Sample mean	$M_{\bar{X}_n}(s) = [M_X(s/n)]^n$
Linear transformation	$M_{aX+b}(s) = e^{sb}\, M_X(as)$

Table 11: Moment Generating Functions (MGFs)

Distribution	$M_X(s)$
Bernoulli (p)	$1 - p + pe^s$
Binomial (n, p)	$(1 - p + pe^s)^n$
Geometric (p)	$\dfrac{pe^s}{1-(1-p)e^s}, \quad s < -\ln(1-p)$
Poisson (λ)	$\exp[\lambda(e^s - 1)]$
Negative Binomial (r, p)	$\left(\dfrac{p}{1-(1-p)e^s}\right)^r, \quad s < -\ln(1-p)$
Discrete Uniform $\{1, \dots, n\}$	$\dfrac{e^s(1-e^{ns})}{n(1-e^s)}$
Exponential (θ)	$\dfrac{1}{1-\theta s}, \quad s < \dfrac{1}{\theta}$
Gamma (α, θ)	$(1-\theta s)^{-\alpha}, \quad s < \dfrac{1}{\theta}$
Chi-square (r)	$(1-2s)^{-r/2}, \quad s < \dfrac{1}{2}$
Normal (μ, σ^2)	$\exp\left(\mu s + \frac{1}{2}\sigma^2 s^2\right)$
Uniform (a, b)	$\dfrac{e^{sb} - e^{sa}}{s(b-a)}$
Beta (α, β)	${}_1F_1(\alpha; \alpha+\beta; s)$ (via confluent hypergeometric function)
Cauchy (m, d)	does not exist (diverges for all $s \neq 0$)

Chapter 7: Estimation & Least Squares

Estimation formalizes the passage from randomness to inference. When quantities of interest cannot be observed directly, they are inferred from noisy data using rules that balance accuracy against stability under uncertainty.
This chapter frames estimation as an optimal approximation problem: selecting functions of the data that are closest to an unknown target under a mean-square criterion. The MMSE estimator emerges as a projection onto available information, while least squares provides its concrete realization in linear models. Together, these ideas form the foundation of modern statistical inference and regression.



1 Estimators & Mean-Square Error

Estimation concerns inferring an unknown quantity from noisy or incomplete observations. Because the data are random, any rule that produces an estimate is itself random. Estimators are therefore random variables whose performance must be assessed probabilistically.

Data and Estimator. Let

$$\mathcal{X}_n = \{X_1, X_2, \dots, X_n\}$$

denote a collection of observed random variables. An *estimator* of a parameter θ is any measurable function of the data,

$$\hat{\theta}_n = g(\mathcal{X}_n) = g(X_1, X_2, \dots, X_n).$$

Since $\hat{\theta}_n$ is constructed from random inputs, it is itself a random variable (or statistic). Different choices of the function g correspond to different estimation rules, each inducing a distinct distribution for $\hat{\theta}_n$.

Random Parameters and Joint Modeling. In many settings, the parameter θ is modeled as a random variable with prior density $f(\theta)$. The observed data then depend on θ through a joint distribution,

$$f(\theta, X_1, X_2, \dots, X_n) = f(\theta \mid X_1, X_2, \dots, X_n)\, f(X_1, X_2, \dots, X_n).$$

This formulation treats estimation as an inference problem on a joint probability space, where uncertainty about θ is updated through observation of the data.

Mean-Square Error as a Performance Criterion. To compare estimators, a notion of optimality is required. The most common and analytically tractable criterion is *mean-square error*

(MSE),

$$\mathrm{MSE}(\hat{\theta}_n) = E\left[(\hat{\theta}_n - \theta)^2\right].$$

Mean-square error penalizes deviations quadratically and admits a geometric interpretation as an expected squared distance. This criterion induces a Hilbert-space structure in which optimal estimators appear as projections, leading directly to the MMSE estimator and the orthogonality principle.

2 MMSE Estimation & Orthogonality Principle

The mean-square error criterion endows the space of random variables with a geometric structure. Within this framework, estimation becomes a projection problem: among all functions of the observed data, the optimal estimator is the one closest—in the mean-square sense, to the unknown quantity.

Minimum Mean-Square Error (MMSE) Estimation. The minimum mean-square error (MMSE) estimator of a parameter θ given data $\mathcal{X}_n$ is defined as

$$\hat{\theta}_n^{\mathrm{MMSE}} = E[\theta \mid \mathcal{X}_n].$$

Equivalently, it is the solution to the optimization problem

$$\hat{\theta}_n^{\mathrm{MMSE}} = \arg\min_{\hat{\theta}(\mathcal{X}_n)} E\left[(\theta - \hat{\theta}(\mathcal{X}_n))^2\right].$$

Thus, the conditional expectation is not merely an average, but the unique estimator that minimizes expected squared error among all measurable functions of the data.

Orthogonality Principle. A defining property of the MMSE estimator is that its estimation error is orthogonal to every function of the observed data:

$$E\left[(\theta - \hat{\theta}_n^{\mathrm{MMSE}})\, g(\mathcal{X}_n)\right] = 0 \quad \text{for all measurable } g.$$

This condition characterizes the MMSE estimator as the projection of θ onto the data space, so the estimation error has no component in any direction spanned by the observations.
The orthogonality principle formalizes optimality: once the MMSE estimator is obtained, no further function of the data can reduce the mean-square error, since any additional adjustment is uncorrelated with the observations and therefore ineffective.

3 Linear Model & Ordinary Least Squares

The MMSE framework identifies the optimal estimator among all functions of the data. In many applications, however, attention is restricted to estimators that are linear in unknown parameters. Ordinary least squares arises as the MMSE solution within this linear class and provides a concrete realization of the orthogonality principle.

Linear Regression Model. Consider the simple linear model

$$Y = \beta_0 + \beta_1 X + \varepsilon,$$

or, for observations indexed by k,

$$Y_k = \beta_0 + \beta_1 X_k + \varepsilon_k,$$

where the error terms satisfy $E[\varepsilon_k] = 0,\ V[\varepsilon_k] = \sigma^2,\ \varepsilon_k$ independent.
Under the additional assumption of Gaussian noise, $\varepsilon_k \sim \mathcal{N}(0, \sigma^2)$, the model admits a probabilistic interpretation consistent with MMSE estimation.

Model Assumptions. The ordinary least squares framework relies on:

1. Linearity in parameters,
2. Zero-mean errors,
3. Constant error variance (homoscedasticity),
4. Independence of error terms.

Least Squares Objective. OLS estimates (β_0, β_1) by minimizing the sum of squared residuals

$$J^{\mathrm{LS}} = \sum_{k=1}^{n} (Y_k - \beta_0 - \beta_1 X_k)^2.$$

Setting the partial derivatives of J^{LS} to zero yields the closed-form estimators.

$$\hat{\beta}_1 = \frac{\sum_{k=1}^{n} (X_k - \bar{X})(Y_k - \bar{Y})}{\sum_{k=1}^{n} (X_k - \bar{X})^2} = \frac{S_{XY}}{S_{XX}},$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X},$$

where

$$\bar{X} = \frac{1}{n} \sum_{k=1}^{n} X_k, \qquad \bar{Y} = \frac{1}{n} \sum_{k=1}^{n} Y_k.$$

Properties of OLS. Under the stated assumptions, the OLS estimator satisfies:

1. *Unbiasedness:* $E[\hat{\beta}] = \beta$,
2. *Consistency:* $\hat{\beta} \to \beta$ as $n \to \infty$,
3. *Efficiency:* minimum variance among all unbiased linear estimators.

Ordinary least squares is the MMSE estimator restricted to linear models. The resulting estimation error is orthogonal to the regressors, reflecting the same projection principle that characterizes MMSE estimation in general.

Part II Stochastic Processes

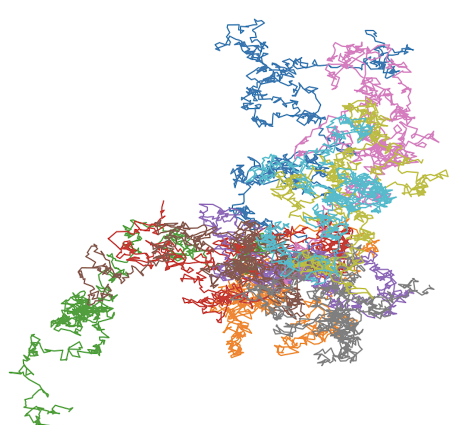
The history of human understanding of the world is a history of recognizing and describing various forms of motion — from the macroscopic movements of celestial bodies to the motion of molecules, even to the movements within the human psyche. We generally call this “change,” meaning something altering over time.

The most successful model humans developed for describing motion is Newton’s celestial mechanics. Determinism is the most significant characteristic of the Newtonian system. Given a position and velocity, the trajectory of motion is fixed. However, science after the 20th century lost the beautiful aura of Newtonian certainty.

This is because when people attempted to describe motion in the real world, which is full of complex and unknown factors, they found that uncertain factors (commonly referred to as noise) are crucial to how things change, and Newtonian methods are almost inapplicable. The best tool we have for describing changes in things is a framework called probability theory. Correspondingly, a completely new method for studying motion emerged — stochastic processes, which provide a refined mathematical description of motion under uncertainty.

We are surrounded by vast amounts of data — the so-called Big Data era. The most fundamental characteristic of this data is that it contains enormous amounts of noise. *Stochastic processes are the weapon for extracting information from this noise.*

The first great triumph of the stochastic process method was Einstein’s work on *Brownian Motion*. Small pollen particles suspended in water exhibit random movement due to constant collisions with water molecules (pollen grains, being small, are easily affected by the thermal fluctuations of water molecules). Studying the tiny movements of these pollen particles might seem a bit naive, yet from it, we found important information about the molecular world. And the disordered and variable trajectory of the pollen provided us with a paradigm for random motion (*Random Walk*).



Computer-generated Brownian motion trajectories for ten particles

If we were to use an analogy, a stochastic process is like a garden of forking paths. Standing at the present point and looking towards the future, there are countless ways things might change. Each possibility continually branches out into other possibilities.

Chapter 8: Random Processes

Random processes extend probability theory from isolated uncertainty to structured randomness indexed over time, space, or other domains. Rather than studying single random variables, we study collections whose components interact, evolve, and accumulate information across an index set. This shift is essential for modeling dynamic systems, where uncertainty is not static but unfolds with structure. In this chapter, we focus on how processes are indexed, how they are classified structurally, how they are observed through different analytical views, and how dependence, second-order structure, and stationarity organize their behavior. The emphasis is on concepts and definitions that recur throughout stochastic modeling, serving as the scaffolding for later chapters on Markov processes, Brownian Motion, and stochastic calculus.

1 What is a Random Process?

A random process generalizes a random variable from a single uncertain quantity to an indexed collection of random variables. Rather than describing randomness at a single point, it models how randomness is organized across an index such as time, space, or another parameter.

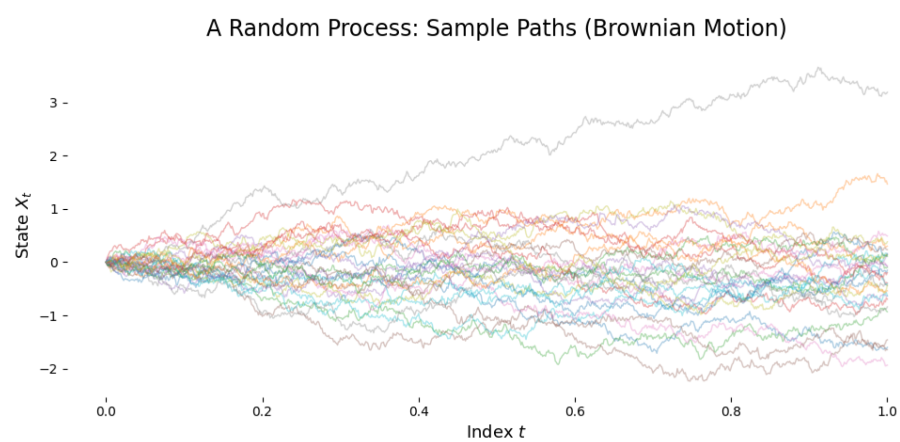
Random Process. A *random* (or *stochastic*) process is a family of random variables indexed by a set $\mathcal{T}$:

$$\{X_t \, : \, t \in \mathcal{T}\},$$

where all X_t are defined on the same underlying probability space.

For each fixed index value $t \in \mathcal{T}$, the quantity X_t is an ordinary random variable. The index set $\mathcal{T}$ specifies where the process is observed and may represent time, space, frequency, or any other ordered or unordered parameter.

A random process is treated as a single mathematical object rather than as a collection of unrelated variables. Its defining structure lies in the joint behavior of $\{X_t\}$ across different indices, which encodes dependence, regularity, and long-run behavior beyond what is visible at any single index.



Realizations of a random process indexed by t.

2 Structural Classification of Random Processes

A random process is structurally determined by two ingredients:

- the *index set*, which specifies where the process is observed,
- the *state space*, which specifies the values the process can take.

Together, these define the geometric framework in which randomness is expressed.

Index Set. The *index set* $\mathcal{T}$ specifies the domain over which the process is defined. Formally, a random process is written as

$$\{X_t \,:\, t \in \mathcal{T}\},$$

where each X_t is a random variable defined on a common probability space.
Two structural cases arise most often.

1. **Discrete Indexing.** A process is said to have a *discrete index set* if

$$\mathcal{T} \subseteq \mathbb{Z} \qquad \text{(or any countable set).}$$

   The process is observed at isolated index values and may be written as a sequence

$$X_0, X_1, X_2, \ldots$$

   Analysis is naturally expressed using differences and sums.

2. **Continuous Indexing.** A process is said to have a *continuous index set* if

$$\mathcal{T} \subseteq \mathbb{R} \qquad \text{(typically an interval such as } [0, T] \text{ or } [0, \infty)\text{).}$$

   The process is defined at every point of a continuum, enabling analysis via limits, derivatives, and integrals.

State Space. The *state space* $\mathcal{X}$ specifies the set of possible values taken by the random variables $\{X_t\}$. For each $t \in \mathcal{T}$,

$$X_t \in \mathcal{X}.$$

Again, two structural cases dominate.

1. **Discrete State Space.** A process is said to have a *discrete state space* if

$$\mathcal{X} \subseteq \mathbb{Z} \qquad \text{(or finite/countable).}$$

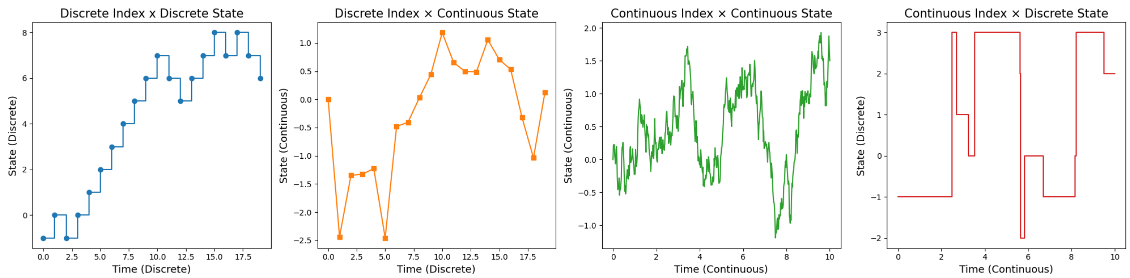
   The process takes values in isolated states, and probabilities are described using mass functions.

2. **Continuous State Space.** A process is said to have a *continuous state space* if

$$\mathcal{X} \subseteq \mathbb{R}^d, \quad d \geq 1.$$

The process varies over a continuum, and probabilistic descriptions are given via densities.

Index–State Classification. Most stochastic models encountered in practice fall into a small number of index–state combinations. This classification is purely structural: it describes *how* randomness is organized, not *how* it behaves.



Classification of stochastic processes by index set and state space.

Table 12: Classification by Index Set and State Space

Index Set	State Space	Description	Examples
Discrete	Discrete	Observed at discrete indices with values in a finite or countable set.	Simple random walk
Discrete	Continuous	Observed at discrete indices with continuously varying values.	Autoregressive models
Continuous	Continuous	Defined at all indices with continuously evolving values.	Brownian motion
Continuous	Discrete	Defined at all indices with state changes occurring through jumps.	Poisson process

3 Analytical Views and Realizations

A stochastic process can be examined by fixing different components of its mathematical structure. This section distinguishes two fundamental perspectives for analysis and introduces the deterministic objects obtained when randomness is resolved.

Analytical Perspectives. Two complementary viewpoints are used throughout stochastic process theory, depending on what is held fixed.

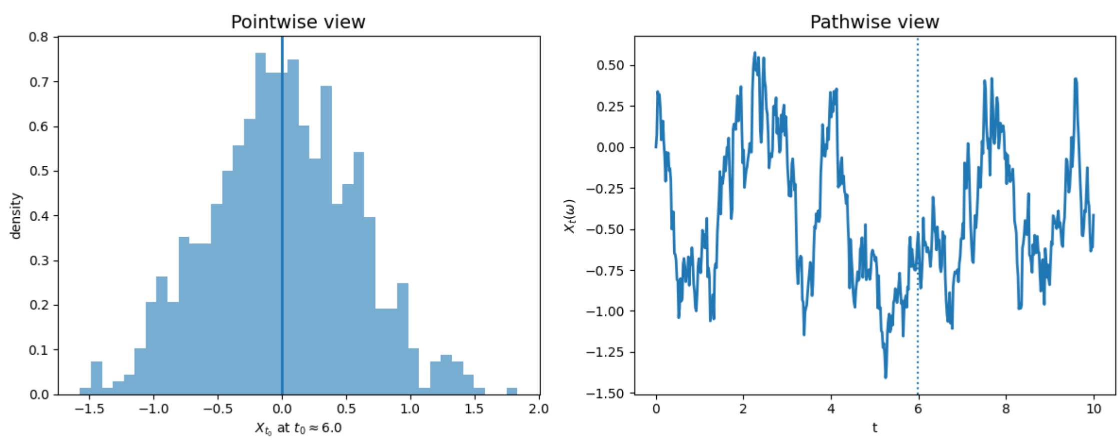
1. **Pointwise View** (Fixed Index). The index $t \in \mathcal{T}$ is fixed and randomness varies. For each t, the quantity X_t is treated as an ordinary random variable with its own distribution, mean, and variance.

$$\text{Study } X_t \text{ as a random variable for fixed } t.$$

This perspective emphasizes marginal and joint distributions at specific index locations.

2. **Pathwise View** (Fixed Outcome). The outcome $\omega \in \Omega$ is fixed and the index varies. The process is examined as a deterministic function of the index,
$$t \mapsto X_t(\omega).$$
This perspective focuses on geometric and regularity properties such as continuity, smoothness, and long-run behavior.

The distinction reflects whether randomness is studied across outcomes or across the index. Both perspectives apply to all indexed random objects; only the geometry of what is viewed changes.



Pointwise and pathwise views of a stochastic process.

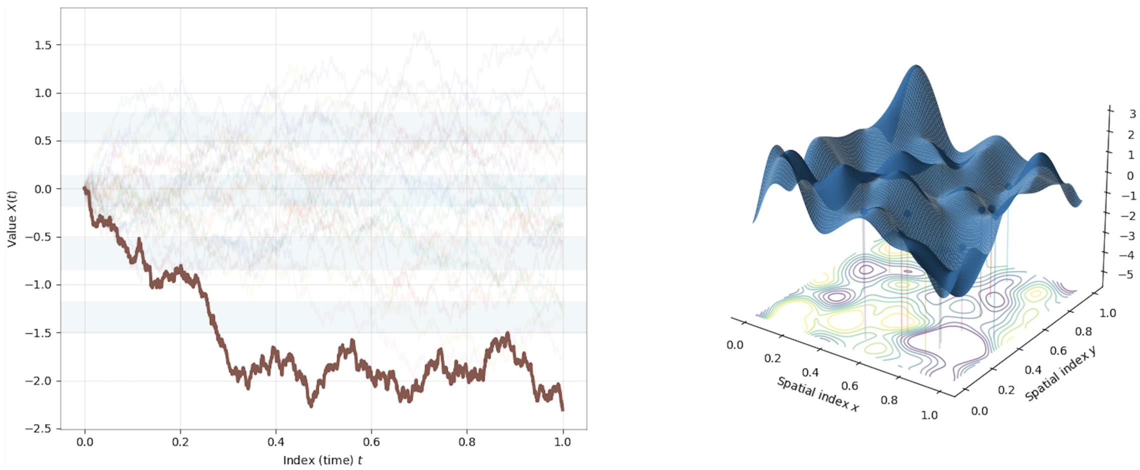
Realizations. A *realization* is the deterministic object obtained by fixing the outcome of the underlying probability space.

1. **Realization of a Random Process.** Let $\{X_t : t \in \mathcal{T}\}$ be a random process defined on $(\Omega, \mathcal{F}, \mathbb{P})$. For a fixed outcome $\omega \in \Omega$, the mapping
$$t \mapsto X_t(\omega), \qquad t \in \mathcal{T},$$
is called a realization.
(*Sample Paths*) If the index set is one-dimensional, $\mathcal{T} \subseteq \mathbb{R}$ or $\mathcal{T} \subseteq \mathbb{Z}$, each realization takes the form of a curve or sequence. In this case, the terms *realization of a random process* and *sample path* refer to the same object.

2. **Realization of a Random Field.** If the index set is higher-dimensional,
$$\mathcal{T} \subseteq \mathbb{R}^d, \quad d \geq 2,$$
realizations no longer resemble paths.
A realization of a random field is a deterministic function defined over a multi-dimensional index domain. Such realizations appear as spatial configurations, such as surfaces or vol-

umes, reflecting the geometry of the underlying index set.

In all cases, a stochastic model is fully specified by the collection of all possible realizations together with their probability structure. Whether realizations appear as paths or fields depends solely on the dimension of the index set.



Realizations of a random process (left) and a random field (right).

4 Second-Order Descriptive Structure

While the full probabilistic structure of a stochastic process is encoded in its finite-dimensional joint distributions, practical analysis often relies on lower-order summaries. The most important of these are second-order descriptors, which capture average behavior and linear dependence across the index set.

Mean Function. The *mean function* describes the expected value of the process at each index. Discrete index:

$$m_k \equiv \mathbb{E}[X_k], \qquad k \in \mathbb{Z}.$$

Continuous index:

$$m(t) \equiv \mathbb{E}[X_t], \qquad t \in \mathbb{R}.$$

The mean function provides a baseline or trend against which fluctuations of the process are measured.

Covariance Function. The *covariance function* quantifies linear dependence between values of the process at different indices.
Discrete index:

$$C(k, \ell) \equiv \mathbb{E}[X_k X_\ell] - m_k m_\ell.$$

Continuous index:

$$C(t, s) \equiv \mathbb{E}[X_t X_s] - m(t)m(s).$$

The covariance function captures how variability is shared across the index set and encodes dependence beyond marginal behavior.

Variance as a Special Case. Variance arises as the diagonal of the covariance function.
Discrete index:

$$\mathrm{Var}(X_k) = C(k, k).$$

Continuous index:

$$\mathrm{Var}(X_t) = C(t, t).$$

Autocovariance and Autocorrelation. For many stochastic processes, dependence depends on relative separation between indices rather than their absolute location. In such cases, covariance can be expressed as a function of the lag between indices, leading to the notions of autocovariance and autocorrelation.

- *Autocovariance function*: expresses covariance as a function of lag
- *Autocorrelation function*: normalizes this dependence by marginal variances.

These descriptors are particularly important for processes whose statistical properties are invariant across the index set.

5 Dependence and Information Structure

Dependence describes how values of a stochastic process interact across the index set. This section classifies common forms of dependence, ranging from complete independence to structured, information-based relationships.

Finite-Dimensional Distributions. A stochastic process is not defined by its individual random variables alone, but by their joint behavior across indices. Formally, for any finite collection $t_1, \dots, t_n \in \mathcal{T}$, the random vector

$$(X_{t_1}, \dots, X_{t_n})$$

has a joint distribution. The collection of all such finite-dimensional joint distributions fully determines the probabilistic structure of the process.
Marginal distributions describe behavior at individual index values, but provide only isolated snapshots. Dependence across the process is encoded entirely in joint distributions.

Independence. The random variables $\{X_{t_1}, \dots, X_{t_n}\}$ are independent if their joint distribution factors as

$$\mathbb{P}(X_{t_1} \in A_1, \dots, X_{t_n} \in A_n) = \prod_{i=1}^{n} \mathbb{P}(X_{t_i} \in A_i).$$

When this factorization fails, the variables are said to be dependent.

Independent and Identically Distributed (IID) Processes. A stochastic process $\{X_t : t \in \mathcal{T}\}$ is *independent and identically distributed (IID)* if all variables are mutually independent and share a common marginal distribution. For any finite index set,

$$p_X(x_1, \dots, x_n; t_1, \dots, t_n) = \prod_{i=1}^{n} p_X(x_i),$$

where the marginal distribution p_X does not depend on t.
IID processes represent repeated draws from a fixed distribution. The index does not encode interaction or temporal structure and serves only as a label.
Key properties of IID processes:

- *Independence:* observations provide no information about one another,
- *Memorylessness:* future behavior is independent of past observations,
- *Shift invariance:* probabilistic behavior is invariant under index shifts,
- *Stationarity:* all marginal distributions are identical.

Markov Dependence. A stochastic process satisfies the *Markov property* if the future depends on the present state only. Formally,

$$\mathbb{P}(X_{n+1} \mid X_1, \dots, X_n) = \mathbb{P}(X_{n+1} \mid X_n).$$

Given the current state, earlier observations provide no additional information about future evolution.
The Markov property introduces structured dependence with limited memory and forms the foundation of many stochastic models. Markov chains are developed in detail in Chapter 9.

Dependence Through Increments. Some processes are characterized by the behavior of changes over intervals rather than by direct state-to-state dependence.

1. **Independent Increments.** Let $\{X(t), t \in [0, \infty)\}$ be a continuous-time random process. We say that $X(t)$ has *independent increments* if, for all $0 \leq t_1 < t_2 < t_3 < \cdots < t_n$, the random variables

   $$X(t_2) - X(t_1), \quad X(t_3) - X(t_2), \quad \dots, \quad X(t_n) - X(t_{n-1})$$

   are independent.
2. **Stationary Increments.** Let $\{X(t), t \in [0, \infty)\}$ be a continuous-time random process. We say that $X(t)$ has *stationary increments* if, for all $t_2 > t_1 \geq 0$ and all $r > 0$, the two random variables

   $$X(t_2) - X(t_1) \quad \text{and} \quad X(t_2 + r) - X(t_1 + r)$$

   have the same distribution. In other words, the distribution of the difference depends only on the length of the interval $(t_1, t_2]$, and not on its exact location on the real line.

Independent and stationary increments are central to counting processes and underlie many fundamental continuous-time models.

Information-Based Dependence. In some processes, dependence is governed by the information revealed as the index evolves. Let $\mathcal{F}_t$ denote the information available up to index t. A central example is the *martingale property*, which states that the conditional expectation of the future given current information equals the present value:

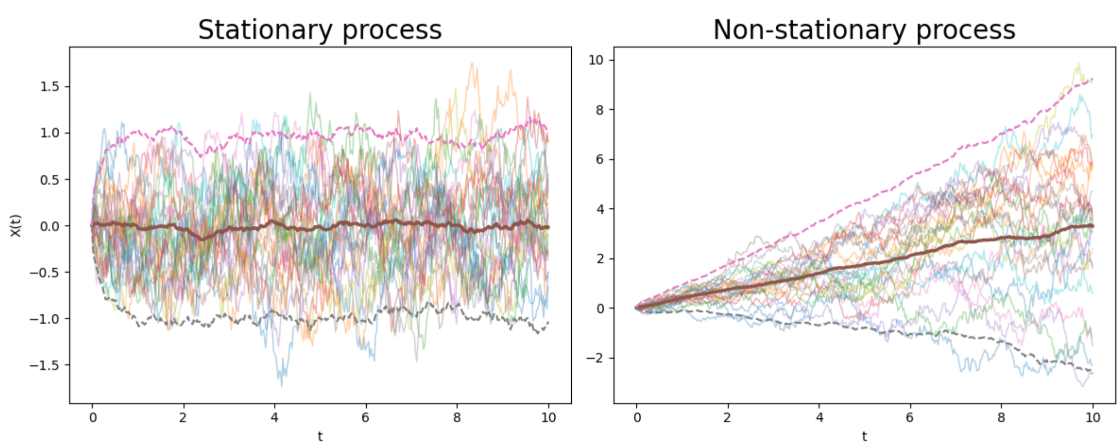
$$\mathbb{E}[X_{t+1} \mid \mathcal{F}_t] = X_t.$$

Martingales exhibit no systematic drift relative to available information and play a fundamental role in stochastic analysis. They are developed in detail in Chapter 11.

6 Stationarity

Stationarity formalizes invariance of a stochastic process under shifts of the index. Informally, a process is stationary if its statistical properties do not depend on absolute index location, allowing behavior observed in one region of the index set to be representative elsewhere.

Stationarity. A stochastic process $\{X_t : t \in \mathcal{T}\}$ is said to be *stationary* if its probabilistic structure is invariant under shifts of the index. That is, translating the index does not alter the laws governing the process.



Stationary process exhibits time-invariant mean and variability; non-stationary shows systematic change over time.

Stationarity can be imposed with varying strength, depending on how much of the distributional structure is required to be invariant:

1. Strict stationarity: all finite-dimensional distributions are invariant,
2. First-order stationarity: only first moments are invariant.
3. Second-order stationarity: only first and second moments are invariant,

Strict Stationarity. A stochastic process $\{X_t\}$ is *strictly stationary* if all of its finite-dimensional distributions are invariant under shifts of the index. Formally, for any $n \geq 1$, any indices $t_1, \ldots, t_n$, and any shift Δ,

$$(X_{t_1}, \ldots, X_{t_n}) \stackrel{d}{=} (X_{t_1+\Delta}, \ldots, X_{t_n+\Delta}).$$

Equivalently, the joint distribution of the process is unchanged by translating the index.

First-Order Stationarity. A process is *first-order stationary* if its mean does not depend on the index:

$$\mathbb{E}[X_t] = \mu, \qquad \text{for all } t.$$

In this case, the process exhibits no systematic trend in its average behavior.

Second-Order Stationarity. A process is *second-order stationary* if, in addition to a constant mean, its second moments depend only on relative index separation:

$$\mathbb{E}[X_t X_s] = R(t-s).$$

Thus, linear dependence is determined by lag rather than absolute index location.

Wide-Sense Stationarity (WSS). A stochastic process is *wide-sense stationary* if it is first- and second-order stationary:

1. $\mathbb{E}[X_t] = m$ (constant mean),
2. $\mathbb{E}[X_t X_s] = R(t-s)$ (lag-dependent second moments).

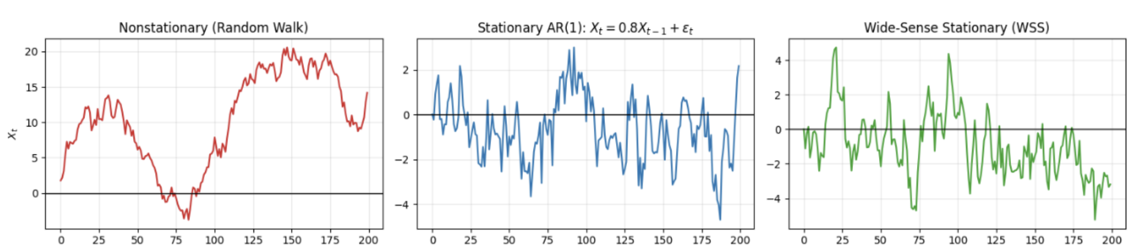
This definition applies in both discrete and continuous index settings.
Note: Strict stationarity implies wide-sense stationarity, but the converse need not hold.
Symmetry of the Autocovariance: If $\{X_t\}$ is wide-sense stationary, its second-order structure depends only on lag. In particular,

$$R(t-s) = R(s-t).$$

Writing $s = t + \tau$ yields

$$R(\tau) = R(-\tau),$$

so the autocovariance function is even.
This symmetry follows from commutativity of multiplication: $X_t X_s = X_s X_t$. At the second-order level, correlation with the future equals correlation with the past.



Chapter 9: Gaussian Processes

Gaussian distributions occupy a central place in probability theory because of both their analytical tractability and the way linear structure interacts with randomness. In one dimension they yield the familiar bell curve; in higher dimensions they form elliptically contoured clouds whose geometry is determined by covariance. This chapter extends these ideas to infinite dimensions, where randomness is carried by entire functions rather than finite vectors.

Viewing a stochastic process as a family of jointly Gaussian random variables leads to the notion of a Gaussian process: a probability law on function spaces fully specified by a mean function and a covariance kernel. This framework unifies random curves, spatial fields, and continuous–time models under the same linear–algebraic structure that governs multivariate normals and underlies both Brownian motion and Gaussian process regression.

1 Gaussian Random Variable

Before introducing Gaussian vectors and Gaussian processes, we first recall the basic properties of the one–dimensional Gaussian (or normal) distribution, which serves as the fundamental building block for all Gaussian models.

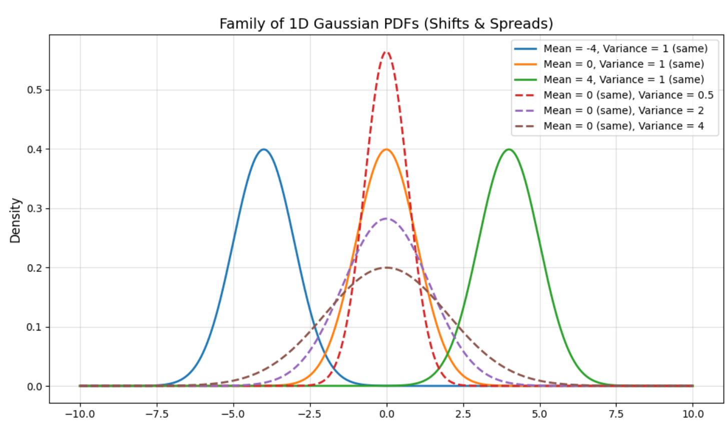
A real–valued random variable X is said to be Gaussian with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$, written

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

if it has probability density function

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}.$$

Mean & Variance. The parameter $\mu = \mathbb{E}[X]$ determines the location of the distribution, while $\sigma^2 = \mathrm{Var}(X)$ controls its spread. The density is symmetric about μ and its level sets are determined by the quadratic form $(x-\mu)^2$.



Gaussian probability density functions with varying means and variances

Moment Generating Function. The MGF of X is given by

$$M_X(t) = \mathbb{E}[e^{tX}] = \exp\left(\mu t + \tfrac{1}{2}\sigma^2 t^2\right), \qquad t \in \mathbb{R},$$

which shows that the Gaussian distribution is completely characterized by its first two moments.

Stability under Linear Transformations. A fundamental property of the Gaussian distribution is stability under linear transformations. If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $a, b \in \mathbb{R}$, then

$$Y = aX + b$$

is also Gaussian, with

$$Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2).$$

This closure under linear operations, together with complete determination by mean and variance, will extend in higher dimensions and ultimately to collections of infinitely many random variables, leading to the notion of Gaussian vectors and Gaussian processes.

2 Gaussian Vectors (Multivariate Gaussian)

We now extend the one–dimensional Gaussian distribution to the joint behavior of several random variables. This leads to the multivariate Gaussian distribution, which describes random vectors whose components may be correlated but whose joint structure remains entirely determined by first and second moments.

A random vector

$$X = (X_1, \dots, X_n)^\top \in \mathbb{R}^n$$

is said to be Gaussian with mean vector $\mu \in \mathbb{R}^n$ and covariance matrix $\Sigma \in \mathbb{R}^{n\times n}$, written

$$X \sim \mathcal{N}(\mu, \Sigma),$$

if its joint probability density function is

$$f_X(x) = \frac{1}{(2\pi)^{n/2}(\det\Sigma)^{1/2}} \exp\left(-\tfrac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)\right), \qquad x \in \mathbb{R}^n,$$

where Σ is symmetric and positive definite.

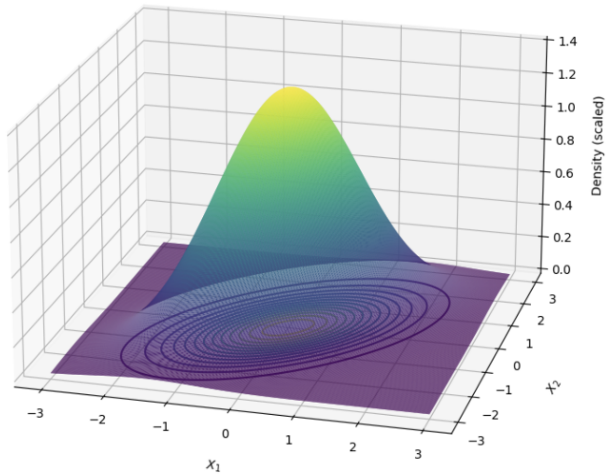
Mean Vector & Covariance Matrix. The components of the mean vector and covariance matrix are given by

$$\mu_i = \mathbb{E}[X_i], \qquad \Sigma_{ij} = \mathrm{Cov}(X_i, X_j) = \mathbb{E}\left[(X_i - \mu_i)(X_j - \mu_j)\right].$$

The diagonal entries Σ_{ii} represent the variances of the individual components, while the off–diagonal entries Σ_{ij} encode their linear dependence. The covariance matrix is always symmetric and positive semidefinite.

A Gaussian vector is completely characterized by first and second moments, and behaves perfectly under linear operations and conditioning.

Bivariate Gaussian Density

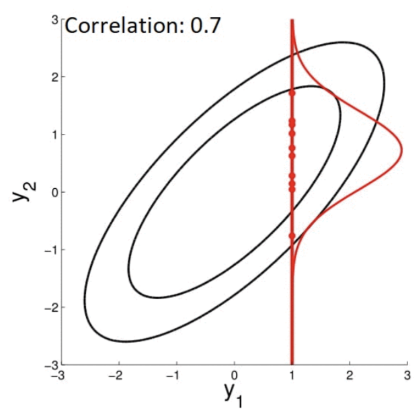


Covariance matrix determines the orientation and spread of probability mass

Correlation Matrix. It is often convenient to separate scale from dependence by normalizing the covariance matrix. The correlation matrix $\rho = (\rho_{ij})$ is defined by

$$\rho_{ij} = \frac{\Sigma_{ij}}{\sqrt{\Sigma_{ii}\Sigma_{jj}}}, \qquad -1 \leq \rho_{ij} \leq 1.$$

While Σ contains both marginal variances and dependence information, ρ captures only the strength of linear association between components.



Elliptical level sets of a bivariate Gaussian and the conditional distribution obtained by fixing one component

How correlation determines the conditional mean and varianceprocess.

Marginal and Conditional Distributions. Any subvector of a Gaussian vector is again Gaussian. Moreover, if X is partitioned as

$$X = (X_A, X_B),$$

then the conditional distribution of X_A given X_B is also Gaussian. This closure under marginalization and conditioning is a defining structural property and will underlie Gaussian process regression.

Independence and Uncorrelatedness. For general random variables, uncorrelatedness does not imply independence. For Gaussian vectors, however, the two notions coincide:

$$X_1, \dots, X_n \text{ are independent} \quad \Longleftrightarrow \quad \Sigma_{ij} = 0 \text{ for all } i \neq j.$$

Thus, a Gaussian vector has independent components if and only if its covariance matrix is diagonal.

Degeneracy. A Gaussian vector is called non–degenerate if $\det(\Sigma) \neq 0$, in which case it admits a density on $\mathbb{R}^n$. If $\det(\Sigma) = 0$, the distribution is degenerate and supported on a lower–dimensional linear subspace, reflecting exact linear relations among the components.

Cholesky Representation. Since Σ is symmetric and positive semidefinite, there exists a matrix A such that

$$\Sigma = AA^\top.$$

If $Z \sim \mathcal{N}(0, I_n)$ is a standard Gaussian vector with independent components, then

$$X = \mu + AZ$$

has distribution $\mathcal{N}(\mu, \Sigma)$. This representation shows that any Gaussian vector can be obtained as a linear transformation of independent standard Gaussians and provides both a geometric interpretation and a practical method for simulation.

The multivariate Gaussian distribution therefore inherits all essential properties of the one–dimensional case: complete characterization by first and second moments, stability under linear operations, and closure under marginalization and conditioning. These features will extend from finite collections of random variables to indexed families, leading naturally to the notion of a Gaussian process.

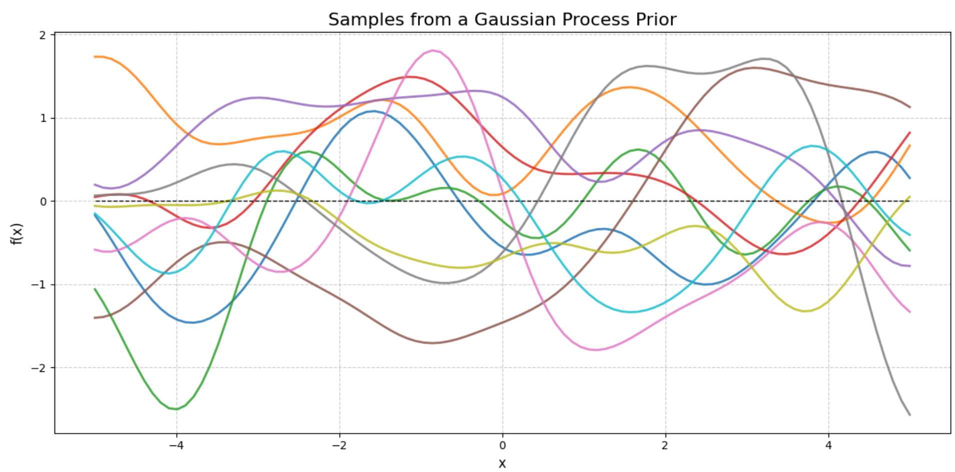
3 Gaussian Processes

We now pass from finite–dimensional Gaussian vectors to collections of random variables indexed by a continuous or discrete parameter set. Such indexed families arise naturally in the modeling of time series, spatial fields, and random functions. The central idea is to extend the multivariate Gaussian structure by requiring that every finite subcollection of the process have a

joint Gaussian distribution, thereby defining an infinite–dimensional object whose local behavior is governed by Gaussian vectors.
A stochastic process $X = \{X_t : t \in \mathcal{T}\}$ is called a *Gaussian process* if, for every finite choice of index points $t_1, \dots, t_n \in \mathcal{T}$, the random vector

$$(X_{t_1}, \dots, X_{t_n})$$

is a Gaussian vector. In other words, all finite–dimensional distributions of the process are multivariate normal.



Sample paths drawn from a zero-mean Gaussian process prior with a squared exponential covariance kernel, illustrating a Gaussian process as a distribution over functions

Mean Function. The mean function of a Gaussian process is defined by

$$m(t) = \mathbb{E}[X_t], \qquad t \in \mathcal{T}.$$

It generalizes the mean vector of a Gaussian random vector and describes the deterministic trend of the process across the index set.

Covariance Function (Kernel). The covariance function, or kernel, is defined by

$$k(s, t) = \mathrm{Cov}(X_s, X_t), \qquad s, t \in \mathcal{T}.$$

It generalizes the covariance matrix and encodes the dependence structure of the process. The kernel is symmetric,

$$k(s, t) = k(t, s),$$

and must be positive semidefinite in the sense that, for any finite collection $t_1, \dots, t_n \in \mathcal{T}$, the matrix

$$K = \big(k(t_i, t_j)\big)_{i,j=1}^{n}$$

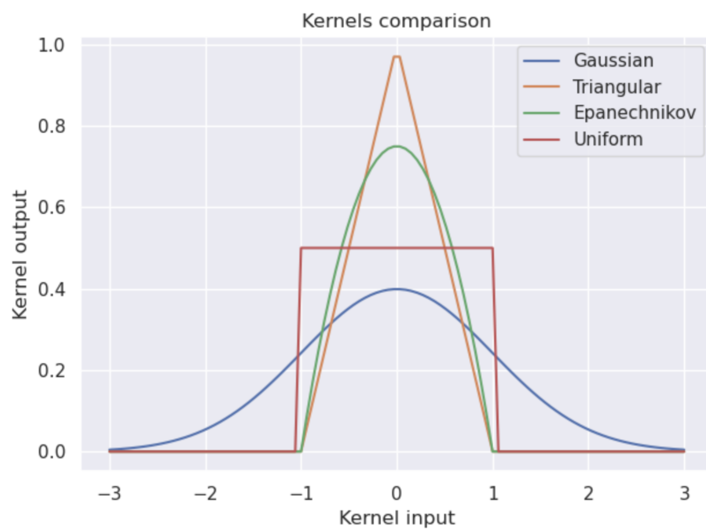
is symmetric and positive semidefinite.

This condition is not merely technical: it is exactly what guarantees that, for every finite set of points, the induced covariance matrix defines a valid multivariate Gaussian distribution.

Kernel as Infinite–Dimensional Covariance Matrix. For each finite subset $\{t_1, \ldots, t_n\} \subset \mathcal{T}$, the kernel induces the covariance matrix

$$K_{ij} = k(t_i, t_j),$$

which coincides with the covariance matrix of the Gaussian vector $(X_{t_1}, \ldots, X_{t_n})$. In this sense, the covariance function plays the role of an infinite–dimensional covariance matrix, whose finite restrictions determine the joint distributions of the process.



Examples of covariance kernels. Different kernel shapes encode different notions of similarity and smoothness, and determine the correlation structure of the associated Gaussian process.

4 Properties of Gaussian Processes

The defining feature of a Gaussian process is that all of its finite–dimensional distributions are Gaussian. As a consequence, many of the structural properties of Gaussian random vectors extend directly to the infinite–dimensional setting. These properties are inherited from the linear and second–order nature of the Gaussian distribution and hold independently of the particular choice of covariance function.

Finite–Dimensional Marginals. For any finite subset $\{t_1, \ldots, t_n\} \subset \mathcal{T}$, the restriction $(X_{t_1}, \ldots, X_{t_n})$ is a Gaussian vector. More generally, restricting a Gaussian process to any subset of the index set yields another Gaussian process. Thus, all local finite collections of values inherit the multivariate Gaussian structure.

Complete Determination by Mean and Covariance. A Gaussian process is fully characterized by its mean function $m(t)$ and covariance function $k(s, t)$. Two Gaussian processes with the same mean and covariance functions have identical finite–dimensional distributions and therefore the same law.

Closure under Linear Combinations. Let $X^{(1)}, \dots, X^{(m)}$ be Gaussian processes defined on the same index set $\mathcal{T}$, and let $a_1(t), \dots, a_m(t)$ be deterministic functions. Then the process

$$Y_t = \sum_{i=1}^{m} a_i(t) X_t^{(i)}$$

is also a Gaussian process. Its mean and covariance functions are obtained by applying the same linear combinations to the mean functions and covariance kernels of the original processes.

Conditional Distributions. Conditioning a Gaussian process on finitely many observed values results in another Gaussian process on the remaining index set. The conditional mean and covariance functions are obtained from the original kernel by the standard formulas for conditioning of multivariate Gaussian distributions. This property underlies inference and prediction for random functions and forms the theoretical basis of Gaussian process regression.

Stationarity. A Gaussian process is called (second–order) stationary if its mean function is constant and its covariance function depends only on the difference of its arguments,

$$m(t) = m, \qquad k(s,t) = k(t-s).$$

In this case, the statistical properties of the process are invariant under shifts of the index variable.

Isotropy. For processes indexed by $\mathbb{R}^d$, a Gaussian process is isotropic if its covariance function depends only on the Euclidean distance between points,

$$k(x, x') = k(\|x - x'\|).$$

Such processes exhibit rotational invariance and are commonly used in spatial modeling.

Regularity of Sample Paths. The smoothness and continuity properties of the sample paths of a Gaussian process are determined by the regularity of its covariance function. Roughly speaking, smoother kernels give rise to smoother sample paths. Precise conditions can be obtained from the Kolmogorov continuity theorem, which relates moment bounds derived from the kernel to almost sure continuity and differentiability of realizations.

Markov Property. Only special classes of Gaussian processes possess the Markov property. For example, Brownian motion and the Ornstein–Uhlenbeck process are Gaussian and Markov, while most Gaussian processes are non–Markovian. The Markov property can be characterized in terms of specific structural forms of the covariance function.

5 Examples of Gaussian Processes

Several fundamental examples of Gaussian processes arise naturally in probability theory and stochastic modeling. Each is fully characterized by its mean and covariance structure, and different choices of covariance functions encode properties such as drift, stationarity, long–range dependence, and conditioning on boundary values. These processes will serve as canonical building blocks for later developments in stochastic calculus and continuous-time modeling.

Brownian Motion. The standard Brownian motion (or Wiener process) $(B_t, t \geq 0)$ is the Gaussian process with mean

$$m(t) = \mathbb{E}[B_t] = 0, \qquad t \geq 0,$$

and covariance function

$$\mathrm{Cov}(B_t, B_s) = \mathbb{E}[B_t B_s] = s \wedge t,$$

where $s \wedge t = \min\{s, t\}$. Moreover, the process admits a version with continuous sample paths, i.e. the mappings $t \mapsto B_t(\omega)$ are continuous for ω in a set of probability one.
More generally, for $\sigma > 0$ and $\mu \in \mathbb{R}$, the process

$$X_t = \sigma B_t + \mu t$$

is a Gaussian process (Brownian motion with drift) with mean $\mathbb{E}[X_t] = \mu t$ and covariance

$$\mathrm{Cov}(X_t, X_s) = \sigma^2 (s \wedge t).$$

Brownian motion describes random continuous motion with no preferred direction, where future changes are independent of the past and variability grows with time.

Brownian Bridge. The Brownian bridge $(Z_t, t \in [0, 1])$ is the Gaussian process with mean

$$\mathbb{E}[Z_t] = 0,$$

and covariance

$$\mathrm{Cov}(Z_t, Z_s) = s(1 - t), \qquad s \leq t,$$

and satisfies $Z_0 = Z_1 = 0$ almost surely. If $(B_t, t \in [0, 1])$ is standard Brownian motion, then the process

$$Z_t = B_t - tB_1, \qquad t \in [0, 1],$$

has the distribution of a Brownian bridge.
A Brownian bridge is a Brownian motion that is forced to start and end at zero, creating random paths that are “pinned down” at both ends.

Fractional Brownian Motion. For $0 < H < 1$, the fractional Brownian motion $(B_t^{(H)}, t \geq 0)$ with Hurst index H is the centered Gaussian process with covariance function

$$\mathrm{Cov}\left(B_t^{(H)}, B_s^{(H)}\right) = \frac{1}{2}\left(t^{2H} + s^{2H} - |t-s|^{2H}\right).$$

The special case $H = \frac{1}{2}$ corresponds to standard Brownian motion, for which the covariance reduces to $s \wedge t$.
Fractional Brownian motion generalizes Brownian motion by allowing long–range dependence, so past movements can positively or negatively influence future ones depending on the Hurst parameter.

Ornstein–Uhlenbeck Process. The Ornstein–Uhlenbeck process $(Y_t, t \geq 0)$ starting from $Y_0 = 0$ is the Gaussian process with mean

$$\mathbb{E}[Y_t] = 0,$$

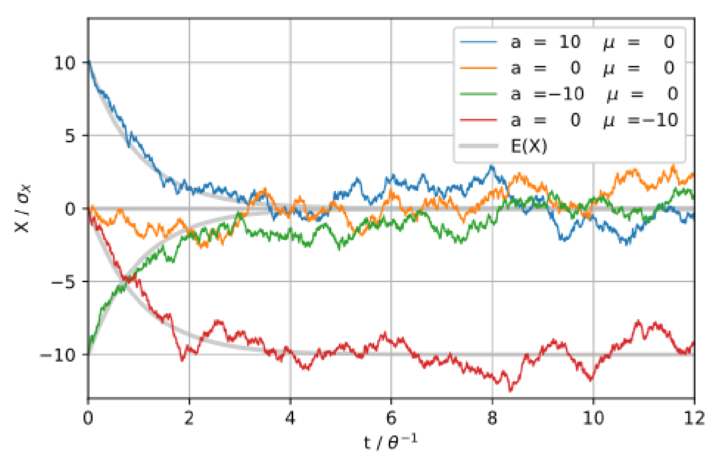
and covariance function

$$\mathrm{Cov}(Y_t, Y_s) = \frac{e^{-2(t-s)}}{2}\left(1 - e^{-2s}\right), \qquad s \leq t.$$

If instead the initial condition Y_0 is Gaussian with mean 0 and variance $1/2$, then the process becomes stationary and the covariance simplifies to

$$\mathrm{Cov}(Y_t, Y_s) = \frac{e^{-2(t-s)}}{2}, \qquad s \leq t,$$

which depends only on the time difference $t - s$.

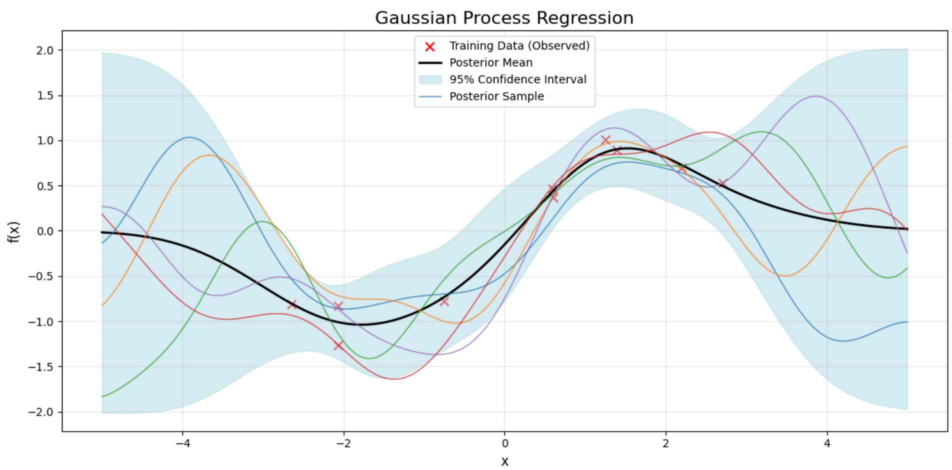


Ornstein–Uhlenbeck process

The Ornstein–Uhlenbeck process models random motion that is pulled back toward a long–run average, so it fluctuates but does not drift off to infinity.

6 Extension: Gaussian Process Regression

Gaussian Process Regression (GPR) is a powerful and flexible non-parametric regression technique used in machine learning and statistics. It is particularly useful when dealing with problems involving continuous data, where the relationship between input variables and output is not explicitly known or can be complex. GPR is a Bayesian approach that can model certainty in predictions, making it a valuable tool for various applications, including optimization, time series forecasting, and more.



We now briefly connect the theory of Gaussian processes to supervised regression. The goal is to infer an unknown function from noisy observations and to quantify uncertainty in predictions at new inputs.
Let $\{(x_i, y_i)\}_{i=1}^n$ be training data with $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$. We model the response as

$$y(x) = h(x)^\top \beta + f(x) + \varepsilon,$$

where $h(x) \in \mathbb{R}^p$ is a vector of known basis functions, $\beta \in \mathbb{R}^p$ is a coefficient vector, $f(x)$ is a latent function drawn from a zero-mean Gaussian process,

$$f(x) \sim \mathcal{GP}\big(0, k(x, x')\big),$$

and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ is independent observation noise.
Thus, conditional on $f(x)$, the observation model is

$$y_i \mid f(x_i), x_i \sim \mathcal{N}\left(h(x_i)^\top \beta + f(x_i),\ \sigma^2\right), \qquad i = 1, \dots, n.$$

In vector form, define

$$X = \begin{pmatrix} x_1^\top \\ x_2^\top \\ \vdots \\ x_n^\top \end{pmatrix}, \qquad y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \qquad H = \begin{pmatrix} h(x_1)^\top \\ h(x_2)^\top \\ \vdots \\ h(x_n)^\top \end{pmatrix}, \qquad f = \begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{pmatrix}.$$

The regression model becomes

$$y = H\beta + f + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_n),$$

so that

$$p(y \mid f, X) = \mathcal{N}(y \mid H\beta + f, \sigma^2 I_n).$$

The latent function values follow the joint Gaussian prior

$$f \mid X \sim \mathcal{N}\big(0, K(X, X)\big),$$

where the kernel (covariance) matrix is

$$K(X, X) = \begin{pmatrix} k(x_1, x_1) & k(x_1, x_2) & \cdots & k(x_1, x_n) \\ k(x_2, x_1) & k(x_2, x_2) & \cdots & k(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ k(x_n, x_1) & k(x_n, x_2) & \cdots & k(x_n, x_n) \end{pmatrix}.$$

The covariance function $k(x, x')$ is parameterized by hyperparameters θ, often written $k(x, x' \mid \theta)$, which control properties such as smoothness, length scale, and amplitude of the latent function.

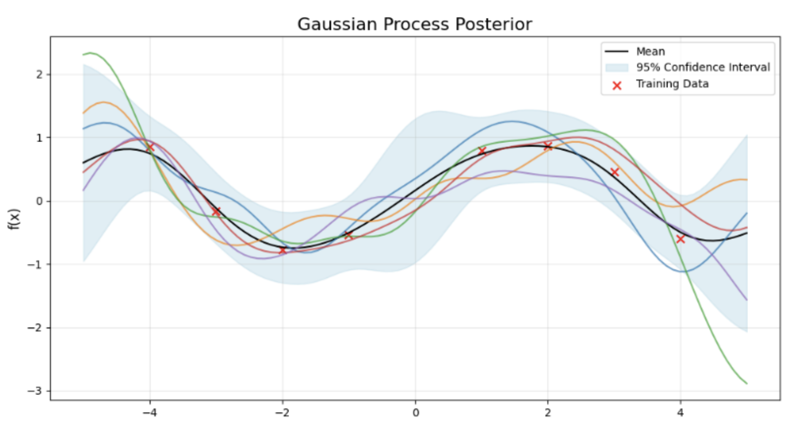
Combining the GP prior with the Gaussian noise model yields the marginal distribution

$$y \mid X \sim \mathcal{N}\left(H\beta,\ K(X, X) + \sigma^2 I_n\right).$$

For a new input x_*, the joint distribution of the observed outputs y and the latent value $f_* = f(x_*)$ is Gaussian, and conditioning yields the posterior predictive distribution

$$p(f_* \mid y, X, x_*),$$

with closed-form mean and covariance given by standard Gaussian conditioning formulas. This distribution provides both a point prediction and a principled measure of uncertainty, which is the defining feature of Gaussian process regression.



Steps in Gaussian Process Regression

1. Data Collection: Gather the input-output data pairs for your regression problem.
2. Choose a Kernel Function: Select an appropriate covariance function (kernel) that suits your problem. The choice of kernel influences the shape of the functions that GPR can model.
3. Parameter Optimization: Estimate the hyperparameters of the kernel function by maximizing the likelihood of the data. This can be done using optimization techniques like gradient descent.
4. Prediction: Given a new input, use the trained GPR model to make predictions. GPR provides both the predicted mean and the associated uncertainty (variance).

Chapter 10: Counting Processes

Counting processes are among the most fundamental building blocks of stochastic process theory, providing a framework for modeling the random occurrence of events over time. At the most general level, a *counting process* records how many events have occurred by time t; when these events are viewed as temporal occurrences, the model is interpreted as an *arrival process*. Additional structure arises with *renewal processes*, which describe systems whose successive interarrival times are independent and identically distributed, capturing repetitive random phenomena such as service completions, failures, or transactions.

At the core of this hierarchy lies the *Poisson process*, a cornerstone of stochastic modeling and the only arrival process with both stationary and independent increments, whose memoryless nature makes it a universal first-order model for randomness in time. Remarkably, a vast range of complex systems—from queueing networks and communication traffic to radioactive decay and financial transactions—can often be understood as refinements, generalizations, or perturbations of the Poisson paradigm.

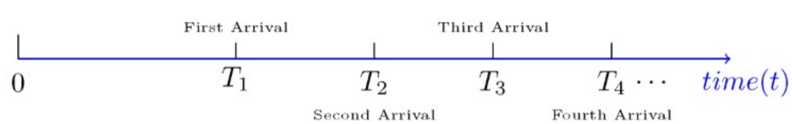
1 Counting Process

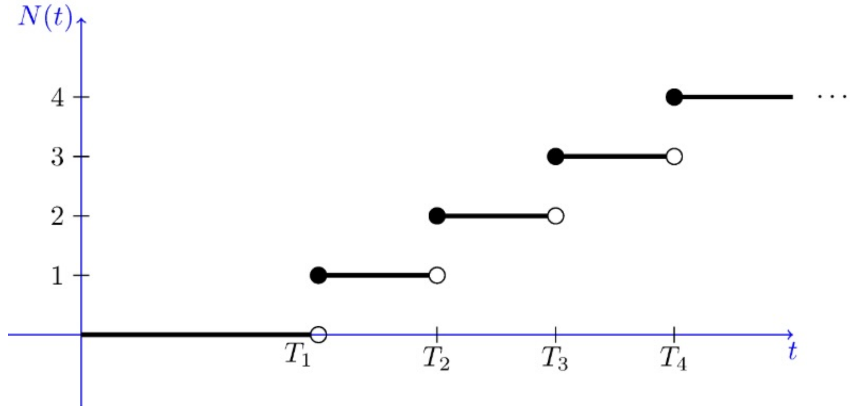
A *counting process* is a way to describe how random events occur over time. It tracks the total number of events that have happened from time 0 up to any time t, increasing by one whenever a new event occurs. The difference $N(t) - N(s)$ indicates how many events took place between times s and t. Counting processes are used to model various types of arrival-like phenomena, such as customers entering a store, phone calls received, or system failures over time.

Counting Process. A random process $\{N(t),\ t \in [0, \infty)\}$ is said to be a *counting process* if $N(t)$ represents the number of events that have occurred from time 0 up to and including time t. For a counting process, we assume:

1. $N(0) = 0$.
2. $N(t) \in \{0, 1, 2, \dots\}$, for all $t \in [0, \infty)$.
3. For $0 \leq s < t$, the increment $N(t) - N(s)$ shows the number of events that occur in the interval $(s, t]$.

Since counting processes are often used to model *arrivals* (such as customer arrivals in a supermarket), each event is typically referred to as an *arrival*. For example, if $N(t)$ denotes the number of accidents in a city up to time t, each accident is treated as an arrival. The figure illustrates a possible realization and corresponding sample path of a counting process.





A sample path of a counting process. Such processes are always right continuous and have a staircase form with jumps of height 1. The randomness is in the times T_i at which whatever we are counting happens

2 Arrival Processes

An arrival process is simply a counting process interpreted in time; it records how many events have occurred by time t and emphasizes the random times at which these events take place. While a counting process $N(t)$ captures only the cumulative number of occurrences, the arrival–process viewpoint highlights the underlying sequence of event times and waiting periods between them. Such processes form the basic input models for queueing systems, communication networks, and traffic flow.

An *arrival process* is a stochastic process $\{N(t), t \geq 0\}$ satisfying

1. $N(t)$ takes values in $\mathbb{Z}_{\geq 0}$
2. $N(0) = 0$
3. $N(t)$ is nondecreasing in t
4. $N(t)$ increases only by jumps of size one

Equivalently, an arrival process may be described by a sequence of strictly increasing nonnegative random variables

$$0 < S_1 < S_2 < \cdots,$$

where S_n denotes the *arrival epoch* of the nth event. The strict ordering $S_{n+1} > S_n$ implies that arrivals occur at distinct times almost surely, so that simultaneous arrivals occur with probability zero. The associated counting process is then

$$N(t) = \max\{n \geq 1 \,:\, S_n \leq t\},$$

which counts the number of arrivals that have occurred by time t. A complete specification of an arrival process is obtained by giving the joint distributions of $(S_1, \ldots, S_n)$ for all $n \geq 1$, or equivalently, the finite–dimensional distributions of $\{N(t), t \geq 0\}$.

Interarrival Times. An equivalent description of an arrival process is given by the *interarrival times* which measure the waiting time between successive arrivals.

$$X_1 = S_1, \qquad X_n = S_n - S_{n-1}, \quad n \geq 2,$$

where X_1 is the waiting time to the first arrival and X_n is the waiting time between the $(n-1)$st and nth arrivals.
Each X_n is a positive random variable, and the arrival epochs can be reconstructed as partial sums,

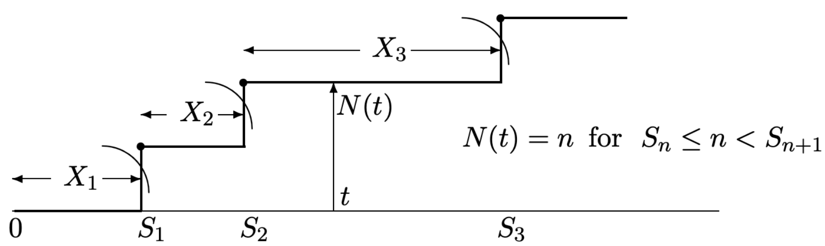
$$S_n = \sum_{i=1}^{n} X_i.$$

In many models of interest, the interarrival times are independent and identically distributed, making this representation particularly convenient.

Counting Process Representation. A third, equivalent representation is the associated *counting process* $\{N(t) : t \geq 0\}$, where $N(t)$ denotes the number of arrivals in $(0, t]$. By definition, $N(0) = 0$ and $N(t)$ is nondecreasing in t. The counting process and arrival epochs are linked by the identities

$$\{S_n \leq t\} = \{N(t) \geq n\}, \qquad \{S_n > t\} = \{N(t) < n\},$$

which allow one to translate statements about arrival times into statements about counts, and vice versa.



Arrival epochs $\{S_n\}$, interarrival times $\{X_n\}$, and the associated counting process $N(t)$.

Equivalent Descriptions. An arrival process may therefore be characterized equivalently by the arrival epochs $\{S_n\}$, the interarrival times $\{X_n\}$, or the counting process $\{N(t)\}$. In principle, specifying the joint distribution of any one of these objects determines the others. In practice, the interarrival–time representation is often the most convenient for modeling, while the counting–process formulation is natural for analyzing the evolution of arrivals over continuous time.

3 Renewal Process

We have seen that a counting process $N(t)$ may be represented through its associated arrival epochs $\{S_n\}$, or equivalently through the interarrival times between successive events. A *renewal process* is a fundamental and widely used special case in which these interarrival times are independent and identically distributed with an arbitrary nonnegative distribution. By relaxing

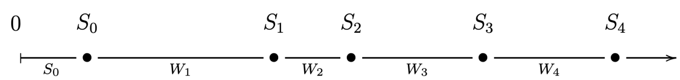
the exponential waiting–time assumption of the Poisson process, the renewal framework models systems in which each event initiates a probabilistic regeneration of the process, while the future evolution depends only on the time elapsed since the most recent renewal.
Let $W_1, W_2, \ldots$ be independent and identically distributed random variables on $[0, \infty[$ with distribution A. We also use the letter A to denote the distribution function of A i.e.

$$A(x) = \mathbb{P}(W_1 \leq x).$$

Let S_0 be another random variable on $[0, \infty[$ with distribution A_0. Suppose that S_0 is independent of $\{W_n\}$. For $i \geq 1$ define

$$S_i = S_0 + W_1 + \cdots + W_i.$$

The process $\mathrm{S} = \{S_n\}_{n \in \mathbb{N}_0}$ is called a *renewal process* with *interarrival* or *waiting–time distribution* A and *delay distribution* A_0.
The interpretation is that S represents the occurrence times of successive events or *renewals.* The times between renewals are assumed to be i.i.d., except that the distribution of the first renewal is allowed to follow a possibly different distribution A_0.



Renewal Counting Process and Renewal Function. Associated with the renewal epochs $\{S_n\}$ is the counting process

$$N(t) = \max\{n \geq 0 \, : \, S_n \leq t\}, \qquad t \geq 0,$$

which counts the number of renewals in the interval $[0, t]$. The *renewal function*

$$H(t) = \mathbb{E}[N(t)], \qquad t \geq 0,$$

gives the expected number of renewals by time t. Since H is nondecreasing and finite on bounded intervals, it may be identified with the distribution function of a measure on $[0, \infty)$, called the *renewal measure.* Using the arrival–time representation, H admits the series expansion

$$H(t) = \sum_{k=0}^{\infty} A_0 * A^{k*}(t),$$

where A^{k*} denotes the k–fold convolution of the interarrival distribution A with itself (that is, the distribution of $W_1 + \cdots + W_k$), and $A^{0*} = \delta_0$ by convention.
The renewal function therefore describes the average growth of the counting process, giving the expected number of regeneration points up to time t and characterizing the typical long–run accumulation of events.

Renewal Equation. In the zero–delayed case $A_0 = \delta_0$, the renewal function $H_0(t) = \mathbb{E}[N_0(t)]$ satisfies the integral equation

$$H_0(t) = 1 + \int_0^t H_0(t-u)\, A(du), \qquad t \ge 0.$$

This *renewal equation* expresses the self–regenerating structure of the process: after each renewal, the future evolution restarts probabilistically from the origin with the same interarrival distribution.

This equation expresses the regenerative structure of the process: after each renewal, the future evolution is probabilistically identical to the original process and is independent of the past before the most recent renewal.

Elementary Renewal Theorem. Assume that the interarrival times have finite mean

$$\mu = \int_0^\infty u\, A(du) < \infty.$$

Then, for any delay distribution A_0, the renewal function satisfies the asymptotic relation

$$\lim_{t\to\infty} \frac{H(t)}{t} = \frac{1}{\mu}.$$

Equivalently,

$$\frac{N(t)}{t} \longrightarrow \frac{1}{\mu} \qquad \text{almost surely and in mean.}$$

Thus, in the long run the renewal process evolves at a constant average rate equal to the reciprocal of the mean interarrival time, and this limiting behavior is independent of the initial delay distribution.

In words, the process eventually settles into a steady rhythm: regardless of how it starts, the average number of renewals per unit time converges to the inverse of the mean waiting time between successive events.

Relation to the Poisson Process. A renewal process generalizes the Poisson process by permitting the interarrival times to follow an arbitrary nonnegative distribution. When the interarrival times are exponential, the renewal process reduces to a Poisson process; the exponential distribution is the unique continuous law with the memoryless property, which in turn implies stationary and independent increments. Consequently, the Poisson process is the only renewal process that is Markovian in continuous time, and thus occupies a central and exceptionally tractable position within the broader class of renewal models.

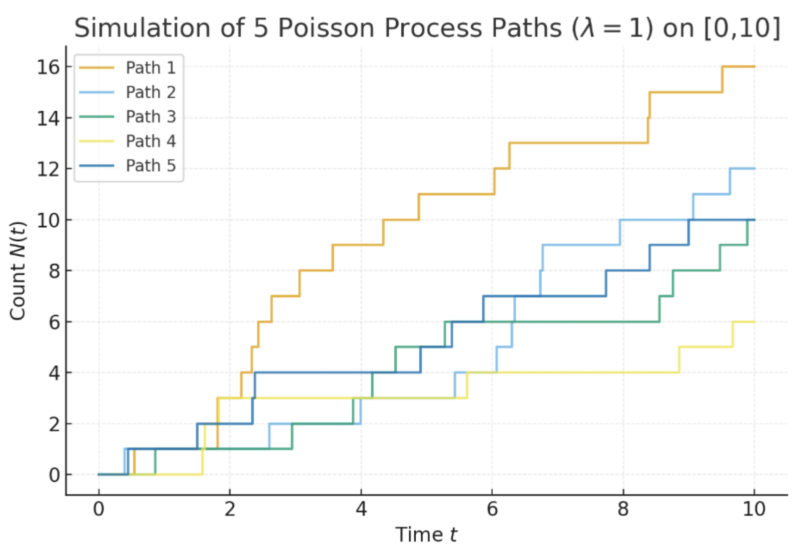
4 Poisson Process

The *Poisson process* occupies a central position in the theory of counting and arrival processes and serves as the fundamental benchmark against which more general models are understood. The development of this chapter naturally leads to it: from abstract counting processes, to arrival epochs, to renewal processes with i.i.d. interarrival times, the Poisson process emerges as the unique case in which the waiting times are exponential and hence memoryless. This special structure endows the process with stationary and independent increments and makes it the only renewal process that is also Markovian in continuous time. Owing to its mathematical tractability and its remarkable ability to model random phenomena ranging from queueing systems and telecommunications to radioactive decay and financial transaction arrivals, the Poisson process is not merely a special case but a cornerstone of stochastic process theory.

Poisson Process. Let $\lambda > 0$ be fixed. A counting process $\{N(t),\ t \geq 0\}$ is called a *Poisson process* with rate λ if it satisfies:

1. $N(0) = 0$;
2. For all $0 \leq s < t$, the increment $N(t) - N(s)$ has a Poisson distribution with parameter $\lambda(t-s)$,
$$N(t) - N(s) \sim \text{Poisson}(\lambda(t-s));$$
3. The process has *independent increments*, that is, the numbers of arrivals in disjoint time intervals are independent random variables.

Property (2) implies that the distribution of the number of arrivals in any interval depends only on the length of the interval and not on its location on the time axis. Consequently, the Poisson process has *stationary increments*. In contrast, a general renewal process does not possess stationary increments unless its interarrival distribution is exponential. Together with the independent–increment property, stationarity of increments characterizes the Poisson process as the unique simple counting process whose increments are both stationary and independent.



A typical sample path of a Poisson process with unit jumps occurring at random times.

Arrival and Interarrival (Jump) Times. A Poisson process $\{N(t), t \geq 0\}$ with rate λ describes random arrivals over time. The *first interarrival time* X_1 is the waiting time until the first event:

$$\mathbb{P}(X_1 > t) = \mathbb{P}(\text{no arrival in } (0, t]) = e^{-\lambda t}, \qquad t \geq 0.$$

These random variables are called the *interarrival times* of the Poisson process and describe how long one must wait for the next event, regardless of the past.

The distribution function of X_1 is

$$F_{X_1}(t) = \begin{cases} 1 - e^{-\lambda t}, & t \geq 0, \\ 0, & t < 0, \end{cases}$$

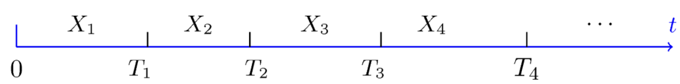
so that $X_1 \sim \text{Exponential}(\lambda)$.

By the independent–increment property, the waiting times between successive arrivals

$$X_i = T_i - T_{i-1}, \qquad i \geq 1,$$

are independent and identically distributed, with

$$X_i \sim \text{Exponential}(\lambda), \qquad i = 1, 2, \ldots$$



Interarrival times $X_1, X_2, \ldots$ and arrival epochs $T_1 < T_2 < \cdots$ of a Poisson process.

The exponential distribution is characterized by the *memoryless property*

$$\mathbb{P}(X > x + a \mid X > a) = \mathbb{P}(X > x),$$

which reflects the "forgetful" nature of Poisson arrivals: the process does not remember how long it has already waited for the next event.

The *arrival epochs* (or *jump times*) $T_1, T_2, T_3, \ldots$ denote the times at which the counting process increases by one, so that $N(t)$ jumps from $k - 1$ to k at time $t = T_k$. As in the general renewal framework, these arrival times admit the sum representation

$$T_k = \sum_{i=1}^{k} X_i,$$

where X_i are the interarrival times. The connection between the arrival epochs and the counting process is

$$\{T_k > t\} = \{N(t) < k\},$$

since the kth arrival occurs after time t if and only if fewer than k arrivals have occurred by time t.

Using the Poisson distribution of $N(t)$,

$$\mathbb{P}(T_k > t) = \sum_{n=0}^{k-1} \frac{(\lambda t)^n e^{-\lambda t}}{n!},$$

so that the distribution function of T_k is

$$F_{T_k}(t) = 1 - \sum_{n=0}^{k-1} \frac{(\lambda t)^n e^{-\lambda t}}{n!}, \qquad t \geq 0.$$

Differentiation yields the density

$$f_{T_k}(t) = \frac{\lambda^k t^{k-1} e^{-\lambda t}}{(k-1)!}, \qquad t > 0,$$

and $f_{T_k}(t) = 0$ for $t < 0$.
Thus,

$$T_k \sim \text{Gamma}(k, \lambda),$$

and each arrival time is the sum of k independent exponential interarrival times:

$$T_k = X_1 + X_2 + \cdots + X_k, \qquad X_i \sim \text{Exp}(\lambda).$$

The Gamma(k, λ) (or *Erlang*) distribution therefore describes the total waiting time until the kth event, with

$$\mathbb{E}[T_k] = \frac{k}{\lambda}, \qquad \text{Var}(T_k) = \frac{k}{\lambda^2}.$$

Although the arrival times $T_1 < T_2 < \cdots$ are not independent, their increments $X_i = T_i - T_{i-1}$ are independent and exponentially distributed. Intuitively, each T_k represents the accumulation of k random waiting periods, and as k increases the total waiting time becomes increasingly regular by the law of large numbers.

Second–Order Properties. In addition to its distributional and renewal structure, the Poisson process admits simple closed–form expressions for its first and second moments.
Mean. For a Poisson process with rate λ,

$$\mathbb{E}[N(t)] = \lambda t,$$

which reflects the interpretation of λ as the average number of arrivals per unit time.
Second moment and autocorrelation. For any $s, t \geq 0$,

$$\mathbb{E}[N(s)N(t)] = \lambda^2 st + \lambda \min\{s, t\}.$$

In particular, if $s \leq t$,

$$\mathbb{E}[N(s)N(t)] = \lambda^2 st + \lambda s.$$

The term $\lambda^2 st$ corresponds to the product of the means, while the additional term $\lambda \min\{s,t\}$ arises from the shared arrivals in the overlapping time interval. This structure reflects the independent–increment property: counts on disjoint intervals are independent, whereas dependence enters only through the portion of time common to both intervals.

Poisson Random Variable. For a Poisson process $\{N(t), t \geq 0\}$ with rate λ, the counting variable $N(t)$ has a Poisson distribution with parameter λt, that is,

$$N(t) \sim \text{Poisson}(\lambda t).$$

Abstracting from the time scale, a random variable X is said to have a *Poisson distribution* with rate $\lambda > 0$ (written $X \sim \text{Poisson}(\lambda)$) if

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad k = 0, 1, 2, \ldots$$

Moments.

$$\mathbb{E}[X] = \lambda, \qquad \text{Var}(X) = \lambda,$$

and in particular,

$$\mathbb{E}[X^2] = \lambda + \lambda^2, \qquad \mathbb{E}[X^3] = \lambda + 3\lambda^2 + \lambda^3.$$

Moment generating function.

$$M_X(t) = \mathbb{E}[e^{tX}] = \exp\big(\lambda(e^t - 1)\big).$$

Additivity. If $X_1 \sim \text{Poisson}(\lambda_1)$ and $X_2 \sim \text{Poisson}(\lambda_2)$ are independent, then

$$X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2),$$

so the sum of independent Poisson random variables is again Poisson, with rate equal to the sum of the individual rates.

Law of Rare Events (Poisson limit of the Binomial). Let $X_n \sim \text{Binomial}(n, p_n)$ with $n \to \infty$, $p_n \to 0$, and $np_n \to \lambda \in (0, \infty)$. Then

$$X_n [n \to \infty] \text{Poisson}(\lambda),$$

that is, the distribution of X_n converges to the Poisson distribution. This expresses the fact that a large number of independent trials with very small success probabilities leads, in the limit, to a Poisson-distributed number of occurrences.

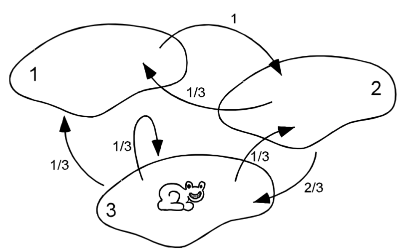
Proof. For fixed k,

$$\mathbb{P}(X_n = k) = \binom{n}{k} p_n^k (1 - p_n)^{n-k} = \frac{n(n-1)\cdots(n-k+1)}{k!}\, p_n^k\, (1 - p_n)^n (1 - p_n)^{-k} \longrightarrow \frac{\lambda^k e^{-\lambda}}{k!}.$$

Chapter 11: Markov Chains

The theory of Markov chains originates in the early twentieth century with the work of Andrey Markov, who studied stochastic sequences in which dependence on the past could be reduced to dependence on the present. His original motivation was neither physics nor finance, but the statistical structure of symbolic sequences, where he showed that meaningful long-run regularity could arise even in the absence of independence. This observation, that global statistical behavior can emerge from strictly local dependence, laid the foundation for a class of stochastic models governed by simple transition rules yet capable of rich asymptotic behavior.

A *Markov chain* is a stochastic process used to model systems that evolve step by step in discrete time. At each time step, the process occupies one of a finite collection of possible states, called the *state space.* Markov chains provide a mathematically precise framework for describing random dynamics in systems where uncertainty unfolds sequentially and transitions are resolved locally in time.



The Markov Frog

The defining assumption underlying Markov chains is the *Markov property.* Informally, this property states that the future evolution of the process depends only on its current state and not on the sequence of states that preceded it. In effect, the process carries no memory beyond the present.

$$\text{future} = f(\text{present}), \qquad \text{i.e., independent of the past}$$

$$f(X_{n+1} \mid X_0, X_1, \dots, X_n) = f(X_{n+1} \mid X_n)$$

This memoryless structure allows the behavior of the entire process to be encoded through local transition rules, while still giving rise to rich long-run phenomena that are the focus of this chapter.

1 Discrete-Time Markov Chain

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A *discrete-time Markov chain* (DTMC) is a stochastic process $\{X_n\}_{n\geq 0}$ indexed by the nonnegative integers that satisfies the Markov property and takes values in a finite set $\mathcal{S} = \{1, 2, \dots, N\}$.

Formally, the Markov property requires that, for all $n \geq 0$ and all states $i_0, i_1, \dots, i_n, j \in \mathcal{S}$,

$$P(X_{n+1} = j \mid X_n = i_n, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i_n).$$

The process $\{X_n\}_{n\geq 0}$ is indexed by $n \in \mathbb{Z}_{\geq 0}$ and takes values in a finite set $\mathcal{S}$, called the *state space*. The index n represents discrete time, with transitions occurring at integer-valued time steps.

Time-Homogeneous Markov Chains. A Markov chain is said to be *time-homogeneous* if the conditional transition probabilities

$$P(X_{n+1} = j \mid X_n = i)$$

do not depend on the time index n. In this case, the transition behavior of the chain is the same at every step.

Time-Inhomogeneous Markov Chains. If the transition probabilities depend explicitly on n, the chain is called *time-inhomogeneous*. In this setting, the probabilistic rules governing state transitions may vary over time.

Unless stated otherwise, all Markov chains considered in this chapter are finite-state, discrete-time, and time-homogeneous.

2 Transition Structure and Evolution

Transition Probabilities. For a time-homogeneous discrete-time Markov chain with state space $\mathcal{S} = \{1, 2, \dots, N\}$, the one-step transition probabilities are defined by

$$p_{ij} \;=\; P(X_{n+1} = j \mid X_n = i), \qquad i, j \in \mathcal{S}.$$

$$i \xrightarrow{P_{ij}(n)} j$$

$P_{ij}(n)$ = probability of moving from state i to j, at time n.

These probabilities specify the conditional distribution of the next state given the current state.

Transition Matrix. The transition probabilities are collected into the *transition matrix*

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1N} \\ p_{21} & p_{22} & \cdots & p_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ p_{N1} & p_{N2} & \cdots & p_{NN} \end{bmatrix}.$$

Each row of P corresponds to a current state, and each entry gives the probability of transitioning to a future state.

Stochastic Matrix Properties. The transition matrix P is row-stochastic:

$$\sum_{j=1}^{N} p_{ij} = 1, \qquad p_{ij} \geq 0, \qquad \forall\, i.$$

Thus, each row of P defines a probability distribution over the state space.

Graph Interpretation. A finite-state Markov chain can be represented as a weighted directed graph whose nodes correspond to states in $\mathcal{S}$. A directed edge from state i to state j exists if $p_{ij} > 0$, with edge weight equal to p_{ij}. This representation emphasizes reachability and path structure without altering the underlying probabilistic model.

Initial State and Initial Distribution. The chain may be initialized in a deterministic state $X_0 = i$ or according to an initial distribution

$$p_0 = (p_{01}, p_{02}, \dots, p_{0N}), \qquad \sum_{i=1}^{N} p_{0i} = 1.$$

The vector p_0 specifies the distribution of X_0 over the state space.

One-Step Evolution. Given an initial distribution p_0, the distribution of the chain after one step is

$$p_1 = p_0 P.$$

This update applies the transition probabilities once, redistributing mass from the current states to the possible next states according to the rows of the transition matrix.

Multi-Step Transitions and Matrix Powers. The distribution of the chain after k steps is given by

$$p_k = p_0 P^k.$$

The (i, j) entry of P^k gives the probability that the chain, starting in state i, is in state j after exactly k transitions. Matrix powers therefore encode the cumulative effect of repeated applications of the transition mechanism.

Evolution of State Distributions. Successive powers of the transition matrix encode the cumulative effect of repeated transitions. Each multiplication by P advances the chain by one time step, and the sequence $\{p_k\}$ describes the full probabilistic evolution of the process.

3 State and Chain Classification

This section collects the main classification concepts used to describe a finite-state Markov chain beyond its one-step transition rules. We first organize the state space using reachability and communication, which reveals how the chain decomposes into classes that may be closed or accessible. We then classify states by their long-run return behavior and timing through recurrence, transience, and periodicity. These structural conditions culminate in the aperiodic–irreducible setting, where the chain admits a unique stationary distribution and converges to equilibrium.

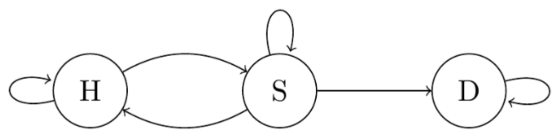
Connectivity and decomposition describe which states can be reached from others and how the state space breaks into mutually accessible subsets.

Reachability. A state $j \in \mathcal{S}$ is said to be *reachable* from a state $i \in \mathcal{S}$ if there exists an integer $k \geq 1$ such that

$$(P^k)_{ij} > 0.$$

In this case, the chain can move from i to j in a finite number of steps with positive probability.

Communication and Communicating Classes. Reachability is directional; mutual reachability leads to the communication relation. Two states $i, j \in \mathcal{S}$ are said to *communicate* if j is reachable from i and i is reachable from j. Communication defines an equivalence relation on the state space, and the resulting equivalence classes are called *communicating classes*. The state space $\mathcal{S}$ can be partitioned uniquely into disjoint communicating classes.



H and S communicate with each other, while D only communicates with itself

Closed and Open Classes. Once the state space is partitioned into communicating classes, the next distinction is whether a class can be exited. A communicating class $C \subseteq \mathcal{S}$ is said to be *closed* if

$$P(X_{n+1} \in C \mid X_n \in C) = 1,$$

that is, once the chain enters C, it cannot leave. A communicating class that is not closed is called *open*. Closed classes are subsets of states from which the chain cannot escape, while open classes may be exited with positive probability.

Irreducibility. A transition matrix $P \in \mathbb{R}^{n \times n}$ is said to be *irreducible* if, for every pair of states $i, j \in \mathcal{S}$, there exists an integer $k \geq 1$ such that

$$(P^k)_{ij} > 0.$$

That is, every state can reach every other state in a finite number of steps with positive probability. Equivalently, P is irreducible if the directed graph associated with it is *strongly connected*. Irreducibility is a global structural property and rules out the presence of isolated or disjoint subsets of states.

A separate long-run classification concerns whether the chain returns to a state once it leaves.

Recurrence and Transience. Each state of a discrete-time Markov chain is classified as either *recurrent* or *transient*. Let $i \in \mathcal{S}$ be a state of the chain $\{X_n\}_{n \geq 0}$.

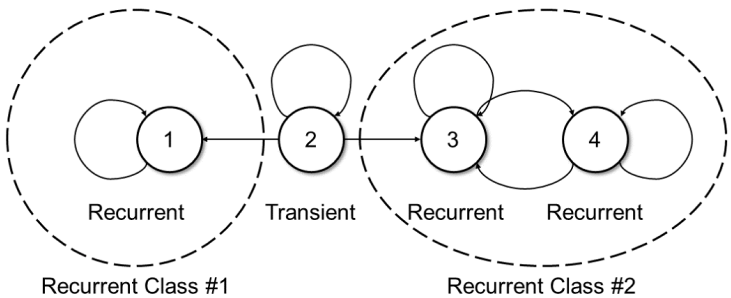
- *Recurrent state:* State i is called recurrent if it can be accessed from any state

$$P(X_n = i \text{ for some } n \geq 1 \mid X_0 = i) = 1.$$

- *Transient state:* If state i is not recurrent, then we can say that the state is transient

$$P(X_n = i \text{ for some } n \geq 1 \mid X_0 = i) < 1.$$

A recurrent state is revisited with probability one, while a transient state may be left and never visited again.



Recurrent vs transient states.

Recurrence and Communicating Classes. Because communication links states through paths in both directions, recurrence and transience are class properties: if two states communicate, then they are either both recurrent or both transient. Consequently, each communicating class is classified as recurrent or transient.

First Return Times. Recurrence can be expressed using the first time the chain returns to its starting state. For a state $i \in \mathcal{S}$, define the first return time

$$T_i = \inf\{n \geq 1 \,:\, X_n = i\}.$$

A state is recurrent if $P(T_i < \infty \mid X_0 = i) = 1$ and transient otherwise.
Beyond whether the chain returns, it is often important to understand *when* returns can occur. This leads to periodicity.

Period of a State. For a state $i \in \mathcal{S}$, define the set of return times

$$\mathcal{N}_i = \{n \geq 1 \,:\, (P^n)_{ii} > 0\}.$$

The *period* of state i is defined as

$$d(i) = \gcd \mathcal{N}_i.$$

Periodicity and Aperiodicity. The period summarizes the arithmetic pattern of possible return times. A state i is called *periodic* if $d(i) > 1$ and *aperiodic* if $d(i) = 1$. All states within a communicating class have the same period. A Markov chain is called *aperiodic* if all of its states are aperiodic.
Periodicity corresponds to cyclic timing of returns, while aperiodicity rules out deterministic oscillations and allows returns to occur at irregular times. The period of a state is the number of

steps it takes to return to itself, and it must always be the same. If a state can only return in 2, 4, 6... steps, its period is 2. If a state can return at irregular intervals, 2, 3, 4 steps, it's aperiodic. With irreducibility providing global connectivity and aperiodicity ruling out cyclic trapping, the chain exhibits a stable long-run behavior.

Aperiodic–Irreducible (AI) Theorem. Let P be the transition matrix of a finite-state, discrete-time Markov chain. If the chain is both *irreducible* and *aperiodic*, then there exists a stationary distribution π_e satisfying the following properties:

1. π_e is invariant under the transition dynamics:
$$\pi_e P = \pi_e.$$
2. The transition matrix converges:
$$\lim_{k\to\infty} P^k = \begin{bmatrix}\pi_e \\ \pi_e \\ \vdots \\ \pi_e\end{bmatrix}.$$
In the long run, each row of P^k converges to π_e, independently of the initial state.
3. For any initial distribution u,
$$\lim_{k\to\infty} uP^k = \pi_e.$$

Irreducibility ensures that all states can be reached from one another, while aperiodicity prevents persistent oscillations. Together, these conditions guarantee convergence to a unique long-run equilibrium distribution, independent of how the chain is initialized.

Existence and Uniqueness. In a finite state space, irreducibility guarantees the existence of a stationary distribution, and aperiodicity ensures that this stationary distribution is unique.

4 Stationary Distributions

A *stationary distribution* π for a Markov chain with transition matrix P is a probability vector satisfying
$$\pi P = \pi, \qquad \sum_{i\in\mathcal{S}} \pi_i = 1, \qquad \pi_i \geq 0.$$

These equations are known as the *global balance equations.* If the chain is initialized with distribution π, then the distribution remains unchanged at all future times.

The evolution equation $p_k = p_0 P^k$ describes how the state distribution changes over time, whereas a stationary distribution is a fixed point of this evolution. If $p_0 = \pi$, then $p_k = \pi$ for all $k \geq 0$.

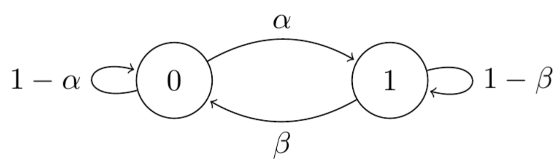
Stationarity characterizes the equilibrium mix of states in the long run. For finite-state Markov chains that are irreducible and aperiodic, the stationary distribution exists, is unique, and attracts all initial distributions. If the chain has multiple closed classes, stationary distributions are not unique; if the chain has transient states, those states receive zero stationary mass.

Stationary distributions are computed by solving the linear system $\pi P = \pi$ together with the normalization condition $\sum_i \pi_i = 1$.

Two-State Chain: Closed-Form Stationary Distribution. Consider a finite-state Markov chain with state space $\{1, 2\}$ and transition matrix

$$P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}, \qquad a, b \in [0, 1].$$

Here $a = P(1 \to 2)$ and $b = P(2 \to 1)$ denote the cross-transition probabilities.



State transition diagram for two-state Markov chain.

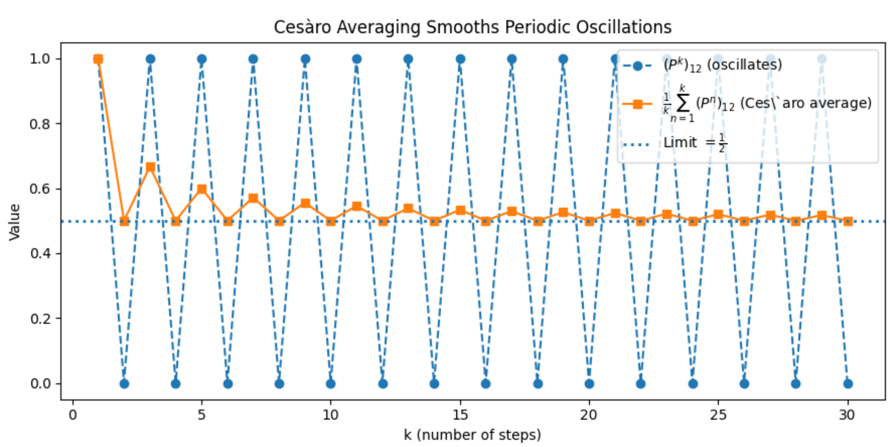
A stationary distribution $\pi = (\pi_1, \pi_2)$ satisfies $\pi P = \pi$ and $\pi_1 + \pi_2 = 1$. Solving,

$$\pi_1 = \frac{b}{a+b}, \qquad \pi_2 = \frac{a}{a+b}, \quad \text{provided } a + b > 0.$$

The stationary mass assigned to a state increases as that state becomes harder to leave relative to how often it is entered.

5 Limiting Behavior and Time Averages

Limiting Probabilities. A natural question is whether the powers of the transition matrix converge as $k \to \infty$. In general, the limit $\lim_{k\to\infty} P^k$ need not exist. In particular, periodic chains may exhibit persistent oscillations. Nevertheless, the *Cesàro averages* of the powers always converge and provide a meaningful description of long-run behavior.

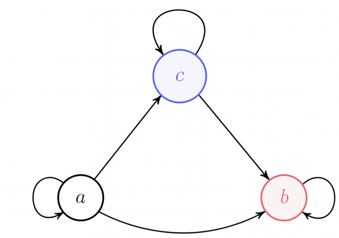


Cesàro averaging of a periodic Markov chain.
Although P^k oscillates, the averaged powers converge to the long-run limiting distribution.

Ergodic class. A set C of states is an *ergodic class* if:

1. Every pair in C communicates (all mutually reachable), and
2. C is *closed*: once you enter C, you never leave ($p_{ij} = 0$ for all $i \in C$, $j \notin C$).

Inside an ergodic class the chain wanders forever; all states in C are recurrent.



In this diagram, state a is *transient* because it can transition to b or c but has no returning paths. States b and c form a *closed ergodic class*: once the chain enters this pair, it remains within them forever, with transitions occurring between b and c or staying in place via their self-loops.

Theorem (Cesàro Limit of Powers).

$$\lim_{l \to \infty} \frac{1}{l} \sum_{k=0}^{l-1} P^k \; = \; P^*.$$

That is, the average of the first l powers of P converges to a limit matrix P^*.

Properties of the Cesàro Limit.

1. $P^* = (p^*_{ij})$ is a transition matrix: $p^*_{ij} \geq 0$ and $\sum_j p^*_{ij} = 1$.
2. p^*_{ij} equals the *long-run fraction of time* the chain is in state j given it started in i.
   Time-averages converge to stationary "visiting frequencies."
3. If i and m lie in the same ergodic class, then $p^*_{ij} = p^*_{mj}$ for all j.
   Within one closed communicating class, the chain forgets the starting state when averaged over time.
4. If the whole chain is a single ergodic class (irreducible), then every row of P^* is identical.
   Everyone shares the same long-run visiting frequencies.
5. If j is transient, then $p^*_{ij} = 0$ for all i.
   Transient states contribute zero long-run time.

Irreducible Case: Cesàro Limit Structure. If the chain is irreducible with stationary distribution $\pi = (\pi_1, \pi_2, \pi_3)$, then

$$\lim_{l \to \infty} \frac{1}{l} \sum_{k=0}^{l-1} P^k = P^* = \begin{bmatrix} \pi_1 & \pi_2 & \pi_3 \\ \pi_1 & \pi_2 & \pi_3 \\ \pi_1 & \pi_2 & \pi_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} \pi_1 & \pi_2 & \pi_3 \end{bmatrix},$$

where (π_1, π_2, π_3) is the left eigenvector of P with eigenvalue 1 and unique stationary distribution satisfying

$$\pi P = \pi, \qquad \pi_i \geq 0, \quad \sum_i \pi_i = 1.$$

Remark (Oscillatory Chains). Consider

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \text{which has period 2.}$$

Then

$$P^{\,l=\text{even}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad P^{\,l=\text{odd}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

so the limit $\lim_{k\to\infty} P^k$ does *not* exist. However,

$$\lim_{l\to\infty} \frac{1}{l} \sum_{k=0}^{l-1} P^k = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Time averaging smooths out periodic oscillations and reveals the correct long-run visiting frequencies.

Remark (Multiple Ergodic Classes and Transient States). Suppose a finite-state Markov chain has two closed communicating classes, $(1, 2, 3, 8)$ and $(4, 5)$, and a set of transient states $\{6, 7\}$. After relabeling, the transition matrix P can be written in block form

$$P = \begin{bmatrix} P_{11} & 0 & * \\ 0 & P_{22} & * \\ 0 & 0 & P_{33} \end{bmatrix},$$

where P_{11} and P_{22} are stochastic (closed-class) blocks and P_{33} is substochastic (transient block). The stationary equations ($\lambda = 1$ left-eigenvector equations) decouple across closed classes:

$$(\pi_1, \pi_2, \pi_3, \pi_8)\, P_{11} = (\pi_1, \pi_2, \pi_3, \pi_8), \qquad (\pi_4, \pi_5)\, P_{22} = (\pi_4, \pi_5),$$

while

$$\pi_6 = \pi_7 = 0.$$

Thus, the P_{33} block is of the form

$$P_{33} = \begin{bmatrix} p_{66} & p_{67} \\ p_{76} & p_{77} \end{bmatrix},$$

$$p_{66} + p_{67} < 1, \quad p_{76} + p_{77} < 1,$$

Since the rows of P_{33} sum to less than one,

$$P_{33}^{l} \;\to\; 0 \quad \text{as } l \to \infty.$$

Thus transient mass vanishes in the long run, and the Cesàro limit P^* assigns positive mass only to closed classes, with zero columns corresponding to transient states and identical rows within each closed class.

Even when P^k fails to converge, Cesàro averages always capture the correct long-run behavior: time spent within each closed class follows its stationary distribution, while transient states disappear from the limiting description.

6 Limit Theorems for Markov Chains

Markov Chain Law of Large Numbers (MCLLN). Let $\{X_k\}_{k\geq 1}$ be a Markov chain with stationary distribution π. Suppose that

- π is the stationary pdf
- The Markov chain is ergodic,
- $f : \mathcal{S} \to \mathbb{R}$ is measurable,
- $\mathbb{E}_\pi[\,|f(X)|\,] < \infty$.

Then, with probability one,

$$\bar{f}_n(x) = \frac{1}{n}\sum_{k=1}^{n} f(X_k) \;\longrightarrow\; \mathbb{E}_\pi[f(X)].$$

Thus, even though the observations $\{X_k\}$ are dependent, their time average converges to the expectation of f under the stationary distribution.

Markov Chain Central Limit Theorem (MC CLT). Under additional regularity conditions, the centered and rescaled time average satisfies

$$\sqrt{n}\left(\bar{f}_n - \mathbb{E}_\pi[f]\right) \;\xrightarrow{d}\; W \sim \mathcal{N}(0, \tau_f^2),$$

provided $\tau_f^2 < \infty$, where

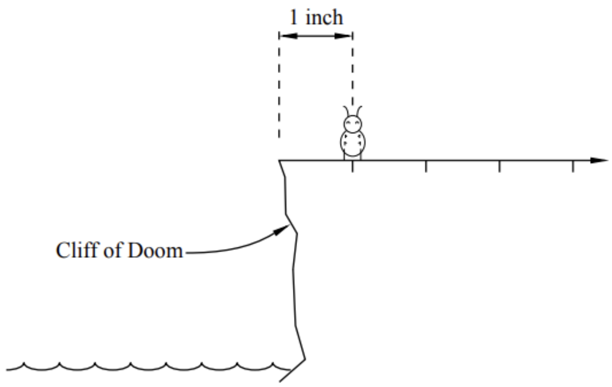
$$\tau_f^2 = \mathrm{Var}_\pi[f(X)] + \mathrm{Cov}_\pi\big(f(X), f(X_1)\big).$$

The asymptotic variance τ_f^2 accounts for serial dependence along the chain and reduces to the usual variance only in the independent case.

These results show that *independence is not required for classical limit behavior*: ergodic Markov chains satisfy laws of large numbers and central limit theorems, with dependence reflected through the asymptotic variance.

Chapter 12: Random Walks

A *random walk* models the cumulative effect of a sequence of random, independent movements. At each step, a particle (or a price, or a flea named *Stencil*) moves one unit to the right or left, determined by chance. Despite the simplicity, random walks capture the essence of uncertainty and form the foundation of modern stochastic processes.



The example above tells the story of *Stencil the flea*. He begins on a plateau one inch away from the “Cliff of Doom.” Each second, he hops left or right with equal probability. If he steps off the edge, his random journey ends abruptly. This toy scenario perfectly encapsulates the idea of randomness under simple, repeatable rules.

1 Definition and Basic Properties

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Discrete-Time Random Walk. A *discrete-time random walk* $\{S_n\}_{n\geq 0}$ is defined by

$$S_0 = 0, \qquad S_n = \sum_{i=1}^{n} X_i,$$

where $\{X_i\}$ are i.i.d. random variables representing the individual steps.
Conceptually: the process starts at S_0 and, at each time step, adds an independent random increment X_i, so the position after n steps is the cumulative sum of all past increments.